A DESIGN APPLICATION HANDBOOK OF INDIRECT BIOPHILIC ATTRIBUTES IN HAWAII HOTEL INTERIOR DESIGN ——COMPARATIVE STUDY ON WAIKIKI 4-STAR HOTELS

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By

Chunya Wu

Dissertation Committee:
Hyoung-June Park, Chairperson
Roger Chen, Advisor
Janice Y.K. Li, Advisor

Keywords: Biophilic Design; Hotel Interior Design; Nature Language Processing; Environmental Psychology; Customer-action Analysis

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ABSTRACT

This study investigates the influence of indirect biophilic design elements on guest experiences in Waikiki four-star hotels. Biophilic design, integrating nature-inspired elements in architecture and interior design, is believed to enhance well-being and satisfaction, particularly in hospitality settings. Focusing on indirect biophilic attributes, this research proposes a novel Biophilic Design Indirect Attributes (BDIA) model to evaluate these elements quantitatively and qualitatively in hotel interiors, such as guest rooms, lobbies, and dining areas. The study employs a mixed-method approach, including natural language processing of customer reviews and expert assessments, to analyze the impact of these designs on customer experience. Theoretical underpinnings from environmental psychology, including Attention Restoration Theory and Stress Reduction Theory, are used to explain customer responses. The findings are expected to provide insights into the effectiveness of biophilic design in the hospitality industry and offer practical recommendations for hotel design and marketing strategies.

Keywords: Biophilic Design; Hotel Interior Design; Nature Language Processing; Environmental Psychology; Customer-Action Analysis

CONTENTS

ACKNOWLEDGMENTS i
ABSTRACT ii
CONTENTS iii
LIST OF TABLES vi
LIST OF FIGURES viii
LIST OF ABBREVIATIONS x
CHAPTER 1. INTRODUCTION 1
1.1 Research Background 1
1.2 Research Objectives 2
1.3 Research Framework 5
CHAPTER 2. LITERATURE REVIEW 7
2.1 Biophilic Design in Hotel interior Design 7
2.1.1 Physiological and Psychological Benefits 8
2.1.2 Economical Benefits 10
2.1.3 Environmental Benefits 12
2.1.4 Research Gap 13
2.2 Hotel Design in The Information Age 16
2.2.1 Machine learning 18
2.2.2 Natural Language Processing (NLP) 21
2.3 Consume as an Integrated Process 24
2.3.1 Environmental Perception 25
2.3.2 Servicescape Evaluation 31
2.3.3 Consumer Decision Making and Behavior 32
CHAPTER 3. BDIA FEATURE ENGINEERING 34
3.1 Version 01 36
3.2 Version 02 40
3.3 Final Version 45

3.4 Conclusion 59
CHAPTER 4. BIG DATA ANALYSIS AND RESULTS 61
4.1 Descriptive Statistical Analysis 61
4.2 BDIA Weight Adjustment 65
4.3 Comments Emotional Tendency 73
4.3.1 Comments Emotional Tendency by Scenarios 78
4.3.2 Comments Emotional Tendency by Keywords 86
4.3.3 Comments Emotional Tendency by Attributes 89
4.4 Discussion and Conclusion 93
CHAPTER 5. MODEL TESTING BY CUSTOMER ONLINE SURVEY 96
5.1 Survey Design 96
5.2 Results 101
5.2.1 Descriptive Statistics 101
5.2.2 Chi-square analysis 102
5.2.3 Correlation Analysis 103
5.2.4 Path Analysis 106
5.3 Discussion 112
5.3.1 Hypothesis Verifying 112
5.3.2 Comparison of correlation analysis results 113
5.3.3 Possible impact mechanism of BDIA 115
5.4 Conclusion 115
CHAPTER 6. MODEL TESTING BY OTHER HAWAII HOTELS 117
6.1 Hotel Selection and Comparisons 117
6.2 Results 124
6.2.1 Descriptive Results 124
6.2.2 Correlation Analysis Results 126
6.3 Discussion 127
6.3.1 BDIA in Hawaii Hotel Interior Design 127
6.3.2 BDIA-Index Validation 128

6.4 Conclusion 129
CHAPTER 7. APPLICATION ADVICE FOR HOTEL DESIGNERS 130
CHAPTER 8. CONCLUSION 142
8.1 BDIA Results 142
8.1.1 BDIA in Hotel Evaluation 142
8.1.2 Possible impact mechanism of BDIA 143
8.1.3 BDIA in Hawaii Hotel Interior Design 143
8.1.4 BDIA-Index as a valid prediction tool 144
8.2 Research Significance and Importance 145
8.3 Limitations and Futural Studies 146
8.3.1 Limitations 146
8.3.2 Futural Studies 147
APPENDIX 149
REFERENCES 150

LIST OF TABLES

Table 1. Indirect Experience of Nature Biophilic Attributes P13

Table 2. Selected Attributes description in the first round P37

Table 3. ANOVA Result in the first round P39

Table 4. Pearson Correlation Result in the first round P40

Table 5. All hotels that meet the requirement and the 6 selected ones P42

Table 6. Selected hotels introduction P43

Table 7. Pearson Correlation Result in the second round P45

Table 8. BDIA Checklist in the final round P46

Table 9. Scoresheet sample in the final round (Wayfinder Waikiki) P58

Table 10. Pearson Correlation Result in the final round P59

Table 11. Selected 4-Stars Hotels in Waikiki P63

Table 12. Descriptive statistical analysis results P63

Table 13. Descriptive statistical analysis results by Scenarios P65

Table 14. Pearson Correlation Analysis Result on 26 Waikiki Hotels P67

Table 15. Machine Learning Parameter Model summary table P68

Table 16. Machine Learning Parameter Model evaluation result P68

Table 17. Weight analysis Result P69

Table 18. Simplified BDIA prediction indicator P70

Table 19. Simplified Machine Learning Parameter Model evaluation result P71

Table 20. Simplified weight analysis Result P72

Table 21. Simplified correlation analysis Result P72

Table 22. Linear regression analysis parameters P73

Table 23. Output example: OUTRIGGER Reef Waikiki Beach Resort P77

Table 27. Room space review emotional tendency ranking list P80

Table 28. Correlation Analysis Result by Design Scenarios - Guestroom P82

Table 29. Lobby space review emotional tendency ranking list P83

Table 30. Correlation Analysis Result by Design Scenarios - Lobby P84

Table 31. Lounge space review emotional tendency ranking list P86

Table 32. Correlation Analysis Result by Design Scenarios - Lounge P87

Table 33. Output example: “Design” as Keyword P89

Table 34. Correlation Analysis Result by Design Keywords P90

Table 35. Comments Attributes Emotional Tendency Result by BDIA P91

Table 36. Analysis results summary by BDIA P97

Table 37. Customer Online Questionnaire Design P100

Table 38. Perception and Evaluation Chi-square analysis results P105

Table 39. S-O-M Model Correlation Analysis results P107

Table 40. Local Identity Correlation Analysis results P107

Table 41. Servicecape Evaluation Correlation Analysis results P108

Table 42. WTPP Correlation Analysis results P109

Table 43. Path analysis model regression coefficient summary P110

Table 44. Analysis results summary for comparison P114

Table 45. Selected Hotels for BDIA-Index Validation Testing P120

Table 46. Selected hotels for BDIA-Testing introduction P121

Table 47. BDIA-Index Correlation Analysis Results for BDIA-Testing Hotels P129

LIST OF FIGURES

Figure 1 Kellert 24 Biophilic Attributes Source: Author 2
Figure 2 Research scope Source: Author 3
Figure 3 Research Framework Source: Author 5
Figure 4 A comparison between Kellert and Brown's Biophilic Design Theory Source: Author 7
Figure 5 Random Forest Algorithm structural diagram Source: Author 20
Figure 6 NLP Scope Source: Author 21
Figure 7 Customer Cognitive Process Source: Author 24
Figure 8 BID-M Application sample 34
Figure 9 Feature engineering process Source: Author 35
Figure 10 Demographic Results in the First Round Source: Author 36
Figure 11 20 Selected Attributes in the first round Source: Author 37
Figure 12 Demographic Results in the Following Round Source: Author 40
Figure 13 The second round of evaluation process Source: Author 44
Figure 14 Crawler data string indication Source: Author 74
Figure 15 Schematic diagram of data preprocessing process Source: Author .. 78
Figure 16 Hotel rooms with the most positive and negative emotional experiences 79
Figure 17 Hotel Lobbies with the most positive and negative emotional experiences 82
Figure 18 Hotel Loundges with the most positive and negative emotional experiences 85
Figure 19 Demographic cross-analysis results Source: Author 101
Figure 20 Perception and Evaluation Chi-square analysis results Source: Author 103
Figure 21 Theoretical model of path analysis Source: Author 107
Figure 22 Modified impact path model Source: Author 110
Figure 24 Structural model results from Literature Review Source:

Yadegaridehkordi, E., Nilashi, M., Nasir, M. H. N. B. M., & Ibrahim, O. (2018). Predicting determinants of hotel success and development using Structural Equation Modelling (SEM)-ANFIS method. *Tourism Management*, *66*, 364-386. 111

Figure 23 Re-specified Loyalty Outcome Model Revealing the Mediating Role of Customer Satisfaction in the Mid-Market Source: Jana, A., & Chandra, B. (2016). Mediating role of customer satisfaction in the mid-market hotels: An empirical analysis. *Indian Journal of Science and Technology*, *9*(1), 1-16.112

Figure 25 Roadmap for sampling to verify the validity of BDIA-Index 117

Figure 26 BDIA-Scenario Mean ± Standard deviation Result by Group..... 124

Figure 27 Average BDIA Scores by Hotel Type and Location 126

LIST OF ABBREVIATIONS

S-O-R Model: The S-O-R Model (Stimulus-Organism-Response) is a fundamental model in psychology that describes how a stimulus (S) through the internal processing of an organism (O) produces a response (R).

ART: ART (Attention Restoration Theory) was proposed by psychologists Stephen and Rachel Kaplan. The theory suggests that natural environments help restore people's attention, reducing fatigue. In architecture and environmental design, applying this theory can guide how to introduce natural elements (such as plants, water features, etc.) to enhance the restorative qualities of a space.

NLP: NLP (Natural Language Processing) is primarily a branch of computer science concerned with enabling computers to understand, interpret, and generate human language. In architectural environmental psychology, NLP can be used to analyze user descriptions and feedback about spaces, thereby gaining insights for design improvements.

DQI: DQI (Design Quality Indicator) is a tool for assessing the quality of design in architectural projects. It covers multiple dimensions, including functionality, build quality, and impact, to help understand the comprehensive quality of a design in a quantifiable way.

BID-M: Biophilic Interior Design Matrix (BID-M) is a tool for integrating nature into interior design, developed through systematic testing and expansion of the original Biophilic Design Matrix to include six elements and 54 attributes. It aids designers in incorporating biophilic principles throughout the design process, enhancing knowledge and application of nature-based features in interiors.

BDIA: The Biophilic Design Indirect Attributes (BDIA) model is a quantitative assessment tool for evaluating the use of indirect biophilic attributes in hotel interiors, using a binary scoring system to quantify the indirect experience of nature. Developed through NLP analysis from more than 35,000 comments in Waikiki 4-star hotel, it integrates principles from the Design Quality Indicator (DQI) and the Biophilic Interior Design Matrix (BID-M) to offer a streamlined approach for predicting hotel

evaluation scores on social platform.

BDIA-Index: This simplified model is used as a prediction model, and the BDIA-Index is proposed as the weighted sum of 16 BDIAs. Its calculation formula is: BDIA-Scenario = SUM(BDIA*Corresponding weight), BDIA-Index = 0.368*Guestroom+0.285*Loundge+0.347*Lobby.

CHAPTER 1. INTRODUCTION

1.1 Research Background

This research explores the intersection of biophilic design in hotel architecture and its impact on customer experience in the context of the modern digital era. Biophilic design, rooted in the human inclination towards nature, has gained significant importance in architecture and interior design. It's based on the concept that integrating natural elements in built environments enhances human health and well-being. The application of biophilic design principles, such as the presence of natural light, use of natural materials like wood and stone, and incorporation of living architecture, has been shown to reduce stress, improve cognitive performance, and enhance mood and preference. These benefits extend to various settings, including workplaces and healthcare facilities, underlining the relevance of biophilic design in creating spaces that promote human health and wellness. However, few research focused on the benefit and possibility of biophilic design, especially the indirect attributes, on leisure hotel customers.

In the era of big data, online hotel reviews have emerged as a crucial resource for architects and designers. The rise of online booking platforms has revolutionized how consumers select hotels, making customer feedback and reviews an invaluable source of information. These platforms provide a wealth of data that can inform design decisions, highlighting which biophilic elements are most appreciated by guests and thus guiding future hotel designs to better meet customer preferences and enhance their overall experience.

Furthermore, the increasing reliance on online booking platforms by consumers has transformed these platforms into a primary channel for hotel selection. The visual presentation of hotels, including photographs and virtual tours, significantly influences customer booking decisions. This shift underscores the need for hotels to effectively showcase their design elements, including biophilic features, to attract potential guests. By analyzing large-scale customer review data from these platforms, this research aims to establish a correlation between biophilic design

elements and hotel business potential, offering insights into how such design can enhance the competitiveness of the hospitality industry, especially in regions like Hawaii where tourism is a key economic driver.

Overall, this research sits at the confluence of architectural innovation, digital technology, and consumer behavior, aiming to leverage big data to understand and enhance the role of biophilic design in the hotel industry. It seeks to provide actionable insights for hotel design that align with contemporary trends and customer preferences, ultimately contributing to the creation of spaces that are both aesthetically pleasing and beneficial to human well-being.

1.2 Research Objectives

The concept of biophilic design in hotels encapsulates a synthesis of nature-inspired elements to enhance the guest experience and well-being. Biophilic design, as theorized by Kellert, emphasizes the 'Indirect Experience of Nature', integrating aspects such as images of nature, natural materials, colors, the simulation of natural light and air, naturalistic shapes and forms, evocations of nature, information richness, the patina of time, natural geometries, and biomimicry into interior spaces. These elements, as identified by Kellert, contribute to a more immersive, enriching, and restorative environment, connecting guests with nature in subtle yet profound ways.

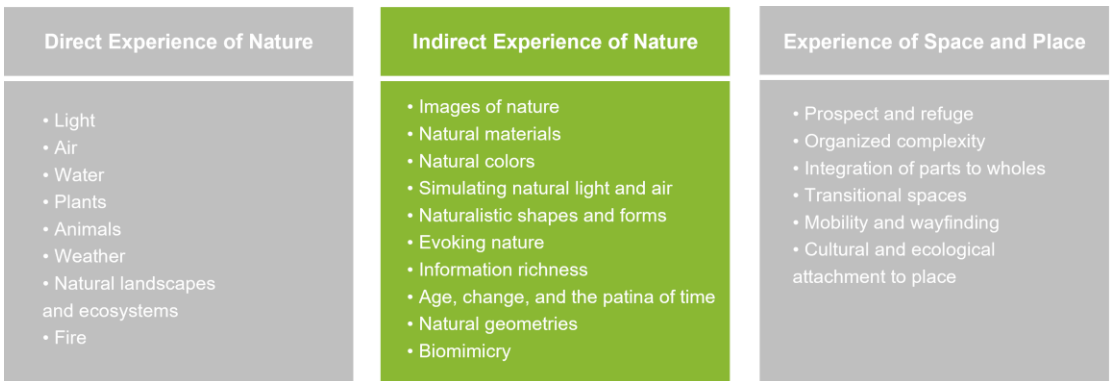


Figure 1 Kellert 24 Biophilic Attributes Source: Author

In the hotel industry, biophilic design has emerged as a pivotal aspect of interior design, contributing significantly to customer satisfaction, peace of mind, and willingness to pay. Studies have shown that biophilic elements in hotel lobbies and rooms can evoke stronger positive emotional responses, enhance pleasure and excitement, and even affect booking intentions. Moreover, the incorporation of biophilic elements has been linked to heightened perceptions of luxury and aesthetic appeal, potentially influencing consumer behaviors and preferences.

However, a notable research gap exists in understanding the full extent of biophilic design's impact, particularly the significance of its indirect natural elements in enhancing guest experiences. While direct elements like outdoor environments are often dependent on geographical location, indirect elements combined with interior design are more controllable by designers and potentially more impactful in a broader

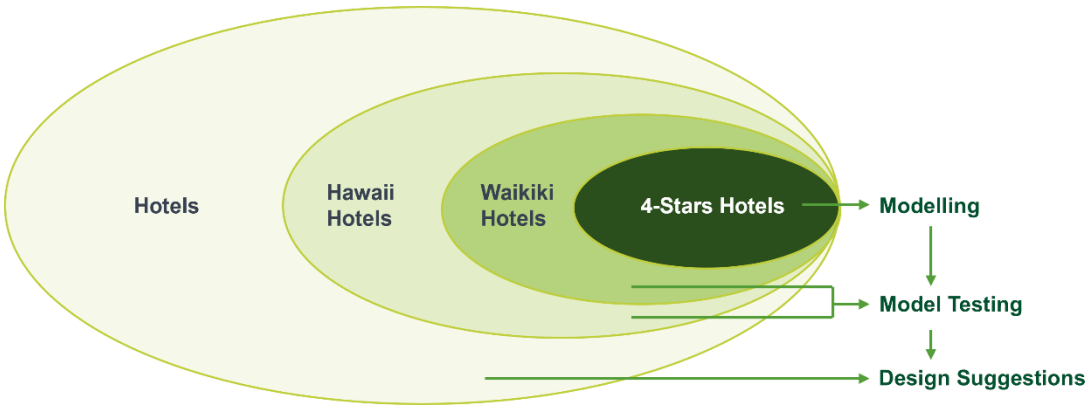


Figure 2 Research scope Source: Author

range of settings. This gap highlights the importance of studying the relationship between Biophilic Design Indirect Attributes (BDIA) and hotel booking and guest experiences. Such a study is crucial in an era where guest expectations are increasingly centered around unique, nature-inspired, and psychologically restful environments.

Waikiki, renowned for its picturesque landscapes and pivotal role in the Hawaiian hotel industry, presents a unique context for exploring biophilic design. The historical evolution of Waikiki's hotel industry, marked by its integration of lush local

flora and catering to affluent guests' tastes, showcases a long-standing commitment to harmonizing nature and luxury. This deep-rooted connection with nature makes Waikiki an ideal location to study the impact of biophilic design on hotel interiors and guest experiences.

To narrow down deviation among research objects, this research only choose exclusively four-star, officially operated hotels in Waikiki. This can reduce the differences between different hotels due to grade rather than design itself. Officially operated four-star hotels generally have mature and relatively unified supporting facilities, such as restaurants and bars, swimming pools, lounges, and lobbies. The size and layout of the guest rooms are also relatively similar.

Meanwhile, a minimum of 100 reviews on Booking.com was settled as a standard to ensure the reliability of NLP results. All selected hotels were provided with official images of the hotel lobby, lounge, and standard double-bed rooms including bathroom, window, and entrance views. The purpose of this is to ensure that a comprehensive perspective can be obtained in the subsequent customer inspection questionnaire and biophilic element richness scoring, and no certain spaces will be missed. After the modeling is completed, in order to test the validity of the results, hotels in other regions and star ratings in Hawaii will be included as tests.

In summary, exploring the role of indirect biophilic elements in the context of a leading and historically rich location like Waikiki provides a novel perspective on enhancing guest experiences through interior design. This research direction is not only pertinent for advancing the field of hotel interior design but also crucial for understanding how these elements can universally enhance guest satisfaction and preference in the hotel industry. Based on the research background and research objects, this study proposes three main objectives:

- To analyze the influence of indirect biophilic attributes in hotel interior design on the guest experience in Waikiki four-star hotels.
- To compare the effectiveness of different indirect biophilic attributes across selected hotels and their influence on consumers’ environmental evaluation and

consuming willingness.

- Insights into the most effective indirect biophilic design elements in Waikiki hotels, generating recommendations for hotel design and marketing strategies to enhance guest experience.

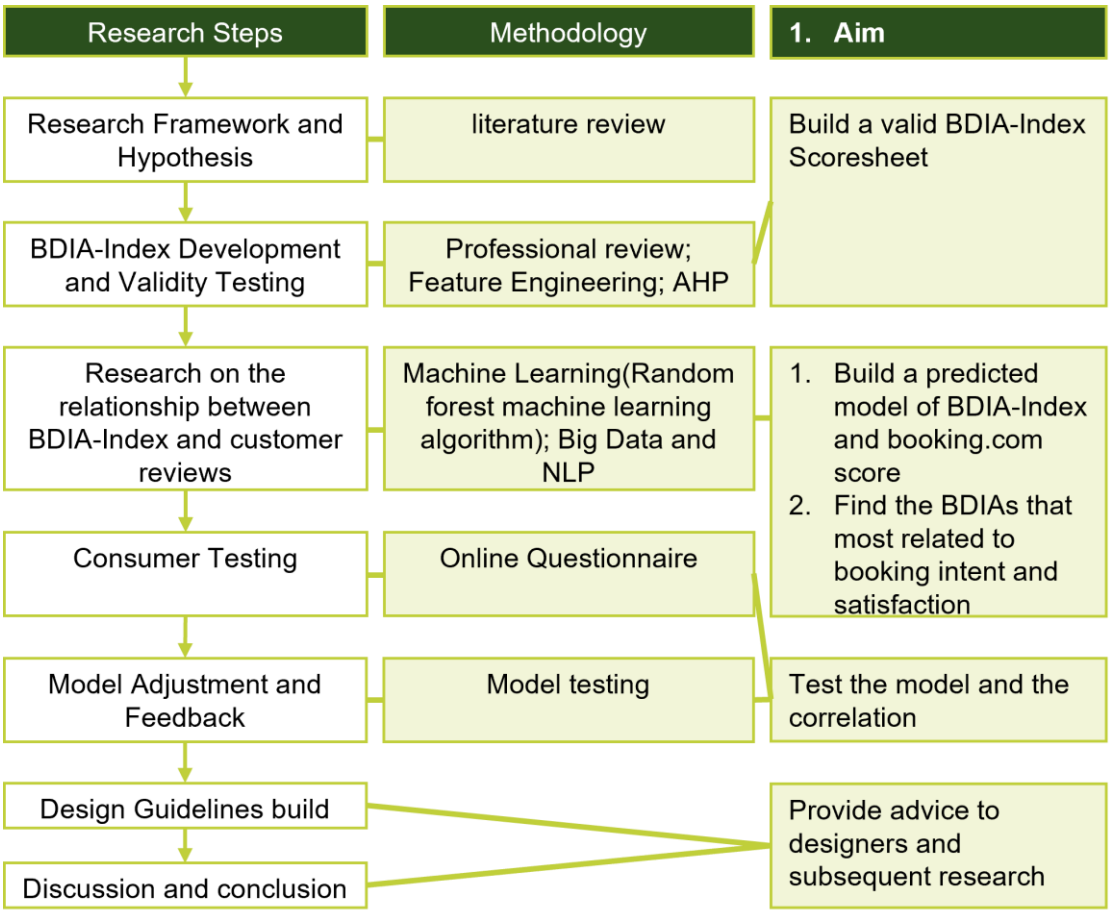


Figure 3 Research Framework Source: Author

1.3 Research Framework

To attain the research goals, this study utilizes a multifaceted approach, combining various methods to offer hotel designers insights into indirect biophilic design. The process begins with a literature review on biophilic design, hotel design, environmental psychology, and hotel consumer behavior, laying the groundwork for the study's theoretical framework and the development of questionnaires.

The next phase involves applying industry-standard metrics for design

evaluation, integrating expert reviews to develop the BDIA-Index framework. This framework is refined through three rounds of evaluation, focusing on enhancing its evaluative accuracy.

The study then employs machine learning for weight analysis and Natural Language Processing (NLP) to explore the relationship between the BDIA-Index and customer feedback, leading to a preliminary predictive model. This model undergoes refinement and testing through customer surveys and the inclusion of additional hotel data.

In the final phase, the study converts the data from the predictive model into visual design recommendations, offering practical guidance for designers. This process involves data visualization techniques to ensure the findings are accessible and applicable to hotel design practices.

CHAPTER 2. LITERATURE REVIEW

2.1 Biophilic Design in Hotel interior Design

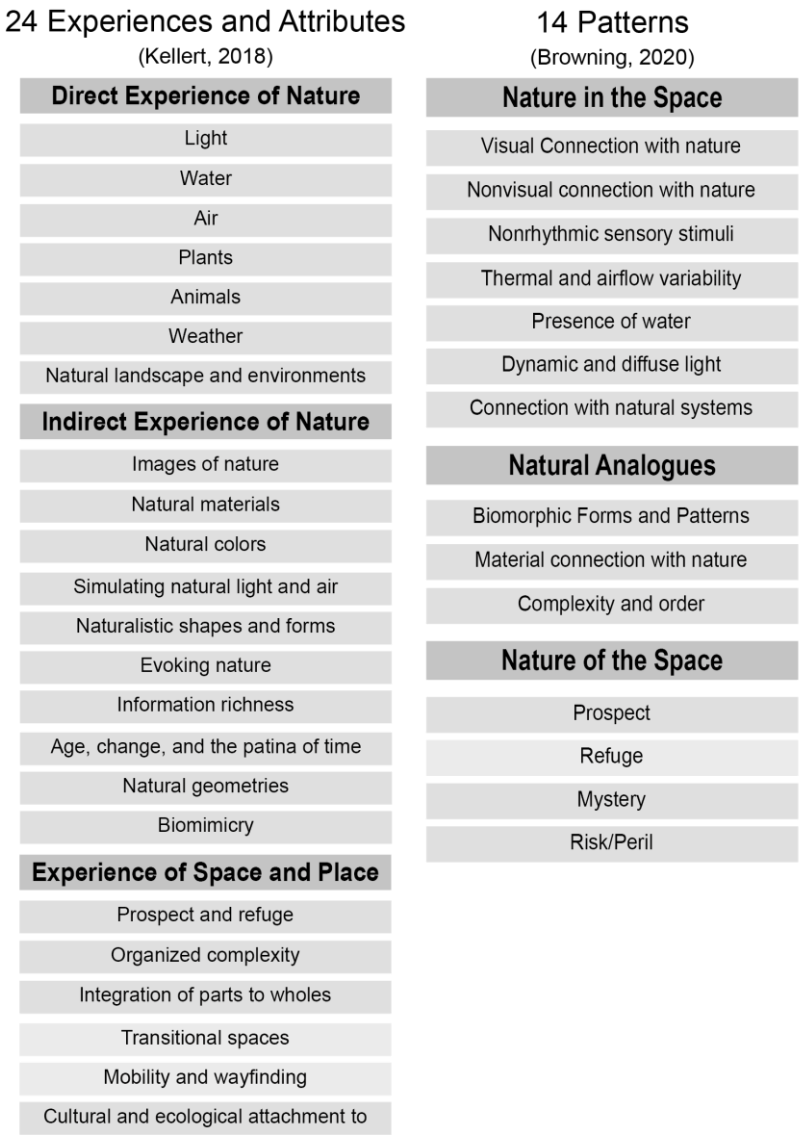


Figure 4 A comparison between Kellert and Brown's Biophilic Design Theory Source: Author

Biophilic design is a concept used within the building industry to increase occupant connectivity to the natural environment using direct nature, indirect nature, and space and place conditions.[1] This theory originating from Edward O. Wilson's biophilia hypothesis, is based on the innate human affinity towards nature (Gupta et

[1] https://en.wikipedia.org/wiki/Biophilic_design

al., 2023; Kumar et al., 2020)[2][3]. There are two important theoretical founders in the practice of biophilic design application in architecture field, the first being Kellert (Kellert, 2015) [4], who systematically broke down biophilic design into 24 basic attributes and explained the historical roots, benefits, significance, prospects, and values of biophilic design. Later, Brown deepened Kellert's theory and reduced the 24 to 14 attributes (Andreucci et al, 2021)[5].

[i]2.1.1 Physiological and Psychological Benefits

The mental health benefits of biophilic design are well-documented in literature. From a psychological perspective, biophilic design has been shown to reduce stress and improve mood and cognitive function. Research represented in studies highlights how exposure to natural elements, such as plants and natural light, can significantly lower stress levels and enhance mental well-being (Li et al, 2022; Jackson et al, 2021)[6][7]. On the physical health front, biophilic design has been linked to numerous benefits. The presence of natural elements can improve air quality, which in turn has a positive impact on respiratory health and overall physical well-being (Africa et al, 2019)[8]. Additionally, natural light, a key component of biophilic design, can improve

[2] Gupta, N., & Lee, S. H. (2023). Trapezoidal interval type-2 fuzzy analytical hierarchy process technique for biophilic element/design selection in lodging industry. Journal of the Operational Research Society, 74(7), 1613-1627.

[3] Kumar, D. S., Purani, K., & Viswanathan, S. A. (2020). The indirect experience of nature: biomorphic design forms in servicescapes. Journal of Services Marketing, 34(6), 847-867.

[4] Kellert, S., & Calabrese, E. (2015). The practice of biophilic design. London: Terrapin Bright LLC, 3, 21-46.

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[6] Li, M., Yin, H., Qu, M., & Trivers, I. (2022). Outdoor Comfort in Public Spaces, a Critical Review.

[7] Jackson, S. M., Singh, L., & Doshi, T. (2021). Valuing Public Perceptions of Biophilia Impact on Human Well-Being: 2 Sustainable Building Case Studies from Greece and India. In Advanced Studies in Efficient Environmental Design and City Planning (pp. 179-195). Springer International Publishing.

[8] Africa, J., Heerwagen, J., Loftness, V., & Ryan Balagtas, C. (2019). Biophilic design and climate change: performance parameters for health. Frontiers in Built Environment, 5, 28.

sleep quality and regulate circadian rhythms, leading to better overall physical health (Ahmad et al., 2023; Yin et al., 2018)[9][10].

Hotels are an important part of urban life and a place for tourists to relax and unwind. The historical development of biophilic design in hotel interiors can be traced through cultural studies in 20th century (Desmond, 1997)[11], which highlight the early adoption of natural elements in commercial spaces. Research has found that biophilic design has a positive effect on commercial buildings (Ortegón-Cortázar & Royo-Vela, 2019; Soderlund & Newman, 2015)[12][13], improving customer perceptions (Esmaili et al., 2021; Rosenbaum et al., 2016)[14][15], resilience (Wang et al., 2019)[16] and consumption (Esan-Ojuri & You, 2021)[17], and reducing employee stress levels (Guzzo & Legendre, 2022)[18]. Hotel spaces evoke similar evocations as biophilic

[9] Ahmad, N. A., & Zakaria, S. A. (2023). Application of biophilic natural lighting to reduce blue spectrum exposure in dark rooms on children's sleep quality. ARTEKS: Jurnal Teknik Arsitektur, 8(3), 387-394.

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[14] Esmaili, N., Golabchi, M., & Ghobadian, V. (2021). Assessing Sustainable Local Identity in the Business Centers with an Emphasis on Biophilic Design (Case Study: Shiraz Enghelab Bazaar). International Journal of Applied Arts Studies (IJAPAS), 6(2), 7-26.

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[16] Wang, T. C., Tsai, C. L., & Tang, T. W. (2019). Restorative quality in tourist hotel marketing pictures: Natural and built characteristics. Current Issues in Tourism, 22(14), 1679-1685.

[17] Esan-Ojuri, O., & You, H. (2021). How does the biophilic design of building projects impact consumers' responses?–Case of retail stores. Journal of Retailing and Consumer Services, 62, 102637.

[18] Guzzo, R. F., Suess, C., & Legendre, T. S. (2022). Biophilic design for urban hotels–prospective hospitality employees' perspectives. International Journal of Contemporary Hospitality Management, 34(8), 2914-2933.

attributes: reducing stress (Robinot & Giannelloni, 2010)[19], relaxing (Song et al., 2022)[20], facilitating communication (Hung & Chang, 2021)[21], and contributing to physical and mental health. Therefore, hotels can be used to study the practical effects of biophilic design. Research shows that the presence of natural elements in hotels can significantly enhance mental well-being and reduce stress for guests (Mody et al., 2020; Suess & Mody, 2018)[22][23]. Some studies demonstrate the significance of natural elements in creating appealing and comfortable hotel environments (Chan et al., 2020; Jeong & Choi, 2005; Bogicevic et al., 2018)[24][25][26].

2.1.2 Economical Benefits

Recent studies have increasingly focused on the economic benefits of biophilic design, particularly in the context of the hospitality industry. This design philosophy, which incorporates natural elements into built environments, has shown to positively impact both hotel performance and guest satisfaction, leading to tangible financial advantages.

Studies delve into how biophilic design elements in hotels, such as the use of

[19] Robinot, E., & Giannelloni, J. L. (2010). Do hotels'"green" attributes contribute to customer satisfaction?. Journal of Services Marketing, 24(2), 157-169.

[20] Song, C., Ali, F., Cobanoglu, C., Nanu, L., & Lee, S. H. J. (2022). The effect of biophilic design on customer's subjective well-being in the hotel lobbies. Journal of Hospitality and Tourism Management, 52, 264-274.

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[22] Mody, M., Suess, C., & Dogru, T. (2020). Restorative servicescapes in health care: Examining the influence of hotel-like attributes on patient well-being. Cornell Hospitality Quarterly, 61(1), 19-39.

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[25] Jeong, M., & Choi, J. (2005). Effects of picture presentations on customers' behavioral intentions on the web. Journal of Travel & Tourism Marketing, 17(2-3), 193-204.

[26] Bogicevic, V., Bujisic, M., Cobanoglu, C., & Feinstein, A. H. (2018). Gender and age preferences of hotel room design. International Journal of Contemporary Hospitality Management, 30(2), 874-899.

natural materials, the inclusion of plants, and maximizing natural light, contribute to an enhanced guest experience (Fidzani & Caughey, 2015)[27]. These studies reveal that hotels incorporating these design elements tend to have higher occupancy rates and can command premium pricing due to the increased demand from guests seeking a more natural and relaxing environment. For example, He (2020) discusses how the aesthetic appeal and unique experience provided by biophilic design can lead to positive word-of-mouth, attracting more guests and improving overall hotel revenue[28].

Furthermore, Zemke et al. (2018) explores the broader implications of biophilic design on hotel performance, suggesting that these design choices not only improve the guest experience but also contribute to operational efficiency[29]. The use of natural light and ventilation, for instance, can reduce energy costs, and the selection of sustainable materials can minimize maintenance expenses over time (Edwards & Torcellini, 2002) [30].

Significantly, Robinot & Giannelloni (2010) focuses on the impact of biophilic design on customer loyalty and perceived value. This research indicates that guests are more likely to return to a hotel that offers a biophilic environment, perceiving a higher value in their stay due to the improved quality of their experience. This aspect of customer loyalty is crucial for long-term business success, as repeat customers often lead to sustained revenue and can serve as brand ambassadors[31]. Similar findings were shown in other experiments (Ortegón-Cortázar & Royo-Vela, 2023;

[27] Fidzani, L. C., & Caughey, C. (2015). First impressions of hotel lobbies as inferences about hotels: a marketing tool.

[28] He, Z. (2020). Mediating Hotel Experiences with Online Photos: How Consumers Are Attracted by a Hotel Through the Perceptions of Mediated Servicescape. Temple University.

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[30] Edwards, L., & Torcellini, P. (2002). Literature review of the effects of natural light on building occupants.

[31] Robinot, E., & Giannelloni, J. L. (2010). Do hotels'"green" attributes contribute to customer satisfaction?. Journal of Services Marketing, 24(2), 157-169.

Sukatmadiredja & Emmywati, 2018)[32][33].

In conclusion, contemporary research underscores that biophilic design is not just an aesthetic choice but a strategic investment. By enhancing guest satisfaction and loyalty, and improving operational efficiency, biophilic design in hotel interiors can lead to significant economic benefits, making it a compelling approach for the hospitality industry.

2.1.3 Environmental Benefits

The synergy between biophilic design and environmental sustainability is a growing area of interest. Joshi and Teller (2021)[34], Ling et al. (2018)[35] illustrates how integrating natural elements like green walls and rooftop gardens not only enhances aesthetic appeal but also boosts sustainability by improving air quality and reducing urban heat. Mirsha & Gupta (2019)[36] and Ko et al. (2023)[37] emphasizes the use of sustainable materials, such as reclaimed wood and bamboo, which capture the essence of biophilic design and promote eco-friendliness[38], thereby reducing the carbon footprint of hotels. Study extends this concept, suggesting that biophilic design fosters environmental consciousness among guests and staff, encouraging a

[32] Ortegón-Cortázar, L., & Royo-Vela, M. (2023). The Effects of the Biophilic Atmosphere on Hedonic Value, Utilitarian Value, and Customer Loyalty. Sustainability, 15(23), 16435.

[33] Sukatmadiredja, N. R., & Emmywati, E. (2018). Effect of Greenery on the Perceived of Value and Customer Loyalty. International Journal of Science and Engineering Investigations, 7(81), 44-49.

[34] Joshi, M. Y., & Teller, J. (2021). Urban integration of green roofs: Current challenges and perspectives. Sustainability, 13(22), 12378.

[35] Ling, T. Y., & Chiang, Y. C. (2018). Well-being, health and urban coherence-advancing vertical greening approach toward resilience: A design practice consideration. Journal of Cleaner Production, 182, 187-197.

[36] Mishra, A., & Gupta, A. (2019). Green hotel servicescape: Attributes and unique experiences. Current Issues in Tourism, 22(20), 2566-2578.

[37] Ko, J. H., Kang, H. S., & Seo, J. W. (2023). Experiential Value, Overall Image, Revisit Intention, and Willingness to Pay a Premium in the Context of Environmentally Certified Hotel Restaurants: The Moderating Effect of Environmental Consciousness. Sustainability, 15(17), 12913.

[38] Wang, Z., Fu, H., Jian, Y., Qureshi, S., Jie, H., & Wang, L. (2022). On the comparative use of social media data and survey data in prioritizing ecosystem services for cost-effective governance. Ecosystem Services, 56, 101446.

shift towards eco-friendly lifestyles (Sharifi & Sabernejad, 2016; Wang et al., 2022)[39][40].

In summary, these studies highlight the significant role of biophilic design in advancing environmental sustainability in the hospitality industry, demonstrating that integrating nature-inspired elements and sustainable practices enhances guest experiences and contributes to ecological responsibility (Esmaili et al., 2021)[41].

2.1.4 Research Gap

Most of the existing studies treat biophilic design as a whole and lack quantitative analysis of each attribute. Even if biophilic design attributes are studied separately, the elements selected are basically directly visible ones, such as water, natural light, plants, etc., and less involve architectural design elements and indirect natural experiences, such as natural-simulating decorations and natural and harmonious color combinations. Substantial evidence exists to support some attributes of Biophilic design (such as the presence of natural elements), while empirical evidence for other attributes (such as the use of natural materials or processes) is lacking (Gillis & Gatersleben, 2015) [42]. However, Kellert emphasizes the 'Indirect Experience of Nature', integrating aspects such as images of nature, natural materials, colors, the simulation of natural light and air, naturalistic shapes and forms, evocations of nature, information richness, the patina of time, natural geometries, and biomimicry into interior spaces. These elements, as identified by

[39] Sharifi, M., & Sabernejad, J. (2016). Investigation of Biophilic architecture patterns and prioritizing them in design performance in order to realize sustainable development goals. European Online Journal of Natural and Social Sciences: Proceedings, 5(3 (s)), pp-325.

[40] Wang, Z., Jie, H., Fu, H., Wang, L., Jiang, H., Ding, L., & Chen, Y. (2022). A social-media-based improvement index for urban renewal. *Ecological Indicators*, *137*, 108775.

[41] Esmaili, N., Golabchi, M., & Ghobadian, V. (2021). Assessing Sustainable Local Identity in the Business Centers with an Emphasis on Biophilic Design (Case Study: Shiraz Enghelab Bazaar). International Journal of Applied Arts Studies (IJAPAS), 6(2), 7-26.

[42] Gillis, K., & Gatersleben, B. (2015). A review of psychological literature on the health and wellbeing benefits of biophilic design. Buildings, 5(3), 948-963.

Kellert, contribute to a more immersive, enriching, and restorative environment, connecting human being with nature in subtle yet profound ways.

Table 1. Indirect Experience of Nature Biophilic Attributes

Attributes	Description
IMAGES OF NATURE	*The image and representation of nature in the built environment—plants, animals, landscapes, water, geological features—can be both emotionally and intellectually satisfying. These images can occur through the use of photographs, paintings, sculptures, murals, video, computer simulations, and other representational means. Single or isolated images of nature typically exert little impact. Representational expressions of nature should be repeated, thematic, and abundant.*
NATURAL MATERIALS	*Natural materials can be especially stimulating, reflecting the dynamic properties of organic matter in adaptive response to the stresses and challenges of survival over time. The transformation of materials from nature frequently elicits positive visual and tactile responses, which few artificial materials can duplicate. Prominent natural building and decorative materials include wood, stone, wool, cotton, and leather, used in a wide array of products, furnishings, fabrics, and other interior and exterior designs.*
NATURAL COLORS	*Humans evolved as daytime animals, and color has long served as an important means for locating food, water, and other resources, as well as facilitating movement and wayfinding. The effective use of color in the built environment can be challenging, given the modern ability to generate artificial, especially bright colors. The effective biophilic application of color should generally favor muted "earth" tones characteristic of soil, rock, and plants. The use of bright colors should be cautiously applied, and emphasize such appealing environmental forms as flowers, sunsets and sunups, rainbows, and certain plants and animals. The occurrence of highly artificial, contrasting, and "vibrating" colors should be avoided.*
SIMULATING NATURAL	*Indoor lighting and processed air have been made possible by advances in building technology and construction. The trade-of*

LIGHT AND AIR	*has often been the occurrence of static conditions that can be physically and psychologically debilitating. Artifcial light can be designed to mimic the spectral and dynamic qualities of natural light. Processed air can also simulate qualities of natural ventilation through variations in airfow, temperature, humidity and barometric pressure.*
NATURALISTIC SHAPES AND FORMS	*The experience of shapes and forms characteristic of the natural world can be especially appealing. These naturalistic forms can be extraordinarily diverse from the leaf-like patterns found on columns, the shapes of plants on building facades, to animal facsimiles woven into fabrics and coverings. The occurrence of naturalistic shapes and forms can transform a static space into one that possesses the dynamic and ambient qualities of a living system.*
EVOKING NATURE	*The satisfying experience of nature can also be revealed through imaginative and fantastic depictions. These representations may not literally occur in nature, but still draw from design principles prominently encountered in the natural world. For example, the "wings" of the Sydney Opera House suggest the qualities of a bird; Notre Dame's stained glass windows, a rose-like fower; while, the skyline of some cities mimic the vertical heterogeneity of a forest. None of these designs actually occurs in nature, but they all draw from design principles and characteristics of the natural world.*
INFORMATION RICHNESS	*The diversity and variability of the natural world are so pronounced, it has been described as the most information-rich environment people will ever encounter. Whether natural or built, people tend to respond positively to information-rich and diverse environments that present a wealth of options and opportunities, so long as the complexity is experienced in a coherent and legible way.*
AGE, CHANGE, AND THE PATINA OF TIME	*Nature is always changing and in fact, life especially refecting the dynamic forces of growth and aging. People respond positively to these dynamic forces and the associated patina of time, revealing nature's capacity to respond adaptively to ever-changing conditions. These dynamic tendencies are often most*

	satisfying when balanced by the complementary qualities of unity and stability. Change and a patina of time can be achieved through such design strategies as naturally aging materials, weathering, a sense of the passage of time, and in other ways.

Meanwhile, most of current studies have small sample sizes, and the common method is to ask participants to compare two virtual scenarios and give ratings, lacking big data support from real projects. For example, studies (Rosenbaum et al., 2019; Ortegón-Cortázar et al., 2019)[43][44] discuss the use of virtual and augmented reality to create immersive natural environments in urban hotel settings. However, with the development of data science technology, it has become possible to collect a large number of real user reviews for research through online platforms.

Therefore, to mix existing gap, this research will focus on indirect natural experiences and use a variety of technical means to conduct big data research and analysis.

2.2 Hotel Design in The Information Age

The emergence of the Industrialization 4.0 era has altered the use of big data as a research tool, indicating a shift in the hospitality industry. In this era, the online environment serves as a channel for communication, marketing, operations, and distribution (Chan et al., 2020)[45].

According to a report by the World Economic Forum, as of 2018, the travel and tourism sector contributed to 10.4% of global GDP and was responsible for sustaining more than 319 million jobs (Calderón and Blanco, 2017)[46]. It is noted that around 75% of international tourists organize their trips using digital platforms,

[43] Rosenbaum, M. S., Ramírez, G. C., & Matos, N. (2019). A neuroscientific perspective of consumer responses to retail greenery. The Service Industries Journal, 39(15-16), 1034-1045.

[44] Ortegón-Cortázar, L., & Royo-Vela, M. (2019). Effects of the biophilic atmosphere on intention to visit: the affective states' mediating role. Journal of Services Marketing, 33(2), 168-180.

[45] Chan, S. H. J., Wu, S., & Vipulakom, K. (2020). Young consumers' behavioural intention based on hotel websites. Anatolia, 31(1), 1-18.

[46] Calderón, F. A. C., & Blanco, M. V. V. (2017). Impacto de Internet en el sector Turístico. Revista UNIANDES Episteme, 4(4), 477-490.

heavily influenced by the online information shared by other travelers (Pencarelli, 2020)[47]. This shift is observable in hotel bookings, where a hotel's digital presence influences customer decisions (Yao et al., 2022) [48]. The visual representation of a hotel, including photos, videos, and 3D models, provides a direct depiction of its environment and design (Giglio et al., 2020)[49]. These visual elements affect potential customers' booking decisions and their perceptions of the hotel's quality, star rating, and comfort (Jeong & Choi, 2005)[50]. The increasing use of online booking platforms indicates a reliance on these digital channels for decision-making (Lin, 2016; Varkaris & Neuhofer, 2017)[51][52]. To foster repeat visits in this sector, aligning with customer expectations is crucial. The role of Big Data analytics in meeting this demand has been significant (Marr, 2016; Gupta et al., 2017)[53][54]. The vast amount of User-Generated Content (UGC) online serves as a valuable resource for businesses and governments in shaping tourism strategies (Li et al., 2019; Xiang et al., 2015)[55][56].

Additionally, the current era has seen customer reviews and feedback becoming important in determining a hotel's reputation (Mauri & Minazzi , 2013)[57]. Customer reviews on various online platforms provide insights into guests' experiences and

[47] Pencarelli, T. (2020). The digital revolution in the travel and tourism industry. Information Technology & Tourism, 22(3), 455-476.

[48] Tianqi, Y. (2022). Integrated models for rocking of offshore wind turbine structures. American Journal of Interdisciplinary Research in Engineering and Sciences, 9(1), 13-24.

[49] Giglio, S., Pantano, E., Bilotta, E., & Melewar, T. C. (2020). Branding luxury hotels: Evidence from the analysis of consumers'"big" visual data on TripAdvisor. Journal of business research, 119, 495-501.

[50] Jeong, M., & Choi, J. (2005). Effects of picture presentations on customers' behavioral intentions on the web. Journal of Travel & Tourism Marketing, 17(2-3), 193-204.

[51] Lin, I. Y. (2016). Effects of visual servicescape aesthetics comprehension and appreciation on consumer experience. Journal of Services Marketing.

[52] Varkaris, E., & Neuhofer, B. (2017). The influence of social media on the consumers' hotel decision journey. Journal of Hospitality and Tourism Technology, 8(1), 101-118.

[53] Marr, B. (2016). How big data and analytics are changing hotels and the hospitality industry. Forbes. com, January, 26.

[54] Gupta, K., Gauba, T., & Jain, S. (2017). Big data in hospitality industry: A survey. International Research Journal of Engineering and Technology, 4(11), 476-479.

[55] Li, Q., Li, S., Zhang, S., Hu, J., & Hu, J. (2019). A review of text corpus-based tourism big data mining. Applied Sciences, 9(16), 3300.

[56] Xiang, Z., Schwartz, Z., Gerdes Jr, J. H., & Uysal, M. (2015). What can big data and text analytics tell us about hotel guest experience and satisfaction?. International journal of hospitality management, 44, 120-130.

[57] Mauri, A. G., & Minazzi, R. (2013). Web reviews influence on expectations and purchasing intentions of hotel potential customers. International journal of hospitality management, 34, 99-107.

satisfaction (He, 2020)[58]. These reviews guide potential customers' choices and serve as data points for analysis. Utilizing big data analytics, hotels can analyze customer feedback patterns and trends, identifying areas for improvement and innovation (Talón-Ballestero et al., 2018; Fuchs et al., 2014)[59][60].

Accordingly, this research uses hotel evaluation data and visual displays from online booking platforms to analyze design elements associated with high ratings and high click volumes.

2.2.1 Machine learning

Machine learning emerges as a pivotal technique, where computational systems are adept at deriving meaningful insights from data autonomously. This advanced form of learning involves algorithms that progressively acquire knowledge from data, empowering these systems to undertake tasks like decision-making (Wang & Yin, 2023)[61], forecasting, and pattern recognition without explicit programming for specific tasks (Sun et al., 2022).[62]

Machine learning has been instrumental in revolutionizing the hotel industry (Jain et al., 2021)[63]. It has been applied in diverse aspects such as evaluating hotel locations, innovating with robotic room services, predicting hotel maintenance needs, distinguishing between authentic and fraudulent online reviews, forecasting hotel

[58] He, Z. (2020). Mediating Hotel Experiences with Online Photos: How Consumers Are Attracted by a Hotel Through the Perceptions of Mediated Servicescape. Temple University.

[59] Talón-Ballestero, P., González-Serrano, L., Soguero-Ruiz, C., Muñoz-Romero, S., & Rojo-Álvarez, J. L. (2018). Using big data from customer relationship management information systems to determine the client profile in the hotel sector. Tourism Management, 68, 187-197.

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[61] Wang, Y., & Lin, Y. S. (2023). Public participation in urban design with augmented reality technology based on indicator evaluation. Frontiers in Virtual Reality, 4, 1071355.

[62] Sun, C., Li, S., Lin, Y., & Hu, W. (2022). From Visual Behavior to Signage Design: A Wayfinding Experiment with Eye-Tracking in Satellite Terminal of PVG Airport. In Proceedings of the 2021 DigitalFUTURES: The 3rd International Conference on Computational Design and Robotic Fabrication (CDRF 2021) 3 (pp. 252-262). Springer Singapore.

[63] Jain, P. K., Pamula, R., & Srivastava, G. (2021). A systematic literature review on machine learning applications for consumer sentiment analysis using online reviews. Computer science review, 41, 100413.

energy demands, anticipating booking cancellations, and projecting hotel room prices (Alotaibi, 2020)[64]. These applications, explored in various studies (Y. Yang, Tang, Luo, & Law, 2015[65]; P.-Y. Yang, Chang, Chang, & Wu, 2018[66]; Athuraliya & Farook, 2018[67]; Antonio, de Almeida, Nunes, Batista, & Ribeiro, 2018[68]; Casteleiro-Roca et al., 2018[69]; Antonio, de Almeida, & Nunes, 2019[70]; Al Shehhi & Karathanasopoulos, 2020[71]; He et al., 2022[72][73]; Lee & Lee, 2020[74]), exemplify the breadth of machine learning's impact in this sector. Furthermore, qualitative studies have delved into the integration of artificial intelligence in hotels (Ananeva, 2019)[75] and the advantages of machine learning in enhancing hospitality services (Pelet, Lick, & Taieb, 2019[76]; Yi,

[64] Alotaibi, E. (2020). Application of machine learning in the hotel industry: a critical review. Journal of Association of Arab Universities for Tourism and Hospitality, 18(3), 78-96.

[65] Yang, Y., Tang, J., Luo, H., & Law, R. (2015a). Hotel location evaluation: A combination of machine learning tools and web GIS. International Journal of Hospitality Management, 47(2015), 14-24.

[66] Yang, P.-Y., Chang, T.-H., Chang, Y.-H., & Wu, B.-F. (2018, 28-30 June). Intelligent Mobile Robot Controller Design for Hotel Room Service with Deep learning Arm-Based Elevator Manipulator. Paper presented at the International Conference on System Science and Engineering (ICSSE), New Taipei City, Taiwan.

[67] Athuraliya, B., & Farook, C. (2018, 1-3 November). “Revyew” Hotel Maintenance Issue Classifier and Analyzer using Machine Learning and Natural Language Processing. Paper presented at the 9th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON), British Columbia, Canada

[68] Antonio, N., de Almeida, A. M., Nunes, L., Batista, F., & Ribeiro, R. (2018). Hotel online reviews: creating a multi-source aggregated index. International Journal of Contemporary Hospitality Management.

[69] Casteleiro-Roca, J.-L., Gómez-González, J. F., Calvo-Rolle, J. L., Jove, E., Quintián, H., Martín, J. F. A., et al. (2018, 20-22 June). Prediction of the energy demand of a hotel using an artificial intelligence-based model. Paper presented at the International conference on hybrid artificial intelligence systems, Oviedo, Spain.

[70] Antonio, N., de Almeida, A., & Nunes, L. (2019). An automated machine learning based decision support system to predict hotel booking cancellations. An automated machine learning based decision support system to predict hotel booking cancellations(1), 1-20.

[71] Karaoglan, K., Temizkan, V., & Findik, O. (2019, 26-28 April). Sentiment Analysis for Hotel Reviews with Recurrent Neural Network Architecture. Paper presented at the International Conference on Advanced Technologies, Computer Engineering and Science, Analya, Turkey.

[72] He, C., Liu, M., Zhang, Y., Wang, Z., Hsiang, S. M., Chen, G., ... & Dai, G. (2023). Space–time–workforce visualization and conditional capacity synthesis in uncertainty. Journal of Management in Engineering, 39(2), 04022071.

[73] He, C., Liu, M., Zhang, Y., Wang, Z., Hsiang, S. M., Chen, G., & Chen, J. (2022). Exploit social distancing in construction scheduling: Visualize and optimize space–time–workforce tradeoff. Journal of Management in Engineering, 38(4), 04022027.

[74] Lee, J. J., & Lee, M. (2020, 5-6 March). Artificial Social Intelligence: Hotel Rate Prediction. Paper presented at the Future of Information and Communication Conference, San Francisco, USA.

[75] Ananeva, D. (2019). Artificial Intelligence as Disruptive Innovation in the Hotel Industry: Finnish Boutique and Lifestyle Hotels Perspective. University of Lapland

[76] Pelet, J.-E., Lick, E., & Taieb, B. (2019, 12-14 June). Internet of Things and Artificial Intelligence in the Hotel Industry: Which Opportunities and Threats for Sensory Marketing? Paper presented at the

2019[77]). These investigations highlight the superiority of machine learning models over traditional methods, especially in areas of forecasting, revenue management, and service improvement. This shift towards data-driven approaches not only propels operational efficiency but also fosters innovation in creating guest-centric and sustainable hotel solutions, mirroring the applications seen in optimizing building layouts and structures with a focus on environmental sustainability, energy efficiency,

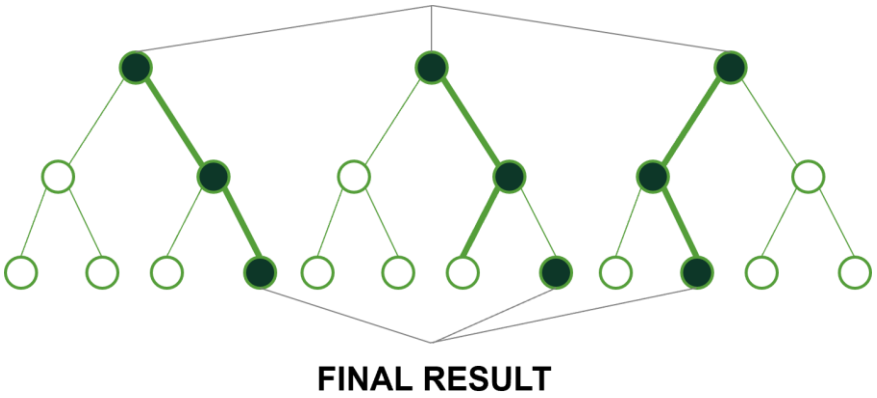


Figure 5 Random Forest Algorithm structural diagram Source: Author

and aesthetic appeal (Liu et al., 2024)[78].

The Random Forest algorithm is a kind of machine learning technique, operates by constructing multiple decision trees during the training phase and outputs the mode of the classes (classification) or mean prediction (regression) of the individual trees. Renowned for its accuracy and robustness, it handles a wide array of data types and is resistant to overfitting, making it a versatile tool in both classification and regression tasks. This algorithm is particularly effective in scenarios where the data set is large and complex, as it can manage thousands of input variables without variable deletion, making it a valuable tool in various domains, including but not

International Conference on Advances in National Brand and Private Label Marketing, Barcelona, Spain

[77] Yi, R. (2019). Research on Development Status and Trend of Unmanned Wisdom Hotel System Based on Artificial Intelligence Service—Take Bangwei Company as an Example. Finance and Market, 4(1), 36-43.

[78] Liu, T., Xu, C., Qiao, Y., Jiang, C., & Chen, W. (2024). News Recommendation with Attention Mechanism. Journal of Industrial Engineering and Applied Science, 2(1), 21-26.

limited to, biomedical, stock market analysis, and e-commerce.[79]

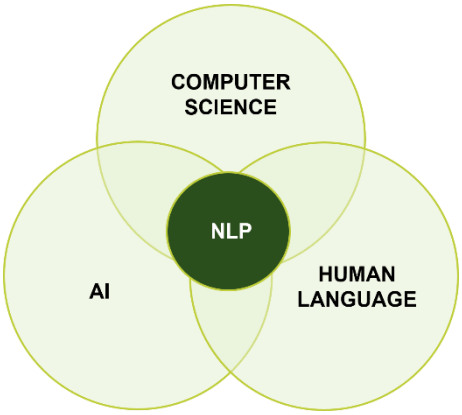


Figure 6 NLP Scope Source: Author

2.2.2 Natural Language Processing (NLP)

Natural Language Processing (NLP) is a pivotal domain within artificial intelligence that focuses on the interaction between computers and human (natural) languages (Liu et al., 2024)[80]. NLP combines computational linguistics—rule-based modeling of human language—with statistical, machine learning, and deep learning models. At its core, NLP strives to enable computers to understand, interpret, and respond to human language in a valuable and meaningful way, this involves a range of tasks such as language translation, sentiment analysis, speech recognition, and text summarization. Recent advancements in Natural Language Processing (NLP) have enhanced tourism strategies through applications such as sentiment analysis, chatbots, and hotel recommendation systems (Putri et al., 2019; García-Pablos et al.,

[79] Hossen, M. S., Jony, A. H., Tabassum, T., Islam, M. T., Rahman, M. M., & Khatun, T. (2021, March). Hotel review analysis for the prediction of business using deep learning approach. In 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS) (pp. 1489-1494). IEEE.

[80] Liu, T., Xu, C., Qiao, Y., Jiang, C., & Yu, J. (2024). Particle Filter SLAM for Vehicle Localization. arXiv e-prints, arXiv-2402.

2016; Prameswari et al., 2017a)[8][18][28][3]. In the last ten years, there has been a significant increase in the publication of research papers focusing on NLP solutions in the tourism industry. Álvarez-Carmona et al. (2022) used PRISMA methodology reviewed 227 relevant tourism studies over the last decade, and find that a novel taxonomy of studies that employ Natural Language Processing in tourism research mainly on 4 topics: Sentiment Analysis, Recommendation systems, Destination branding, and Travel or assisting travel[84].

Although NLP is currently an effective evaluation tool mainly used in the product field (Zhao et al., 2021)[85], NLP can also be used as an effective evaluation tool for design and operation, providing guidance for the updated design of buildings of the same type. One common usage is using NLP to analyze large quantities of textual complaints and detail the various aspects of them, Aspect-Based Sentiment Analysis was looked to as an ideal framework to take on the task (Sann & Lai, 2020).[86] Linking customer evaluation keywords with their satisfaction is also one of the ways NLP is applied to hotel customer evaluations (Chiny et al., 2021).[87] Those applications normally including the following steps (Chang et al., 2020)[88]:

Initially, the process begins with the meticulous collection of data from various

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[81] Putri, F. P., Meidia, H., & Gunawan, D. (2019, December). Designing intelligent personalized chatbot for hotel services. In Proceedings of the 2019 2nd International Conference on Algorithms, Computing and Artificial Intelligence (pp. 468-472).

[82] García-Pablos, A., Cuadros, M., & Linaza, M. T. (2016). Automatic analysis of textual hotel reviews. Information Technology & Tourism, 16(1), 45-69.

[83] Prameswari, P., Surjandari, I., & Laoh, E. (2017, October). Opinion mining from online reviews in Bali tourist area. In 2017 3rd International Conference on Science in Information Technology (ICSITech) (pp. 226-230). IEEE.

[84] Álvarez-Carmona, M. Á., Aranda, R., Rodríguez-Gonzalez, A. Y., Fajardo-Delgado, D., Sánchez, M. G., Pérez-Espinosa, H., ... & Díaz-Pacheco, Á. (2022). Natural language processing applied to tourism research: A systematic review and future research directions. Journal of King Saud University-Computer and Information Sciences, 34(10), 10125-10144.

[85] Zhao, L., Alhoshan, W., Ferrari, A., Letsholo, K. J., Ajagbe, M. A., Chioasca, E. V., & Batista-Navarro, R. T. (2021). Natural language processing for requirements engineering: A systematic mapping study. ACM Computing Surveys (CSUR), 54(3), 1-41.

[86] Sann, R., & Lai, P. C. (2020). Understanding homophily of service failure within the hotel guest cycle: Applying NLP-aspect-based sentiment analysis to the hospitality industry. International Journal of Hospitality Management, 91, 102678.

[87] Chiny, M., Bencharef, O., Hadi, M. Y., & Chihab, Y. (2021). A client-centric evaluation system to evaluate guest's satisfaction on AirBNB using machine learning and NLP. Applied Computational Intelligence and Soft Computing, 2021, 1-14.

[88] Chang, Y. C., Ku, C. H., & Chen, C. H. (2020). Using deep learning and visual analytics to explore hotel reviews and responses. Tourism Management, 80, 104129.

online platforms, such as TripAdvisor and Booking.com. This phase emphasizes the importance of gathering a diverse array of reviews, encapsulating positive, negative, and neutral sentiments across different hotels, ensuring a comprehensive dataset.

Once the data is amassed, it undergoes a rigorous preprocessing phase. This step is crucial in refining the data for analytical precision. It involves cleaning the data by removing irrelevant elements like HTML tags and special characters, and normalizing the text to a uniform format, such as converting all letters to lowercase and eliminating redundant spaces. Tokenization then breaks down sentences into individual words or phrases. Further refinement is achieved by removing stopwords, which are frequent yet insignificant words that offer little value to sentiment analysis. Additionally, stemming or lemmatization techniques are applied to bring words back to their base form.

The subsequent stage is the extraction of features, a pivotal process that transforms text into a numerical format that machine learning models can interpret. Techniques such as Bag of Words and Term Frequency-Inverse Document Frequency (TF-IDF) are commonly employed, alongside more sophisticated methods like Word Embeddings.

With the data suitably preprocessed and features extracted, the focus shifts to the development of the sentiment analysis model. Various machine learning models are considered, ranging from Naive Bayes and Support Vector Machines (SVM) to Random Forests and advanced deep learning models. These models are trained using the prepared data to discern the underlying sentiment of the reviews, categorizing them into positive, negative, or neutral.

The model's performance is then rigorously evaluated and optimized using metrics such as accuracy, recall, and F1 score. Techniques like cross-validation are employed to prevent overfitting, and model parameters are fine-tuned for enhanced accuracy.

Finally, the trained model is applied to new hotel reviews for sentiment analysis. The outcomes of this analysis are not just numerical scores; they offer insightful

interpretations, such as identifying prevalent positive or negative factors in the reviews, or comparing sentiment across different hotels or time periods. This iterative process may require several rounds of adjustment and optimization to achieve optimal results, and specific adaptations might be necessary to address linguistic and cultural variances in reviews from different regions or languages (He et al., 2023).[89]

2.3 Consume as an Integrated Process

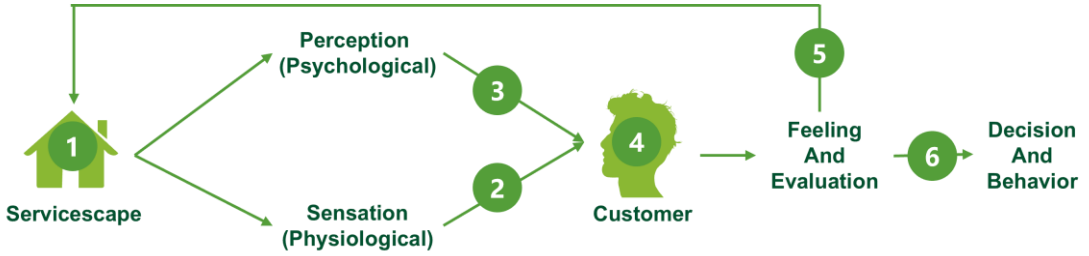


Figure 7 Customer Cognitive Process Source: Author

To accurately quantify and objectively assess how customers perceive their environment, and to understand the impact of this perceptual experience on their decision-making process is of paramount importance. Each individual's unique experience of their surroundings shapes their thoughts, emotions, and behaviors. In this phenomenon, the body and mind are not separate entities; a person's perception and reaction to their environment are intricately interwoven with the physical structure of their body in a unique manner. The architectural environment is comprehended through mechanisms of perception (Psychological)and sensation (Physiological) (Kwon et al., 2023)[90]. Physiologically, it involves sensory input being received and processed by the brain, leading to neural responses. Psychologically, this input is interpreted based on prior knowledge, experiences, and emotions, forming perceptions and memories. For example, the effectiveness of attention, influenced by

[89] He, C., Liu, M., Zhang, Y., Wang, Z., Hsiang, S. M., Chen, G., ... & Dai, G. (2023). Space–time–workforce visualization and conditional capacity synthesis in uncertainty. Journal of Management in Engineering, 39(2), 04022071.

[90] Kwon, J., Linihan, S., Iedema, A., Schmidt, A., Luo, C., & Marrufo, K. (2023). How interior design responds to neurodiversity: implementing wearable technologies in neurodesign processes. Frontiers in Built Environment, 9, 1211519.

previous memories, significantly influences the initiation, continuation, and success of any desired mental activity. The degree to which attention is engaged directly impacts the outcome of these cognitive processes (Ikromova, 2023) [91]. This process is dynamic, with continuous interaction between physiological sensory processing and psychological interpretation, shaping our understanding and response to the world around us.

2.3.1 Environmental Perception

Incorporating insights from environmental psychology, it becomes evident that the customer's response to biophilic design in hotel settings is intricately linked to deeper cultural, cognitive, and psychological factors. Environmental psychology explores the dynamic relationship between individuals and their surroundings, emphasizing how the built environment can affect behavior, emotions, and well-being. This approach is crucial for understanding the potential pathways through which customers react to biophilic design in hotels.

The current mainstream perception theory of consumption landscape is Stimulus, Organism, Response Framework (S-O-R Model), is a classic theory for studying consumer purchasing decisions (Mehrabian & Russell, 1974)[92]. The S-O-R Model is an evolution of the classical Stimulus-Response (SR) approach, consists of three components: Stimulus, Organism, and Response(Al-Adwan et al., 2022)[93]. This model posits that external environmental factors serve as stimuli influencing an individual's psychological reactions (cognitive or emotional) and, consequently, their behavioral responses. In this framework (Xu et al., 2023)[94], individuals' emotional responses to environments are encapsulated by three variables: pleasure, arousal,

[91] Ikromova, S. A. (2023). Cognitive processes and their description in psychology. TECHNICAL SCIENCE RESEARCH IN UZBEKISTAN, 1(5), 115-133.

[92] Mehrabian, A., & Russell, J. A. (1974). An approach to environmental psychology. the MIT Press.

[93] Al-Adwan, A. S., Al-Debei, M. M., & Dwivedi, Y. K. (2022). E-commerce in high uncertainty avoidance cultures: The driving forces of repurchase and word-of-mouth intentions. Technology in Society, 71, 102083.

[94] Xu, J., Liao, W., Lim, N. H. H., Fong, C. S., Wang, H., & Xu, Y. OPERATIONAL AND ENVIRONMENTAL ASSESSMENT FOR NIGHT MARKETS UNDER THE INFORMAL ECONOMY IMPACTS: INSIGHTS FROM THE CASE STUDY OF SELANGOR, MALAYSIA.

and dominance (Kumar et al., 2020)[95]. These factors can trigger either an approach or an avoidance response. Arousal, particularly in hedonic contexts, is noted to influence satisfaction significantly. Pleasure and arousal are critical in eliciting affective responses to environments, while dominance is less influential in determining approach or avoidance behaviors. Recent research has focused less on dominance in studying approach or avoidance behaviors(Nanu et al., 2020; Lee, 2019)[96][97].

Extensive research employing the S-O-R Model has been conducted to explore the interplay between environmental stimuli, internal organismic states, and behavioral responses within the context of website marketing environments. For instance, Parboteeah et al.[98] utilized the SOR model to investigate the impact of website interface design cues on user internal mechanisms and impulse buying intentions. Chen et al. [99]examined, through the SOR theory, the influence of website stimuli (e.g., media format, visual appeal) on consumers' online impulsive buying behaviors and further discussed cognitive and emotional responses in this process. The SOR theory has also been applied to study the effects of biophilic attributes on consumers, with findings indicating that while the introduction of leafy elements in a store setting may not evoke excitement, they are found to contribute to pleasure and stress relief in 'complex' in-store environments (Brengm et al., 2012)[100]. Other studies have discovered that customers experience more joy and excitement in biophilic environments, especially those with plants and water features (Lee, 2019)[101], and

[95] Kumar, D. S., Purani, K., & Viswanathan, S. A. (2020). The indirect experience of nature: biomorphic design forms in servicescapes. Journal of Services Marketing, 34(6), 847-867.

[96] Nanu, L., Ali, F., Berezina, K., & Cobanoglu, C. (2020). The effect of hotel lobby design on booking intentions: An intergenerational examination. International Journal of Hospitality Management, 89, 102530.

[97] Lee, S. H. (2019). Effects of biophilic design on consumer responses in the lodging industry. International Journal of Hospitality Management, 83, 141-150.

[98] Parboteeah, D. V., Valacich, J. S., & Wells, J. D. (2009). The influence of website characteristics on a consumer's urge to buy impulsively. Information systems research, 20(1), 60-78.

[99] Chan, T. K., Cheung, C. M., & Lee, Z. W. (2017). The state of online impulse-buying research: A literature analysis. Information & Management, 54(2), 204-217.

[100] Brengman, M., Willems, K., & Joye, Y. (2012). The impact of in‐store greenery on customers. Psychology & Marketing, 29(11), 807-821.

[101] Lee, S. H. (2019). Effects of biophilic design on consumer responses in the lodging industry.

that biophilic design impacts satisfaction and emotions across different age groups (Nanu et al., 2020)[102]. Moreover, the relationship between biomorphic service landscape design and landscape preferences has been explored, explaining how biomorphic design helps users connect better with service landscapes through the mediating roles of attention restoration and place identity (emotional and cognitive), as explained by attention restoration theory (Kumar et al., 2020)[103].

Research has delineated the impact of the environment on consumers into three distinct stages (Shang et al., 2023)[104]. In the first stage, the compatibility of the background with the consumer's context positively influences their cognitive process. Specifically, perceived trust is largely influenced by how well the background anchors the consumer, while perceived value is more impacted by the fit between the product and its background. The second stage sees consumers' cognitive processing influencing their emotional responses. Here, some researchers posit that Attention Restoration Theory (ART) and Place Identity Theory act as mediators in this cognitive process, enhancing pleasure and arousal through increased place attachment and attention restoration (Kumar et al., 2020)[105]. Notably, perceived value exerts a stronger positive effect on perceived pleasure than perceived trust, though the latter is essential for enhancing perceived value. The final stage involves the emotional process further driving consumers' purchase intentions, solidifying the sequential impact of environmental factors on consumer behavior.

Kaplan and Kaplan's Attention Restoration Theory (ART), suggests that exposure to natural elements can replenish mental resources depleted by cognitive overload, which is particularly relevant in the context of hotel design. Biophilic design

International Journal of Hospitality Management, 83, 141-150.

[102] Nanu, L., Ali, F., Berezina, K., & Cobanoglu, C. (2020). The effect of hotel lobby design on booking intentions: An intergenerational examination. International Journal of Hospitality Management, 89, 102530.

[103] Kumar, D. S., Purani, K., & Viswanathan, S. A. (2020). The indirect experience of nature: biomorphic design forms in servicescapes. Journal of Services Marketing, 34(6), 847-867.

[104] Shang, Q., Ma, H., Wang, C., & Gao, L. (2023). Effects of Background Fitting of e-Commerce Live Streaming on Consumers' Purchase Intentions: A Cognitive-Affective Perspective. Psychology Research and Behavior Management, 149-168.

[105] Kumar, D. S., Purani, K., & Viswanathan, S. A. (2020). The indirect experience of nature: biomorphic design forms in servicescapes. Journal of Services Marketing, 34(6), 847-867.

elements, by incorporating aspects of nature, could provide guests with a restorative environment, aiding in mental recovery and relaxation. Attention Restoration Theory posits that exposure to natural environments counteracts mental fatigue and enhances concentration. This theory is grounded in the idea that the brain's capacity for focused attention is finite, leading to a state known as 'directed attention fatigue'. Natural settings, according to ART, promote a more effortless mode of brain function, aiding in the restoration and rejuvenation of directed attention capabilities. Kaplan identifies four essential qualities of natural environments that contribute to this restorative effect: the sense of being immersed (Extent), a feeling of escape from routine activities (Being Away), elements that captivate attention effortlessly (Soft Fascination), and a personal desire for and appreciation of the environment (Compatibility). Among these, Soft Fascination is deemed pivotal, with the other elements serving to enhance or sustain this fascination (Ohly et al., 2016)[106]. ART is widely used to explain the mechanism of action of biophilic design. Research has found (Jackson et al., 2021)[107] that commercial spaces can reduce stress, have health benefits, thereby promoting personal and social well-being, and thus can restore people's attention; incorporating restorative qualities into commercial spaces can also help promote consumption (Rosenbaum et al., 2016)[108]. At the same time, through the ART framework, researchers have found that biophilic direct elements are related to mental health and well-being, including but not limited to feeling positive emotions and happiness, attention restoration, preference, feelings of relaxation and reduction of anger. In addition, it also includes Increase cognitive function and performance. However, research has found few psychological effects of indirect nature experiences (Hung & Chang, 2021)[109]. At the same time, some

[106] Ohly, H., White, M. P., Wheeler, B. W., Bethel, A., Ukoumunne, O. C., Nikolaou, V., & Garside, R. (2016). Attention Restoration Theory: A systematic review of the attention restoration potential of exposure to natural environments. Journal of Toxicology and Environmental Health, Part B, 19(7), 305-343.

[107] Jackson, S. M., Singh, L., & Doshi, T. (2021). Valuing Public Perceptions of Biophilia Impact on Human Well-Being: 2 Sustainable Building Case Studies from Greece and India. In Advanced Studies in Efficient Environmental Design and City Planning (pp. 179-195). Springer International Publishing.

[108] Rosenbaum, M. S., Otalora, M. L., & Ramírez, G. C. (2016). The restorative potential of shopping malls. Journal of Retailing and Consumer Services, 31, 157-165.

[109] Hung, S. H., & Chang, C. Y. (2021). Health benefits of evidence-based biophilic-designed

theoretical concepts of ART are vague (e.g., soft charm) and lack clear operability (Joye & Dewitte, 2018)[110], and more research is still needed (Gillis & Gatersleben, 2015)[111].

Furthermore, Ulrich's Stress Reduction Theory (SRT) (Ulrich, 1984)[112] posits that natural environments have a stress-reducing effect on individuals. This theory supports the idea that biophilic design in hotels can create a calming and stress-reducing ambiance, which is likely to be positively received by guests seeking respite from their daily routines. Cultural factors also play a significant role in shaping guests' responses. Different cultures may have varying perceptions and valuations of nature and natural elements, which can influence their reaction to biophilic design. Hofstede's cultural dimensions theory can be applied to understand these cultural variances and how they might affect guests' preferences and expectations regarding hotel environments. From a cognitive perspective, the Concept of Place Attachment also becomes relevant. This concept suggests that emotional bonds are formed with places that are significant to individuals, and these bonds can influence their behaviors and choices. In hotel settings, biophilic elements that create a sense of attachment or belonging can lead to higher guest satisfaction and a preference for returning to the hotel (Hofstede, 1984)[113].

Place identity, on the other hand, a concept rooted in environmental psychology as identified by Proshansky et al. (1983)[114], refers to the alignment between an individual's personality and their environmental setting, fostering a sense of attachment, as described by Rosenbaum and Montoya (2007)[115]. Place identity

environments: A review. Journal of People, Plants, and Environment, 24(1), 1-16.

[110] Joye, Y., & Dewitte, S. (2018). Nature's broken path to restoration. A critical look at Attention Restoration Theory. Journal of environmental psychology, 59, 1-8.

[111] Gillis, K., & Gatersleben, B. (2015). A review of psychological literature on the health and wellbeing benefits of biophilic design. Buildings, 5(3), 948-963.

[112] Ulrich, R. S. (1984). View through a window may influence recovery from surgery. science, 224(4647), 420-421.

[113] Hofstede, G. (1984). Culture's consequences: International differences in work-related values (Vol. 5). sage.

[114] Twigger-Ross, C., Bonaiuto, M., & Breakwell, G. (2003). Identity theories and environmental psychology. na.

[115] Rosenbaum, M. S., & Montoya, D. Y. (2007). Am I welcome here? Exploring how ethnic

refers to the consistency between the consumer's self-identity and the place. This concept is closely associated with related constructs such as 'sense of place', defined by Brocato et al. (2015) as one's personal interpretation of their setting, and 'place attachment'[116], as articulated by Kyle et al. (2007)[117], which is the emotional bond developed with an environment over time, culminating in place bonding. These interrelated constructs collectively underscore the deep connection between individuals and their environmental contexts. Places that are perceived as highly restorative are also thought to create strong place attachment; as servicescapes play a crucial role in service brand experiences, place attachment becomes critical and contributes Brand attachment (Devine-Wright & Howes, 2010).[118]

In conclusion, the application of theories such as the Stimulus-Organism-Response Model, Attention Restoration Theory, and Stress Reduction Theory, along with the understanding of concepts like place identity and cultural dimensions, provides a rich, nuanced perspective on how consumers interact with their surroundings. These interactions are not merely superficial or aesthetic; they are deeply rooted in cultural, cognitive, and psychological realms, influencing emotional and behavioral responses in significant ways. The biophilic design in hotels, therefore, emerges as a powerful tool, not just for aesthetic appeal but as a means to enhance guest satisfaction, well-being, and emotional connection to the space. This multifaceted approach, blending theoretical insights with practical applications, underscores the potential of biophilic design to create environments that resonate with guests on multiple levels, fostering positive experiences, restorative effects, and lasting impressions.

consumers assess their place identity. Journal of Business Research, 60(3), 206-214.

[116] Brocato, E. Deanne, Julie Baker, and Clay M. Voorhees. "Creating consumer attachment to retail service firms through sense of place." Journal of the Academy of Marketing Science 43 (2015): 200-220.

[117] Kyle, G., & Chick, G. (2007). The social construction of a sense of place. Leisure sciences, 29(3), 209-225.

[118] Devine-Wright, P., & Howes, Y. (2010). Disruption to place attachment and the protection of restorative environments: A wind energy case study. Journal of environmental psychology, 30(3), 271-280.

2.3.2 Servicescape Evaluation

"Servicescape" is a term in business and marketing, specifically used in the field of service marketing. It refers to the service environment - the physical space where service delivery occurs. Servicescape encompasses all physical aspects related to the service interaction experience, such as design, layout, decor, music, temperature, etc. This concept was introduced by Booms and Bitner in the 1980s to explain how environmental factors influence customer behavior and service experience (Ezeh & Harris, 2007)[119]. Service landscape not only serves as a cue of expected service quality, but also affects customers' evaluations of other factors that determine perceived service quality. Service landscape has direct and indirect effects on perceived service quality, resulting in service landscape having a higher overall effect (Reimer & Kuehn, 2005).[120] According to Levitt's explanation (1981), since the hotel industry provides a high level of intangible products such as services, consumers are likely to make judgments and evaluations through tangible aspects such as appearance. In this case, the service landscape is not only an important component in the formation of customer impressions, but also an important source of evidence for the overall assessment of the service landscape itself and the entire service organization.[121] Servicescape has the potential to positively shape the emotional responses of hotel customers, which can lead to increased satisfaction. This enhancement in satisfaction can significantly boost the likelihood of positive behavioral intentions from the customers (Park et al., 2019).[122]

In order to measure customers' preference for certain Servicescape, some scales were designed and widely applied based on the theoretical mechanism explained in the previous chapter. A common method is to conduct questionnaire

[119] Ezeh, C., & Harris, L. C. (2007). Servicescape research: a review and a research agenda. The Marketing Review, 7(1), 59-78.

[120] Reimer, A., & Kuehn, R. (2005). The impact of servicescape on quality perception. European Journal of marketing, 39(7/8), 785-808.

[121] Levitt, T. (1981). Marketing intangible products and product intangibles. Cornell Hotel and Restaurant Administration Quarterly, 22(2), 37-44.

[122] Park, J. Y., Back, R. M., Bufquin, D., & Shapoval, V. (2019). Servicescape, positive affect, satisfaction and behavioral intentions: The moderating role of familiarity. International Journal of Hospitality Management, 78, 102-111.

surveys on target customers (Lee et al.)[123]. For example, perceived quality was measured by three items (poor-excellent, poor-excellent, unfavorable-favorable) in a 7-point semantic differential scale were adapted by previous studies (e.g., Wakefield & Blodgett, 2016[124]; Ha and Jang, 2012[125]; Tifferet and Vilnai- Yavetz, 2017[126]). Overall satisfaction with the servicescape had a significant impact on behavioral intention variables such as repeat business, word of mouth, and desire to stay in the environment. Satisfaction is also a main indicator of customer perception of the service landscape and is often measured through open-ended questions and Likert scales (Lin & Mattila, 2015).[127]

2.3.3 Consumer Decision Making and Behavior

Consumer decision-making and behavior is a complex, multifaceted process influenced by various factors. The Engel-Kollat-Blackwell (EKB) model, extending John Dewey's original five-stage problem-solving process to consumer behavior, offers a widely accepted explanation. This model outlines five core stages in the decision-making process: problem recognition, search, alternative evaluation, purchase choice, and outcomes (Darley et al., 2010)[128]. Some research also suggests that consumer purchase behavior can be viewed from three perspectives: decision-making, experience, and behavioral influence (Duong et al., 2022)[129]. The decision-making view posits that buying behavior results from consumers engaging in a problem-solving task involving a series of stages. From the experiential

[123] Lee, C. J., Wang, Y. C., & Cai, D. C. (2015). Physical factors to evaluate the servicescape of theme restaurants. Journal of Asian Architecture and Building Engineering, 14(1), 97-104.

[124] Wakefield, K. L., & Blodgett, J. (2016). Retrospective: The importance of servicescapes in leisure service settings. Journal of Services Marketing, 30(7), 686-691.

[125] Ha, J., & Jang, S. (2012). The effects of dining atmospherics on behavioral intentions through quality perception. Journal of services marketing, 26(3), 204-215.

[126] Tifferet, S., & Vilnai-Yavetz, I. (2017). Phytophilia and service atmospherics: The effect of indoor plants on consumers. Environment and Behavior, 49(7), 814-844.

[127] Lin, I. Y., & Mattila, A. S. (2010). Restaurant servicescape, service encounter, and perceived congruency on customers' emotions and satisfaction. Journal of hospitality marketing & management, 19(8), 819-841.

[128] Darley, W. K., Blankson, C., & Luethge, D. J. (2010). Toward an integrated framework for online consumer behavior and decision making process: A review. Psychology & marketing, 27(2), 94-116.

[129] Duong, N. T. H., Chi, N. K., Nguyen, H. T., Nguyen, N. T. K., Nguyen, C. P., & Nguyen, U. T. T. (2022). WTPP for ecotourism: the impact of intention, perceived value, and materialism. Journal of Hospitality and Tourism Insights, 5(5), 1034-1045.

perspective, consumers buy products to create feelings, experiences, and emotions, rather than merely solving problems. The behavioral influence approach suggests that, in some cases, consumer actions are responses to environmental pressures.

Given the complexity of the decision-making process, measuring the specific consumer landscape's ability to arouse potential customers' interest can be achieved by simulating their real decision-making. The metrics for measurement include two main aspects: reservation and repurchase intentions, and willingness to pay. Studies have confirmed the importance of satisfaction, quality, switching costs, and commitment in explaining repeat purchase intentions (Han & Ryu, 2009; Jones & Chen, 2011)[130][131]. Repeat business is key to long-term company success (Han & Kim, 2010)[132]. Additionally, other research has explored the relationship between biophilic attributes and willingness to pay, finding that moderately implemented direct biophilic attributes can enhance perceptions of aesthetics and attractiveness (Nanu & Pahman, 2023)[133], but excessive implementation may diminish these effects (Guzzo et al., 2022)[134].

[130] Han, H., & Ryu, K. (2009). The roles of the physical environment, price perception, and customer satisfaction in determining customer loyalty in the restaurant industry. Journal of hospitality & tourism research, 33(4), 487-510.

[131] Jones, P., & Chen, M. M. (2011). Factors determining hotel selection: Online behaviour by leisure travellers. Tourism and Hospitality research, 11(1), 83-95.

[132] Han, H., & Kim, Y. (2010). An investigation of green hotel customers' decision formation: Developing an extended model of the theory of planned behavior. International journal of hospitality management, 29(4), 659-668.

[133] Nanu, L., & Rahman, I. (2023). The biophilic hotel lobby: Consumer emotions, peace of mind, willingness to pay, and health-consciousness. International Journal of Hospitality Management, 113, 103520.

[134] Guzzo, R. F., Suess, C., & Legendre, T. S. (2022). Biophilic design for urban hotels–prospective hospitality employees' perspectives. International Journal of Contemporary Hospitality Management, 34(8), 2914-2933.

CHAPTER 3. BDIA FEATURE ENGINEERING

The emerging field of sustainable architecture and interior design continually seeks innovative metrics to integrate natural elements within built environments. In this context, in this research, Biophilic Design Indirect Attributes (BDIA) is proposed as a pioneering model to quantitatively assess biophilic elements in hotel interiors. Focusing on the indirect experience of nature, it aims to quantify and enhance the presence of nature-inspired design, thereby improving the human experience within these spaces.

	Chiman Preschool	Multipurpose Hall	Cafeteria	Classroom 1	Classroom 2	Classroom 3	Classroom 4	Classroom 5	Classroom 6	Corridors	Total Subscores
(1) Environmental features											
1	Color	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
2	Water	×	✓	×	×	×	×	×	×	×	1
3	Air	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
4	Sunlight	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
5	Plants	×	×	×	×	×	×	×	×	×	0

Figure 8 BID-M Application sample

Source: Mohammed, I., Onur, Z., & Çağnan, Ç. (2023). An Exploration of Biophilic Design Features within Preschool Interiors. Sustainability, 15(15), 11913.

BDIA is inspired and informed by two foundational frameworks: the Design Quality Indicator (DQI) and the Biophilic Interior Design Matrix (BID-M). Initiated in 1999 by the Construction Industry Council (CIC), the DQI offers a holistic methodology for assessing and improving building design quality (Gann et al., 2003)[135]. This framework was developed to meet the construction industry's need for a systematic evaluation tool, focusing on key aspects like completion timelines, financial efficiency, and safety. The methodical nature of the DQI's approach to building design assessment provides a solid methodological base for the BDIA. The Biophilic Interior Design Matrix (BID-M), crafted by McGee and Marshall-Baker, on

[135] Gann, D., Salter, A., & Whyte, J. (2003). Design quality indicator as a tool for thinking. Building research & information, 31(5), 318-333.

the other hand, brings a detailed quantification to biophilic design. It assesses the presence and absence of 52 biophilic design attributes as defined by Kellert, within interior environments (Marte et al., 2020)[136]. This Matrix is dynamic and requires adjustments for each type of space to accurately reflect its unique needs.

Distinct from the broader and more intricate frameworks of the DQI and the BID-M, the BDIA introduces a targeted and streamlined scoring system tailored for hotel interiors. In this model, interior design professionals evaluate biophilic attributes in key hotel areas—guestrooms, lobbies, and lounges—using a straightforward binary (YES/1 or NO/0) scoring method. This approach, focusing on building elements such as walls, windows, and furniture, differs from the multi-dimensional assessment of DQI and the extensive attribute checklist of BID-M. The resulting BDIA-Index, derived from the average of these binary scores, offers a concise yet effective measure of a hotel's biophilic design quality.



Figure 9 Feature engineering process Source: Author

To improve the versatility and accuracy of BDIA, BDIA was first optimized for 3 rounds of feature engineering before it was officially put into establishing the online booking prediction model. Feature engineering is a critical process in the field of machine learning and data science, involving the transformation and creation of data inputs that can be effectively used in predictive models. Essentially, it's the act of

[136] Marte, E., Calumpit, A., de Sá Bessa, B., Toledo, A., Fadda, R., & Skoler, T. (2020). Testing reliability of biophilic design matrix within urban residential playrooms. Frontiers in Psychology, 11, 570099.

extracting more useful information from raw data and transforming it into formats that are more suitable for modeling. This process often includes selecting relevant features, constructing new features based on existing data, and transforming variables to improve the model's performance. The goal of feature engineering is to enhance the predictive power of machine learning algorithms by providing them with high-quality, relevant input data, which ultimately leads to more accurate and insightful outcomes. The effectiveness of a machine learning model is often significantly influenced by the quality of feature engineering performed on the dataset (Turner et al., 1999)[137]. In each round of optimization, more than 20 experts were used to review the rating scale to increase the objectivity of the results, and the reliability and validity of the optimization results were analyzed to ensure that the results of the scale were consistent among different raters.

3.1 Version 01

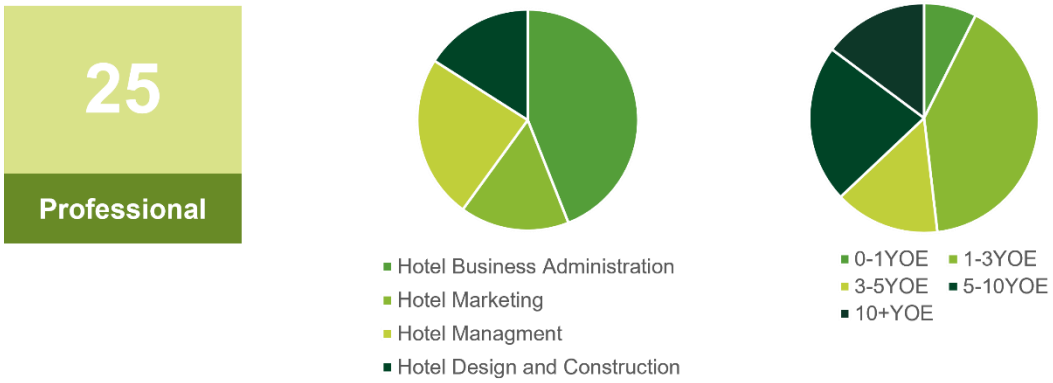


Figure 10 Demographic Results in the First Round Source: Author

In the initial round of feature engineering, eight vacation resorts located on the island of Molokai in Hawaii were randomly selected for study. Additionally, 25 professionals from the hotel industry were invited to evaluate the preliminary version of the BDIA. The initial design of the BDIA did not concentrate on specific elements; instead, it incorporated both Kellert's 24 biophilic design elements and Brown's 16

[137] Turner, C. R., Fuggetta, A., Lavazza, L., & Wolf, A. L. (1999). A conceptual basis for feature engineering. Journal of Systems and Software, 49(1), 3-15.

biophilic elements. These attributes were categorized into descriptions relevant to particular architectural settings and divided into three distinct parts: indoor, outdoor, and abstract spatial features. The quality of each element was measured using a 5-point Likert scale, with higher scores indicating better quality.

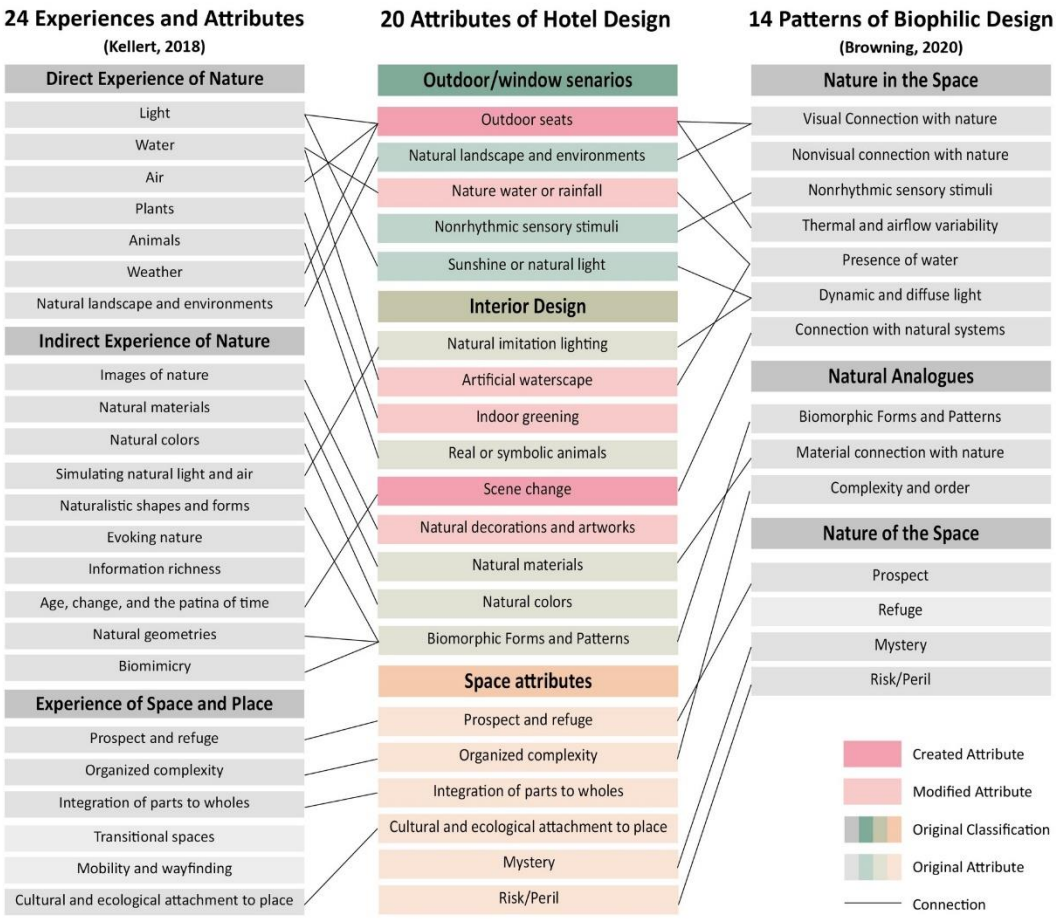


Figure 11 20 Selected Attributes in the first round Source: Author

Table 2. Selected Attributes description in the first round

Attribute	Description
Outdoor	**Elements of a scene that are outside or perceived to be outside (5)**
Seats and chairs	Comfortable outdoor seating, rest stool and other facilities.
Terrace, balcony or roof garden	The hotel has outdoor platforms that rises above the outdoor floor and overlooks the surrounding environment.
Artificial or natural water	Artificial water such as tanks, sprayers or small fountains. Natural water including streams, waterfalls or seascapes.
Nonrhythmic	Views, views and surroundings are perceived through Windows

sensory stimuli	or outdoor facilities.
Sunshine or natural light	The hotel is exposed to direct or indirect sunlight.
Interior design	**Environmental elements that occur primarily indoors. (8)**
Natural imitation lighting	Indoor lighting that simulates natural spectrum or natural light phenomena.
Indoor greening	Real or simulated plants grown indoors, including vines, potted plants, flowers or fake flowers, green walls, etc.
Natural decorations and artworks	A decoration or artwork with natural imagery or abstract features, such as a hanging picture with a motif of a tropical rainforest.
Real or symbolic animals	A physical or image, sculpture, and abstract symbol of an animal.
Natural materials	The main area is made of naturally produced materials, such as wood, stone, fabric and leather.
Artificial natural materials	The main area of the material is made from natural materials, reminiscent of nature, such as terrazzo, exposed red brick, rough plaster walls, etc.
Natural colors	Colors from nature, or direct exposure of natural materials.
Biomorphic Forms and Patterns	A shape or pattern derived from a naturally formed order, such as a spiral, tree, etc.
Space attributes	**The abstract characteristics of space and the psychological feelings it gives people. (3)**
Prospect and refuge	A private area that provides a sense of security, good visibility or privacy.
Cultural and ecological attachment to place	A space or design style characterized by native Hawaiian art, culture, ecology, or architecture.
Mystery	Mysterious, ambiguous Spaces that have the potential to be explored and arouse people's curiosity.

According to this questionnaire, 25 experts submitted their questionnaire responses through an online anonymous platform. SPSS was used to calculate reliability and validity based on the results, and Cronbach α=0.929; KMO=0.776. This means that the reliability of the questionnaire is high (Cronbach α>0.8), but the validity is low (KMO α<0.8), that is, the evaluation results between different experts vary greatly, and the consistency of the scale is low. Analysis of variance showed that the scale was effective in distinguishing different hotel groups, but the results within the group were different. We conducted a correlation analysis between the results

and Booking.com's review scores and found that the results of correlation analysis showed that the attributes in the first version of the scale could not reflect and predict the score well. In summary, the design of the first round of BDIA questionnaire indicators cannot effectively quantify the biophilic characteristics of hotels and requires further optimization.

Table 3. ANOVA Result in the first round

ANOVA										
	HOTEL NUMBER (Mean±Std. Deviation)								F	p
	HOTEL01 (n=25)	HOTEL02 (n=25)	HOTEL03 (n=25)	HOTEL04 (n=25)	HOTEL05 (n=25)	HOTEL06 (n=25)	HOTEL07 (n=25)	HOTEL08 (n=25)		
Seats and chairs	58.56±26.49	80.12±11.27	77.60±15.26	63.88±23.52	68.32±22.45	71.44±19.95	63.64±24.06	65.96±20.26	3.067	0.004**
Terrace, balcony or roof garden	64.72±27.14	84.68±14.77	82.40±16.66	61.76±24.98	74.40±22.41	72.96±22.25	61.76±27.15	72.56±18.47	3.915	0.001**
Artificial or natural water	76.96±19.99	83.12±12.32	80.64±15.13	62.28±25.10	73.12±23.82	78.88±21.51	66.60±25.94	73.60±17.84	2.928	0.006**
	77.48±18.56	83.40±13.67	81.16±11.16	62.92±23.77	69.40±22.10	75.72±16.82	68.32±23.16	71.48±17.79	3.382	0.002**
Sunshine or natural light	82.04±18.40	80.12±17.08	81.52±15.72	64.20±22.96	75.20±22.05	77.28±16.35	69.72±21.27	77.60±13.64	2.732	0.010*
Natural imitation lighting	61.80±30.42	64.36±26.86	73.12±23.56	53.20±26.42	66.60±27.59	68.36±26.90	59.48±26.28	64.48±27.72	1.228	0.289
Indoor greening	54.28±28.30	71.48±20.11	77.28±14.32	54.28±31.03	65.20±25.58	74.44±20.75	58.48±25.68	61.28±26.41	3.364	0.002**
Natural decorations and artworks	67.44±27.81	82.72±12.50	73.28±22.55	56.04±24.83	75.84±21.14	76.04±15.84	62.28±25.59	64.00±26.39	3.756	0.001**
Real or symbolic animals	43.56±33.05	71.08±22.51	69.08±25.38	48.00±31.45	59.52±27.02	64.68±26.09	56.44±29.32	58.36±26.43	2.977	0.005**
Natural materials	68.76±26.68	79.64±15.06	72.96±19.35	54.00±28.15	74.96±20.85	75.16±17.75	61.28±22.86	68.48±21.17	3.629	0.001**
Artificial natural materials	70.88±22.57	79.84±16.06	75.40±16.96	63.20±24.52	70.72±24.48	77.92±18.31	61.52±26.48	71.32±21.69	2.243	0.033*
Natural colors	76.20±23.09	78.24±17.46	77.80±16.55	57.20±25.51	73.08±20.05	73.96±17.62	62.00±24.34	70.96±23.28	3.245	0.003**
Biomorphic Forms and Patterns	61.12±28.82	69.28±24.50	70.76±22.55	57.40±28.67	68.72±23.45	66.32±22.43	60.96±26.84	63.96±24.05	0.877	0.525
Prospect and refuge	55.52±35.07	73.60±22.53	67.84±24.24	51.40±28.66	62.76±30.48	64.44±25.95	53.48±28.69	61.92±24.82	1.842	0.081
Cultural and ecological attachment to place	73.00±21.41	80.64±13.11	78.08±16.83	56.44±27.93	72.40±18.64	74.60±18.07	61.24±26.85	67.28±25.11	3.693	0.001**
Mystery	59.32±28.26	73.20±21.59	64.96±27.42	51.40±27.43	62.88±26.88	63.68±25.46	52.16±28.86	58.04±26.45	1.795	0.09

* p<0.05 ** p<0.01

Table 4. Pearson Correlation Result in the first round

Pearson Correlation								
	Overall	Facilities	Comfort	Comment Number	Value of Price	Price	People	Average Price
Seats and chairs	0.499	0.468	0.222	-0.412	0.357	-0.544	0.04	-0.353
Terrace, balcony or roof garden	0.465	0.383	0.204	-0.151	0.48	-0.609	-0.076	-0.265
Artificial or natural water	0.507	0.356	0.351	0.301	0.456	-0.56	-0.374	0.086
	0.53	0.479	0.489	0.354	0.443	-0.465	-0.483	0.262
Sunshine or natural light	0.388	0.193	0.326	0.562	0.401	-0.497	-0.561	0.327
Natural imitation lighting	0.345	-0.101	-0.009	0.116	0.134	**-0.745***	-0.45	0.069
Indoor greening	0.507	0.277	0.118	-0.249	0.246	-0.676	-0.148	-0.222
Natural decorations and artworks	0.194	0.073	-0.014	-0.016	0.13	-0.637	-0.174	-0.177
Real or symbolic animals	0.453	0.408	0.245	-0.494	0.252	**-0.766***	-0.071	-0.356
Natural materials	0.254	0.09	0.073	0.088	0.246	**-0.707***	-0.247	-0.138
Artificial natural materials	0.597	0.463	0.304	0.12	0.599	-0.484	-0.059	-0.217
Natural colors	0.34	0.119	0.177	0.429	0.348	-0.54	-0.434	0.163
Biomorphic Forms and Patterns	0.239	0.005	-0.07	-0.118	0.138	**-0.754***	-0.248	-0.159
Prospect and refuge	0.489	0.423	0.27	-0.168	0.475	-0.661	-0.072	-0.301
Cultural and ecological attachment to place	0.36	0.173	0.157	0.277	0.307	-0.565	-0.365	0.074
Mystery	0.376	0.309	0.168	0.027	0.385	-0.529	-0.103	-0.194

* p<0.05 ** p<0.01

Interviews were conducted with subject experts who participated in the first round of feature engineering to obtain suggestions for further modifications. According to feedback, the first round of questionnaires mainly had shortcomings in the following three aspects: (1) Too many attributes: "16 elements are a bit too many, and the scoring starts to get messy at the end, so the descriptive/keyword extraction

method is more accurate.” (2) Professional barrier: Hotel management and marketing professionals said that abstract spatial elements were difficult for them to understand and they were unable to give accurate scores (e.g., outdoor terrace or rooftop garden, because the selected hotel did not have a lobby or high-rise, many people would interpret this as a balcony; imitation of natural lighting, which was difficult to see in the photos, and the judges were confused during the scoring process), and that they paid more attention to marketing components (e.g., quality of photos, angles, color palette, presentation elements and activities), and they paid more attention to marketing elements. (e.g. photo quality, angle, color palette, elements and activities presented) rather than the room design itself. (3) The subjects were too similar: Because of the choice of location, the number of hotels and the differences were small and difficult to distinguish.

Therefore, in the design of the second round of BDIA, the following three changes were made: (1) The scope of the study was reduced, reducing the 24 biophilic elements to only 10 items that studied indirect natural experiences. (2) Change the general description into a specific room + part form, focusing on the local area. (3) Change the Likert scale to a yes/no binary question to make it easier to judge.

3.2 Version 02

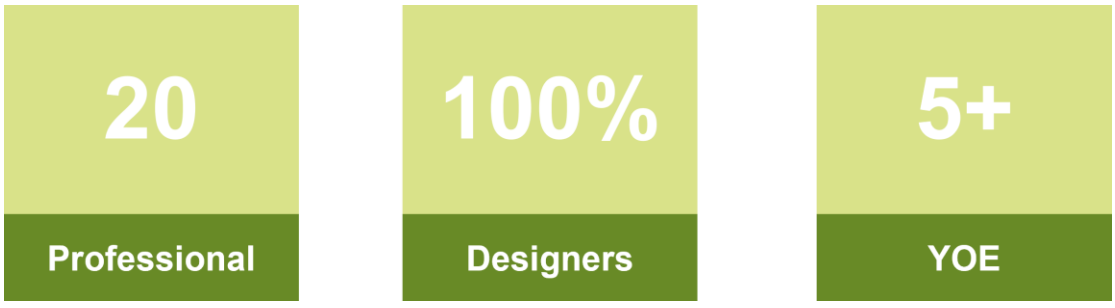


Figure 12 Demographic Results in the Following Round Source: Author

In the second round of feature engineering, the types of review experts included were changed. First of all, since BDIA is a design indicator, hotel management, operations and economic calculation experts will be excluded from this round, and

only design professionals, including interior designers, architectural designers and landscape designers, will be included. At the same time, all people with more than five years of experience are selected to ensure the professionalism of the conclusions.

Secondly, the criteria for selecting hotels have been optimized. In the first round of testing, there was less hotel data selected, and the hotels were relatively similar, making it difficult to differentiate. Therefore, the hotels selected in the second round of feature engineering were all located in Waikiki, and were limited to 4-star hotels. The included pictures and data were also After strict qualifications, the specific standards are as follows:

1) Availability of official images of the hotel lobby, lounge, and standard double-bed rooms including bathroom, window, and entrance views. The purpose of this is to ensure that a comprehensive perspective can be obtained in the subsequent customer inspection questionnaire and biophilic element richness scoring, and no certain spaces will be missed.

2) Exclusively four-star, officially operated hotels in Waikiki (not privately sublet). This can reduce the differences between different hotels due to grade rather than design itself. Officially operated four-star hotels generally have mature and relatively unified supporting facilities, such as lounges and bars, swimming pools, lounges, and lobbies. The size and layout of the guest rooms are also relatively similar.
3) A minimum of 100 reviews to ensure the reliability of NLP results.
4) Presence of each selected indirect biophilic element in at least one hotel, with one other lacking the same element for contrast.
5) Diverse decoration styles range from unique designs to more business-oriented styles.

Table 5. All hotels that meet the requirement and the 6 selected ones

	Hotel Name	Comment Number	Overall Score	Services	Facilities	Cleaness	Comfort	Value of Money	Location	WiFi	Average Price
1	Hilton Garden Inn Waikiki Beach	2638	8.1	8.6	8.2	8.5	8.6	7.9	9.1	7.3	278.06
2	Wayfinder Waikiki	531	8.5	8.9	8.7	8.8	8.8	8.2	9	7.5	224.31
3	Sheraton Waikiki Beach Resort	2063	8.6	8.9	8.8	9	9.1	7.7	9.7		896.37
4	Hyatt Centric Waikiki Beach	321	8.6	9.3	8.7	9.1	9.1	8.1	9.3	8.8	355.67
5	Prince Waikiki	705	8.8	9.3	9.1	9.2	9.3	8.3	8.9		445.71
6	Hilton Hawaiian Village Waikiki Beach Resort	2904	7.8	8.4	8.1	8.2	8.3	7	8.9	7.5	536.32
7	The Surfjack Hotel & Swim Club	695	8.5	9.3	8.5	8.7	8.7	8.4	9.1		197
8	Queen Kapiolani Hotel	2941	8.4	9	8.5	8.8	8.9	8.1	9.3		349.95
9	Hilton Waikiki Beach	679	8.1	8.7	8.2	8.5	8.5	7.7	8.9		311.83
10	Moana Surfrider, A Westin Resort & Spa, Waikiki Beach	862	8.5	8.9	8.6	8.9	9	7.5	9.7		656.42
11	The Equus	673	7.8	9.1	7.7	8	8.1	7.9	8.6	8.7	117.67
12	Shoreline Hotel Waikiki	772	8.1	9.2	8	8.4	8.4	7.9	9.4	8.8	227
13	Hyatt Regency Waikiki Beach Resort & Spa	1178	8.3	8.6	8.3	8.6	8.7	7.6	9.4	8.2	524.52
14	Aston Waikiki Sunset	1380	7.8	8.4	7.7	8	8	7.9	8.8		281.37
15	Waikiki Beach Marriott Resort & Spa	506	8.5	8.8	8.7	9	9	7.7	9.3	8.5	348.52
16	Hotel La Croix	913	7.5	8.7	7.5	7.9	8	7.5	8.7	7.3	249
17	Hilton Grand Vacations Club Grand Waikikian Honolulu	153	8.8	8.9	9	9.3	9.2	8.1	9.1		470.38
18	Embassy Suites by Hilton Waikiki Beach Walk	679	8.7	9.2	8.7	8.9	8.9	8.2	9.4	7.7	599.03
19	Hilton Grand Vacation Club The Grand Islander Waikiki Honolulu	106	8.6	8.9	9	9.2	9.2	8.1	9.1		388.5
20	OUTRIGGER Waikiki Beach Resort	514	8.7	9.2	8.6	8.9	9.1	8	9.7		896.29
21	OUTRIGGER Waikiki Beachcomber Hotel	913	8.7	9.2	8.7	9.1	9.2	8.1	9.4	7.9	470.93
22	Halepuna Waikiki by Halekulani	1519	9	9.4	9.2	9.5	9.6	8.3	9.4	9.2	425
23	The Laylow, Autograph Collection	221	8.4	9.2	8.7	8.9	9.1	7.9	9.3		474.86
24	OUTRIGGER Reef Waikiki Beach Resort	441	8.7	9.5	8.5	9.2	9.3	8.1	9.3	8.5	578.06
25	Hilton Vacation Club The Modern Honolulu	0	7.6	8.2	8	8	8	7.4	8.5	8.3	854.42
26	Aston Waikiki Beach Tower	0	9.4	9.7	9.3	9.5	9.5	8.5	9.8	9.8	666

The purpose of this is to obtain a large amount of reliable real review data, while including hotels with different design styles and BDIA levels. A total of 26 hotels met the standards, and 6 of them were selected for testing. The basic introduction and reasons for choosing the six hotels are as follows:

Table 6. Selected hotels introduction

Hotel Name	Description
Wayfinders Waikiki 	**Brief Introduction:** Wayfinder Newport is a boutique hotel created by a group of local chefs, bartenders, artists, designers, and entrepreneurs who are proud to call Newport, RI home. A shared passion project, we've banded together to create a fresh take on the traditional Newport hotel experience[138]. **Select Reason:** designer hotels with all kinds of indirect biophilic design interior attributes except Biomimicry, and also very Hawaiian.

[138] https://www.wayfinderhotels.com/

Shoreline Hotel Waikiki 	**Brief Introduction:** Shoreline is a tropical boutique hotel located a block from the beach in the heart of Waikiki in Honolulu, Hawaii. From our colorful lobby and comfy rooms to the rooftop pool, experience Shoreline Hotel Waikiki's vibrant and playful vibe, giving Hawaiian style a modern and colorful twist[139]. **Select Reason:** A designer hotel without Natural Materials and colors, good comparison group
Prince Waikiki 	**Brief Introduction:** Experience Waikiki's only oceanfront hotel offering all ocean views and oceanfront rooms and suites[140]. **Select Reason:** Commercial-style hotel without any Simulating Natural Light and Air. It can be used to test if local climate control facilities (Balcony, numerical or horizontal shade, fully open floor sash) can influence customers' reactions.
Hilton Vacation Club The Modern Honolulu 	**Brief Introduction:** Located in the heart of Waikiki, our hotel is less than a mile from Waikiki Beach and downtown shopping and dining. Honolulu Airport is eight miles away and Diamond Head State Monument is four miles. Our modern coastal rooms include a balcony, microwave, and Keurig. Enjoy our poolside bar and lounge, along with activities like hula lessons, lei-making, or morning yoga[141]. **Select Reason:** This hotel is in the standard chain hotel style and lacks indirect biophilic interior design features such as natural colors, shapes, and bionic and Hawaiian features.
Hilton Garden Inn Waikiki Beach 	**Brief Introduction:** Our hotel is just two blocks from the warm sand of Waikiki Beach. We're within walking distance from the International Market Place and surrounded by city dining, with Honolulu Zoo five minutes away. Daniel K. Inouye International Airport is 11 miles from the door. Kick back in our rooftop pool with a sundeck and enjoy our fitness center.[142] **Select Reason:** This hotel is a standard chain hotel style, using natural colors and materials, but lacks natural shape, bionic, evoking nature, and other indirect biophilic interior design features.

[139] https://shorelinehotelwaikiki.com/

[140] https://www.princewaikiki.com/

[141] https://www.hilton.com/en/hotels/hnlmogv-hilton-vacation-club-the-modern-honolulu/

[142] https://www.hilton.com/en/hotels/hnlkugi-hilton-garden-inn-waikiki-beach/

Moana Surfrider, A Westin Resort & Spa, Waikiki Beach 	**Brief Introduction:** Guest rooms at this resort offer tranquil island tones and include a flat-screen LCD TV and a mini-refrigerator. Rooms and suites offer balconies with ocean or city views.[143] **Select Reason:** The nostalgic style of the hotel is rich in biophilic design elements and can be used as the group with the highest content of biophilic design elements.

Interior design professionals were asked to evaluate the BDIA-Index of selected hotels. They will read the instructions about BDIA, then for each space(guestroom, lobby, lounge) evaluate if the BDIA attributes are presented in the required spaces (wall; window/balcony; ceiling/floor/carpet; furniture, decorations), answer by YES(1)/NO(0) only. The average score of each attribute and overall scores from this

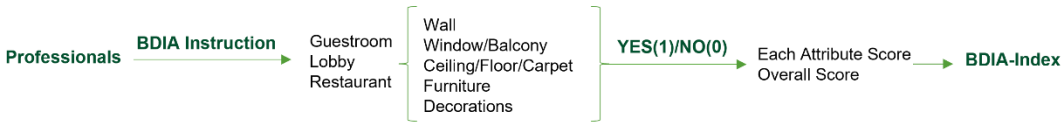


Figure 13 The second round of evaluation process Source: Author

evaluation will be used as the BDIA index of the hotel.

The results of the correlation analysis showed that may be due to sample scale, the attributes in the second version of the scale could not reflect and predict the score well.

Pearson Correlation								
	Booking Overall Score	Booking Facilities Score	Booking Comfort	Booking Value for money	Google Score	Expedia Amenities Score	Expedia Property conditions & facilities Score	Expedia Overall Score
Images of nature	0.309	0.093	0.278	-0.307	0.596	0.351	0.195	0.329
Natural materials	0.666	0.538	0.62	0.35	0.448	0.506	0.491	0.545
Natural colors	0.495	0.609	0.524	0.283	0.241	0.362	0.501	0.405
Simulating natural light and air	0.031	-0.234	-0.113	0.321	0	0	-0.175	0.058
Naturalistic shapes and forms	0.506	0.305	0.461	-0.101	0.634	0.466	0.353	0.48
Evoking nature	**0.823***	0.604	0.729	0.466	0.725	0.72	0.623	0.78
Information richness	0.65	0.685	0.761	0.162	0.621	0.673	0.712	0.547
Age, change, and the patina of time	0.298	0.213	0.164	0.15	0.092	0.069	0.092	0.272
Natural geometries	0.47	0.249	0.516	-0.115	0.569	0.482	0.345	0.33
Biomimicry	0.366	0.318	0.463	-0.37	0.59	0.432	0.391	0.305
Overall	0.626	0.451	0.598	0.061	0.631	0.549	0.471	0.547
* p<0.05 ** p<0.01								

[143] https://www.marriott.com/en-us/hotels/hnlwi-moana-surfrider-a-westin-resort-and-spa-waikiki-beach/overview/

Table 7. Pearson Correlation Result in the second round

As in the previous round of review, interviews were conducted with participating experts to obtain feedback: (1) Vague definition: Many raters found it difficult to distinguish and relate Naturalistic shapes and forms, Evoking nature, Information richness, Natural geometries, and Biomimicry. (2) Frequency of occurrence: Some elements, such as "bionic" and "information richness", are mainly used to describe architectural form and appear less frequently in hotel interior design. The 10 indirect natural experience elements were reduced to 7 elements closely related to hotel design. Evoking nature, Biomimicry, Information richness was deleted. At the same time, the rooms were replaced with specific descriptions, and were paired with specific illustrations to facilitate definition by experts. For example, description such as “Adjustable sash windows allow for natural ventilation.” were used to measure the attributes of “Simulating natural light and air”.

3.3 Final Version

In the third round of BDIA standard design, all indicators were presented to experts in a format similar to the design guide using text description plus illustration to facilitate their understanding. At the same time, the answering device is limited to a computer to prevent the details from being unrecognizable due to pictures that are too small. As in the previous round, the judges will score the three indoor spaces of the same hotel: the lobby, standard rooms and lounge. Depending on whether the specific scene meets the following description, a score of 0 (does not match) or 1 (matches) will be given. Each scenario involves 29 scoring rules. The resulting score table was used in subsequent BDIA studies.

Table 8. BDIA Checklist in the final round

BDIA Checklist
For each scenarios (lobby, standard rooms and lounge), use this checklist to check if the space meets the description
A. Images of nature
The image and representation of nature in the built environment—plants, animals, landscapes, water, geological features—can be both emotionally and

intellectually satisfying. These images can occur through the use of photographs, paintings, sculptures, murals, video, computer simulations, and other representational means. Single or isolated images of nature typically exert little impact. Representational expressions of nature should be repeated, thematic, and abundant.	
	A01. Wall paintings or decorative paintings contain direct natural scenes (such as landscape photos). [144]
	A02. Carpets, pillows, curtains and other fabrics contain direct natural scenes (such as cartoon cat pillows). [145]

[144] https://www.travelandleisure.com/hotels-resorts/beach-hotels/kaimana-beach-hotel-waikiki-design

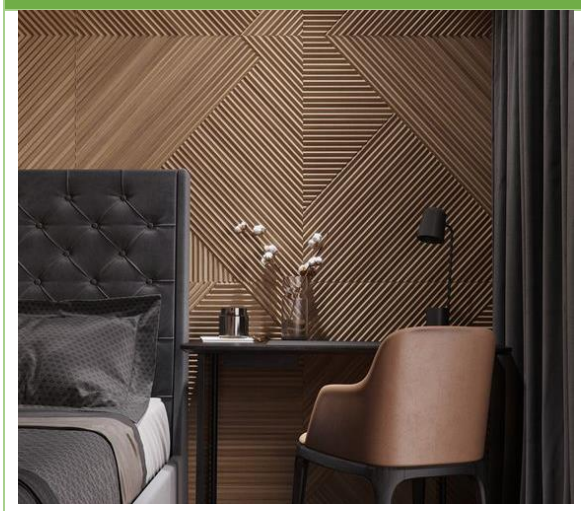
[145] https://www.etsy.com/listing/481569273/bali-wallpaper-with-tropical-print-for?click_key=7aff73111d9312c6ef2a35af75a6dbf87150c067%3A481569273&click_sum=835b7e2b&ref=sold_out-11&pro=1&frs=1



A03. Use decorations with natural forms (such as ceramic rockeries).[146]

B. Natural materials

Natural materials can be especially stimulating, reflecting the dynamic properties of organic matter in adaptive response to the stresses and challenges of survival over time. The transformation of materials from nature frequently elicits positive visual and tactile responses, which few artificial materials can duplicate. Prominent natural building and decorative materials include wood, stone, wool, cotton, and leather, used in a wide array of products, furnishings, fabrics, and other interior and exterior designs.



B01. Background walls, columns and other hard wall decorations are made of natural materials.[147]

[146] https://www.brintons.net/collection/vantage-by-stacy-garcia

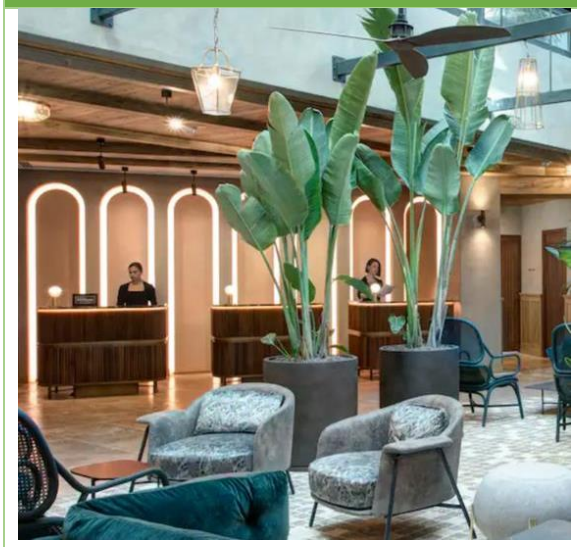
[147] https://www.behance.net/gallery/48360833/3D-rendering-of-hotel-rooms

	**B02.** Window frames, sun visors and balcony railings are made of natural materials.[148]
	**B03.** The ceiling is made of natural materials.[149]
	**B04.** The floors are made of natural materials.[150]

[148]https://www.tumblr.com/visualvocabulary33/675579686651314176/httpswwwosullivanskoufoglou com

[149] https://www.pinterest.jp/pin/4503668371378072/

[150] https://www.pinterest.jp/pin/294282156913167938/

	**B05.** Decorations are made of natural materials.[151]
C. Natural colors Humans evolved as daytime animals, and color has long served as an important means for locating food, water, and other resources, as well as facilitating movement and wayfinding. The effective use of color in the built environment can be challenging, given the modern ability to generate artificial, especially bright colors. The effective biophilic application of color should generally favor muted "earth" tones characteristic of soil, rock, and plants. The use of bright colors should be cautiously applied, and emphasize such appealing environmental forms as flowers, sunsets and sunups, rainbows, and certain plants and animals. The occurrence of highly artificial, contrasting, and "vibrating" colors should be avoided.	
	**C01.** Background walls, columns and other hard wall decorations are made of natural **colors**.[152]

[151] https://uncomn-projects.com/insights-restaurant-design-trends/

[152] https://hotelinteriordesigns.eu/doubletree-by-hilton-rome-monti-biophilic-design-at-its-best/?utm_source=pinterest&utm_medium=pin&utm_content=hotelinteriordesigns-doubletreebyhiltonromemontibi&utm_campaign=inspiration&utm_term=dsoares

	**C02.** Window frames, sun visors and balcony railings are made of natural **colors**.[153]
	**C03.** The ceiling is made of natural **colors**.[154]
	**C04.** The floors are made of natural **colors**.[155]

[153] https://www.pinterest.jp/pin/10977592834306710/

[154] https://www.pinterest.jp/pin/2462974790402874/

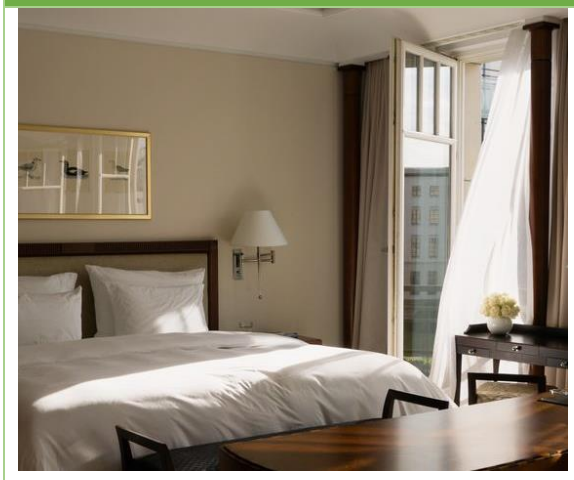
[155] https://www.pinterest.jp/pin/593982638353586741/



C05. Decorations are made of natural **colors**.[156]

D. Simulating natural light and air

Indoor lighting and processed air have been made possible by advances in building technology and construction. The trade-of has often been the occurrence of static conditions that can be physically and psychologically debilitating. Artifcial light can be designed to mimic the spectral and dynamic qualities of natural light. Processed air can also simulate qualities of natural ventilation through variations in airfow, temperature, humidity and barometric pressure.



D01. Adjustable, openable sash windows allow for natural ventilation.[157]

[156] https://www.pinterest.jp/pin/312226186680358017/

[157] https://www.rgdaily.com/the-adlon-kempinski-berlins-most-iconic-hotel/

	**D02.** Adjustable interior or exterior sun visors or blinds, or balconies.
	**D03.** Use indirect ambient lighting.[158]
	**D04.** Lamp with soft lighting effect. [159]

E. Naturalistic shapes and forms

The experience of shapes and forms characteristic of the natural world can be especially appealing. These naturalistic forms can be extraordinarily diverse from the leaf-like patterns found on columns, the shapes of plants on building facades, to animal facsimiles woven into fabrics and coverings. The occurrence of naturalistic shapes and forms can transform a static space into one that possesses the dynamic and ambient qualities of a living system.

[158] https://www.archdaily.com/628507/passo-novita-oska-and-partners

[159] https://i.pinimg.com/originals/ee/d2/e3/eed2e3ad94a5a580ba7d23dfa9bc1eb1.jpg

	**E01.** The furniture adopts irregular natural forms.[160]
	**E02.** Hard decoration (walls, ceilings, etc.) adopts irregular natural forms.[161]
	**E03.** The decorations adopt irregular natural forms.[162]

F. Age, change, and the patina of time

Nature is always changing and in fact, life especially refecting the dynamic forces of growth and aging. People respond positively to these dynamic forces and the

[160] https://noralishdesign.com/reclaimed-wood-furniture/?feed_id=1194&_unique_id=6563957791f5f

[161] https://www.pinterest.jp/pin/901001469177731286/

[162] https://www.pinterest.jp/pin/637541509598835 71/

associated patina of time, revealing nature's capacity to respond adaptively to ever-changing conditions. These dynamic tendencies are often most satisfying when balanced by the complementary qualities of unity and stability. Change and a patina of time can be achieved through such design strategies as naturally aging materials, weathering, a sense of the passage of time, and in other ways.
 **F01.** There are nostalgic elements in the decoration style (such as 90s posters).[163]
 **F02.** The decorative style includes nostalgic craftsmanship (such as bamboo weaving and hand binding).[164]

[163] https://www.pinterest.jp/pin/74309462594694069/

[164] https://www.properhotel.com/austin/



F03. Use recycled materials, furniture or decor.[165]



F04. Use materials that will show natural weathering or aging (such as copper).[166]

G. Natural geometries

The satisfying experience of nature can also be revealed through imaginative and fantastic depictions. These representations may not literally occur in nature, but still draw from design principles prominently encountered in the natural world. For example, the “wings” of the Sydney Opera House suggest the qualities of a bird; Notre Dame’s stained glass windows, a rose-like fower; while, the skyline of some cities mimic the vertical heterogeneity of a forest. None of these designs actually occurs in nature, but they all draw from design principles and characteristics of the natural world.

[165] https://i.pinimg.com/originals/6f/5e/ab/6f5eab41a0bfb0a197a37b3e280da33b.jpg

[166] https://i.pinimg.com/originals/ba/05/bc/ba05bc4dd1dd6ce708fb2ca2d2fd5408.jpg

	**G01.** The furniture adopts abstract and geometric patterns that imitate nature. [167]
	**G02.** Hard decoration (walls, ceilings, etc.) adopts abstract and geometric patterns that imitate nature.[168]
	**G03.** Decorations and soft furnishings adopt abstract and geometric patterns that imitate nature.[169]
Notes: The same scene can meet multiple scoring criteria at the same time. For example, the original color wooden wall can meet B01 "Natural Wall Material" and	

[167] https://www.pinterest.jp/pin/188306828163032672/

[168] https://newravenna.photoshelter.com/gallery-image/Baths/G0000TEgcb8zgTxY/I0000xJSFb24tacc/P0000wXDcW0TFRUY

[169] https://www.sarahdegheselle.com/en/photography/

C01 "Natural Color of Wall". If there are beach photos on the wall, A01 "Wall paintings or decorative paintings contain direct natural scenes" can also be satisfied.

20 professional judges used this standard to conduct correlation analysis and reliability and validity analysis on the ratings of 6 case hotels to verify its effectiveness. Each expert will score in the following table based on the actual situation of the hotel. Five experts will be included in each case hotel. The hotel's name, actual ratings, reviews, average price and other information are kept confidential from the experts. Only the hotel photos are included in the questionnaire to prevent the experts' evaluation results from being affected by the hotel's operation strategy.

Table 9. Scoresheet sample in the final round (Wayfinder Waikiki)

BDIA	**Guestroom**	**Lobby**	**Lounge**
Images of nature	3	2	0
Wall paintings or decorative paintings contain direct natural scenes (such as landscape photos).	1	1	
Carpets, pillows, curtains and other fabrics contain direct natural scenes (such as cartoon cat pillows).	1	1	
Use decorations with natural forms (such as ceramic rockeries).	1		
Natural materials	3	4	5
Background walls, columns and other hard wall decorations are made of natural materials.		1	1
Window frames, sun visors and balcony railings are made of natural materials.			1
The ceiling is made of natural materials.		1	1
The floors are made of natural materials.	1		
The floors are made of natural materials.	1	1	1
Decorations are made of natural materials.	1	1	1
Natural colors	4	5	6
The hard decoration of the background wall, pillars and other walls are in natural colors.	1	1	1
Window frames, sun visors and balcony railings are in natural colours.			1
The ceilings are in natural colours.		1	1
The floors are in natural colours.	1	1	1
Furniture is in natural colours.	1	1	1
Decorations are in natural colors.	1	1	1

Simulating natural light and air	2	2	3
Adjustable sash windows allow for natural ventilation.	1	1	1
Adjustable sun visors or blinds.		1	1
Use indirect ambient light.			
Lamp with soft lighting effect.	1		1
Naturalistic shapes and forms	2	2	0
The furniture adopts irregular natural forms.	1	1	
Hard decoration (walls, ceilings, etc.) adopts irregular natural forms.			
The decorations adopt irregular natural forms.	1	1	
Age, change, and the patina of time	4	4	3
There are nostalgic elements in the decoration style (such as 90s posters).	1	1	1
The decorative style includes nostalgic craftsmanship (such as bamboo weaving and hand binding).	1	1	1
Use recycled materials, furniture or decor.	1	1	
Use materials that will show natural weathering or aging (such as copper).	1	1	1
Natural geometries	2	2	3
The furniture adopts abstract and geometric patterns that imitate nature.	1	1	1
Hard decoration (walls, ceilings, etc.) adopts abstract and geometric patterns that imitate nature.			1
Decorations and soft furnishings adopt abstract and geometric patterns that imitate nature.	1	1	1
	20	**21**	**20**
Overall Score	**61**		

The results of the correlation analysis showed that the attributes in the second version of the scale could reflect and predict the score well. ANOVA result shows that different professionals rate the same scenario differently. Kappa =0.79, indicating high consistency. Therefore, it can be considered that this result of the round test meet the requirements of reliability and validity, can achieve high uniformity of results, and the questionnaire can be used for subsequent analysis.

Table 10. Pearson Correlation Result in the final round

Pearson Correlation								
	Booking Overall Score	Booking Facilities Score	Booking Comfort	Booking Value for money	Google Score	Expedia Amenities Score	Expedia Property conditions & facilities Score	Expedia Overall Score
A01	0.13	0.086	0.167	0.1	0.159	0.188	0.146	0.1
A02	0.124	0.084	0.143	-0.057	0.261*	0.193	0.147	0.151
A03	-0.103	-0.092	-0.096	-0.173	-0.034	-0.089	-0.089	-0.089
B01	**0.264***	**0.301****	**0.310****	**0.098**	**0.306****	**0.314****	**0.331****	**0.269***
B02	0.113	0.13	0.117	0.075	0.099	0.111	0.126	0.118
B03	0.129	0.15	0.115	0.136	0.084	0.109	0.132	0.144
B04	**0.287***	**0.357****	**0.325****	**0.081**	**0.344****	**0.331****	**0.372****	**0.320****
B05	0.026	0.082	0.04	0.041	-0.033	0.005	0.051	0.012
B06	0.106	0.048	0.125	0.124	0.155	0.174	0.118	0.103
C01	0.164	0.131	0.161	0.089	0.186	0.179	0.156	0.17
C02	-0.021	0.008	-0.048	0.084	-0.095	-0.063	-0.032	-0.009
C03	0.159	0.152	0.171	0.068	0.227	0.208	0.191	0.188
C04	0.018	0.027	0.05	-0.04	0.112	0.088	0.073	0.039
C05	0.045	0.08	0.04	0.071	0.011	0.032	0.06	0.058
C06	-0.008	-0.037	0.003	-0.008	0.031	0.025	-0.005	-0.01
D01	0.093	0.028	0.093	0.025	0.168	0.137	0.081	0.103
D02	0.126	0.073	0.136	0.047	0.176	0.163	0.119	0.12
D03	**0.229***	**0.266***	0.227	**0.312****	0.068	0.18	0.227	0.197
D04	**0.255***	0.2	**0.263***	0.196	0.212	**0.255***	0.225	0.215
E01	0.02	-0.03	0.068	-0.1	0.12	0.09	0.038	-0.01
E02	-0.095	-0.037	-0.077	-0.084	-0.1	-0.097	-0.059	-0.095
E03	-0.002	-0.057	0.031	-0.11	0.05	0.027	-0.017	-0.049
F01	0.127	0.074	0.113	-0.076	0.225	0.14	0.102	0.153
F02	-0.093	-0.145	-0.102	-0.029	-0.045	-0.062	-0.11	-0.083
F03	0.046	0.018	0.056	0.035	0.126	0.109	0.071	0.073
F04	**0.279***	**0.243***	**0.257***	0.187	**0.297***	**0.288***	**0.266***	**0.311****
G01	**0.250***	0.2	**0.239***	0.169	**0.254***	**0.259***	0.228	**0.254***
G02	0.226	0.215	**0.238***	0.163	0.215	**0.242***	**0.236***	0.219
G03	0.02	-0.004	0.034	-0.054	0.061	0.04	0.019	0.008
* p<0.05 ** p<0.01								

3.4 Conclusion

In this chapter, we have successfully established a robust and credible BDIA assessment system and scoring standards through three rounds of feature engineering and the critical review of over 20 experts. This comprehensive system requires evaluators to first familiarize themselves with the BDIA guidelines and scoring instructions presented in section 3.3. Once they have thoroughly understood these guidelines, they are tasked with evaluating the given scenarios. The BDIA criteria encompass 29 distinct indicators, each evaluated using a binary scoring model of Yes (1) or No (0). Evaluations are conducted across three different

scenarios in each hotel: the standard guest room, lobby, and lounge. It's important to note that elements not related to interior design, such as swimming pools, the architectural design of the hotel, and landscape facilities, are intentionally excluded from this standard. This exclusion ensures the focus remains on interior design elements, enhancing the uniformity and applicability of the BDIA standard. This chapter's findings not only pave the way for a standardized approach in BDIA evaluations but also contribute to providing a foundation for data analysis and further research into the impacts of interior design on guest experience and satisfaction.

CHAPTER 4. BIG DATA ANALYSIS AND RESULTS

In this chapter, the research uses an interdisciplinary mixed methods approach to analyze and quantify the actual impact on customer experience of the BDIA standards established in the previous chapter. First, 26 four-star hotels in Waikiki were selected and rated by BDIA experts, and weights were assigned to the predictive ability of different BDIAs on hotel ratings through a random forest algorithm. Afterwards, more than of the hotel were collected through big data, and natural language processing technology (NLP) was used to quantify customers' emotional tendencies and evaluations of BDIA. Finally, by comparing and analyzing the two, a structural equation model (SEM) was established to establish a prediction model of BDIA-Index for real hotel space evaluation.

4.1 Descriptive Statistical Analysis

The study first used crawler technology to obtain a list of all four-star hotels in Waikiki, and a total of 34 hotel data were crawled. The data are all sourced from the public data of booking.com and Google maps. Customer private information (such as name, country and region, etc.) will not be collected. Afterwards, screening was carried out based on the hotel selection criteria determined in 3.2. Some hotels were excluded from the BDIA review because they did not meet the requirements of more than 100 customer reviews and the actual star rating was lower than 4 stars. The remaining 26 hotels were included in the review. Each selected hotel will be scored by 3 judges under double-blind conditions according to the BDIA standards of the final round of feature engineering. For each feature, the result will be the mode of the judges. For the selected hotels, the number of reviews, comprehensive ratings and rating subdivisions, and average nightly price are included in the collection. The overall rating, facilities, comfort, and value of money ratings are used to measure the interior design quality in customer perceptions, and Service, cleanliness, and Wifi are used as control indicators.

Table 11. Selected 4-Stars Hotels in Waikiki

Hotel Number	Hotel Name	Comment Number	Overall Score	Services	Facilities	Cleaness	Comfort	Value of Money	Location	WiFi	Average Price
1	Hilton Garden Inn Waikiki Beach	2638	8.1	8.6	8.2	8.5	8.6	7.9	9.1	7.3	278.06
2	Wayfinder Waikiki	531	8.5	8.9	8.7	8.8	8.8	8.2	9	7.5	224.31
3	Sheraton Waikiki Beach Resort	2063	8.6	8.9	8.8	9	9.1	7.7	9.7		896.37
4	Hyatt Centric Waikiki Beach	321	8.6	9.3	8.7	9.1	9.1	8.1	9.3	8.8	355.67
5	Prince Waikiki	705	8.8	9.3	9.1	9.2	9.3	8.3	8.9		445.71
6	Hilton Hawaiian Village Waikiki Beach Resort	2904	7.8	8.4	8.1	8.2	8.3	7	8.9	7.5	536.32
7	The Surfjack Hotel & Swim Club	695	8.5	9.3	8.5	8.7	8.7	8.4	9.1		197
8	Queen Kapiolani Hotel	2941	8.4	9	8.5	8.8	8.9	8.1	9.3		349.95
9	Hilton Waikiki Beach	679	8.1	8.7	8.2	8.5	8.5	7.7	8.9		311.83
10	Moana Surfrider, A Westin Resort & Spa, Waikiki Beach	862	8.5	8.9	8.6	8.9	9	7.5	9.7		656.42
11	The Equus	673	7.8	9.1	7.7	8	8.1	7.9	8.6	8.7	117.67
12	Shoreline Hotel Waikiki	772	8.1	9.2	8	8.4	8.4	7.9	9.4	8.8	227
13	Hyatt Regency Waikiki Beach Resort & Spa	1178	8.3	8.6	8.3	8.6	8.7	7.6	9.4	8.2	524.52
14	Aston Waikiki Sunset	1380	7.8	8.4	7.7	8	8	7.9	8.8		281.37
15	Waikiki Beach Marriott Resort & Spa	506	8.5	8.8	8.7	9	9	7.7	9.3	8.5	348.52
16	Hotel La Croix	913	7.5	8.7	7.5	7.9	8	7.5	8.7	7.3	249
17	Hilton Grand Vacations Club Grand Waikikian Honolulu	153	8.8	8.9	9	9.3	9.2	8.1	9.1		470.38
18	Embassy Suites by Hilton Waikiki Beach Walk	679	8.7	9.2	8.7	8.9	8.9	8.2	9.4	7.7	599.03
19	Hilton Grand Vacation Club The Grand Islander Waikiki Honolulu	106	8.6	8.9	9	9.2	9.2	8.1	9.1		388.5
20	OUTRIGGER Waikiki Beach Resort	514	8.7	9.2	8.6	8.9	9.1	8	9.7		896.29
21	OUTRIGGER Waikiki Beachcomber Hotel	913	8.7	9.2	8.7	9.1	9.2	8.1	9.4	7.9	470.93
22	Halepuna Waikiki by Halekulani	1519	9	9.4	9.2	9.5	9.6	8.3	9.4	9.2	425
23	The Laylow, Autograph Collection	221	8.4	9.2	8.7	8.9	9.1	7.9	9.3		474.86
24	OUTRIGGER Reef Waikiki Beach Resort	441	8.7	9.5	8.5	9.2	9.3	8.1	9.3	8.5	578.06
25	Hilton Vacation Club The Modern Honolulu	0	7.6	8.2	8	8	8	7.4	8.5	8.3	854.42
26	Aston Waikiki Beach Tower	0	9.4	9.7	9.3	9.5	9.5	8.5	9.8	9.8	666
27	Park Shore Waikiki		8.1	9.1	8	8.3	8.2	7.8	9.5	8.2	376.16
28	Luana Waikiki Hotel & Suites		8.4	9.1	8.5	8.6	8.7	8.3	9.1		207
29	Hilton Grand Vacations Club Hokulani Waikiki Honolulu		9.2	9.6	9.3	9.7	9.7	8.9	9.5		385.33
30	The Twin Fin Hotel		8	8.7	7.8	8.1	8.2	7.6	9.5	8.6	314.75
31	Hyatt Place Waikiki Beach		7.8	8.9	7.8	8.4	8.5	7.7	8.8	8.6	303.9
32	DoubleTree by Hilton Alana - Waikiki Beach		7.3	7.8	7.4	8.1	8.3	7	8.3		356.81
33	Courtyard by Marriott Waikiki Beach		7	8.4	7.2	7.3	7.5	6.8	8.5	7.9	208.2
34	Waikiki Resort Hotel		8.1	8.8	7.9	8.4	8.6	8.1	9.4	8.4	205.92

Table 12. Descriptive statistical analysis results

BDIA		Mean	SD
A01	Nature Image Finishing	0.769	0.424
A02	Nature Image Furnishing	0.346	0.479
A03	Nature Image Decorations	0.744	0.439
B01	Natural materials Walls	0.795	0.406
B02	Natural materials Window Frame	0.423	0.497
B03	Natural materials Ceiling	0.41	0.495
B04	Natural materials Floor	0.679	0.47
B05	Natural materials Furnitures	0.949	0.222
B06	Natural materials Decorations	0.744	0.439
C01	Natural colors Walls	0.821	0.386
C02	Natural colors Window Frame	0.5	0.503
C03	Natural colors Ceiling	0.41	0.495
C04	Natural colors Floor	0.897	0.305
C05	Natural colors Furnitures	0.949	0.222
C06	Natural colors Decorations	0.872	0.336
D01	Adjustable Windows	0.756	0.432
D02	Adjustable Sunshades	0.654	0.479

D03	Indirect Lighting	0.231	0.424
D04	Soft Lighting Lamps	0.667	0.474
E01	Naturalistic shapes and forms Furnishing	0.282	0.453
E02	Naturalistic shapes and forms Finishing	0.141	0.35
E03	Naturalistic shapes and forms Decorations	0.526	0.503
F01	Nostalgic Decorations	0.538	0.502
F02	Nostalgic Craftsmanship	0.487	0.503
F03	Recycled Materials	0.179	0.386
F04	Weathered Materials	0.141	0.35
G01	Natural geometries Furnitures	0.59	0.495
G02	Natural geometries Finishing	0.615	0.49
G03	Natural geometries Furnishing	0.744	0.439

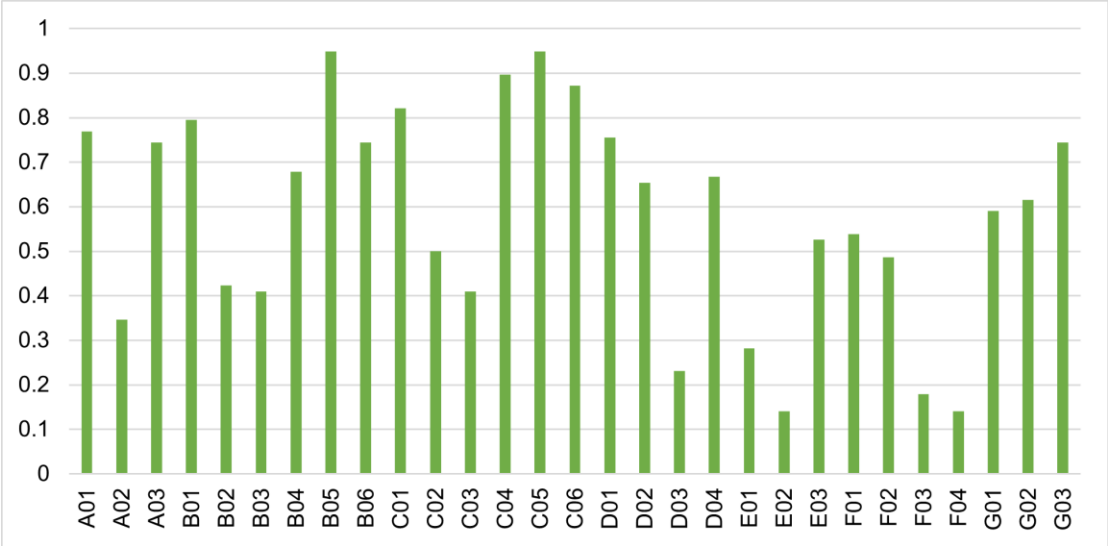


Figure 14. Descriptive statistical analysis results Source: Author

Survey results show that in the design of four-star hotels in the Waikiki area, the most frequently occurring BDIA design elements are B04 (Natural materials Floor), C05 (Natural colors Furnitures), C04 (Natural colors Floor), and C06 (Natural colors Decorations). , whose BDIA average value is >0.5, indicating that most hotels have adopted these designs. A02 (Nature Image Furnishing), D02 (Adjustable Sunshades), E02 (Naturalistic shapes and forms Finishing), F03 (Recycled Materials), and F04 (Weathered Materials) appear less often, with the average number below 0.3, indicating that only a small number of them appear. Hotels employ these design strategies.

Classified by category, sorted by frequency of strategy use: C (Natural colors) 0.74, B (Natural materials) 0.67, G (Natural geometries) 0.65, A (Nature Image) 0.62, D (Simulating natural light and air) 0.58, F (Age, change, and the patina of time) 0.34,

E (Naturalistic shapes and forms) 0.32, that is to say, most hotel interior designs use more natural colors and natural materials, natural geometric patterns and natural Pictures are also a common design element. Design elements with natural shapes and elements with a sense of time appear less frequently, and are used in only one-third of the scenes.

Table 13. Descriptive statistical analysis results by Scenarios

BDIA Mean		Lobby	Guestroom	Lounge
A01	Nature Image Finishing	0.769	0.808	0.731
A02	Nature Image Furnishing	0.308	0.462	0.269
A03	Nature Image Decorations	0.846	0.615	0.769
B01	Natural materials Walls	0.808	0.731	0.846
B02	Natural materials Window Frame	0.538	0.231	0.5
B03	Natural materials Ceiling	0.538	0.115	0.577
B04	Natural materials Floor	0.808	0.5	0.731
B05	Natural materials Furnitures	0.923	1	0.923
B06	Natural materials Decorations	0.923	0.577	0.731
C01	Natural colors Walls	0.808	0.769	0.885
C02	Natural colors Window Frame	0.615	0.346	0.538
C03	Natural colors Ceiling	0.5	0.192	0.538
C04	Natural colors Floor	0.885	0.962	0.846
C05	Natural colors Furnitures	0.923	1	0.923
C06	Natural colors Decorations	0.923	0.885	0.808
D01	Adjustable Windows	0.615	0.923	0.731
D02	Adjustable Sunshades	0.577	0.808	0.577
D03	Indirect Lighting	0.308	0.077	0.308
D04	Soft Lighting Lamps	0.692	0.885	0.423
E01	Naturalistic shapes and forms Furnishing	0.5	0.269	0.077
E02	Naturalistic shapes and forms Finishing	0.308	0	0.115
E03	Naturalistic shapes and forms Decorations	0.692	0.423	0.462
F01	Nostalgic Decorations	0.615	0.308	0.692
F02	Nostalgic Craftsmanship	0.538	0.385	0.538
F03	Recycled Materials	0.269	0.077	0.192
F04	Weathered Materials	0.192	0.077	0.154
G01	Natural geometries Furnitures	0.731	0.577	0.462
G02	Natural geometries Finishing	0.808	0.615	0.423
G03	Natural geometries Furnishing	0.692	0.885	0.654

By classifying and summarizing the BDIA averages according to different scenarios, it can be found that some BDIA appear more in public places, while some are more used in guest room design. For example, all tested hotels use B05 (Natural materials Furnitures) and C05 (Natural colors Furnitures) in the guest room design,

and use them less frequently in the lobby and lounge; no hotel uses E02 (Naturalistic Furnitures) in the guest rooms. shapes and forms Finishing), but is partially used in the lobby. C (Natural colors), E (Naturalistic shapes and forms), F (Nostalgic and weathering) are significantly more common in public spaces, especially lobbies. The difference between public spaces (lobby and lounge) is not as significant as between public and private spaces (guest rooms and others).

4.2 BDIA Weight Adjustment

Before conducting machine learning weight analysis, experts conducted a correlation analysis between the BDIA scores of the 26 selected hotels and the hotel's online public data to initially evaluate the prediction of BDIA for the hotel's various scores. The results show that, while controlling for irrelevant variables, most BDIA indicators are highly correlated with customer-perceived hotel environment quality scores. Among them, the ones with the highest positive correlation with the overall evaluation are B01 (Natural Material Walls), C06 (Nature Color Decorations), F02 (Nostalgic Craftsmanship Decorations), G01 (Natural Geometries Furniture), G03 (Natural Geometries Decorations), that is, these BDIA Presence is associated with higher overall ratings and higher facility ratings. Some BDIA indicators are not necessarily related to quality perception, such as B02 (Natural Material Window Frame), D02 (Adjustable Sunshades), and E02 (Naturalistic shapes and forms Walls). Although some BDIAs are not directly related to the overall score, they are related to one or more of the facilities, comfort, and cost-effectiveness scores. This may be because the data volume (N=26) is small, making the moderately small correlation Sexuality cannot show significance, which may also be because the relationship between this item and quality perception is not significant, and it still needs to be confirmed in subsequent research. To sum up, it can be considered that BDIA has certain predictability for hotel network ratings, so the random forest algorithm is used to conduct weight analysis to determine the main influencing factors.

Table 14. Pearson Correlation Analysis Result on 26 Waikiki Hotels

Pearson Correlation					
	Overall Score	Facilities	Comfort	Value of Money	Average Price
A01	**0.317****	**0.284****	**0.314****	0.107	**0.202***
A02	0.153	**0.158***	0.092	-0.034	0.139
A03	**0.192***	0.096	**0.205***	0.082	0.008
B01	**0.323****	**0.304****	**0.337****	**0.201***	-0.076
B02	0.022	0.006	0.033	-0.038	0.06
B03	**0.195***	**0.227****	**0.216****	**0.313****	**-0.205***
B04	0.044	0.143	0.106	-0.133	**0.235****
B05	**0.201***	**0.240****	**0.243****	0.071	0.105
B06	**0.239****	**0.275****	**0.307****	**0.161***	**0.169***
C01	**0.217****	**0.240****	**0.256****	0.117	0.036
C02	0.131	**0.162***	0.156	0.103	0.125
C03	0.147	**0.221****	**0.176***	0.143	-0.001
C04	0.138	**0.202***	**0.169***	-0.061	**0.254****
C05	**0.227****	**0.202***	**0.243****	0.157	-0.054
C06	**0.309****	**0.292****	**0.342****	**0.236****	-0.05
D01	**0.161***	0.11	0.142	0.09	0.069
D02	-0.018	-0.053	0.014	-0.054	0.045
D03	**0.245****	**0.304****	**0.281****	0.137	**0.264****
D04	**0.204***	**0.213****	**0.224****	0	**0.210****
E01	0.111	0.180*	0.156	0.230**	-0.131
E02	0.022	0.016	0.045	-0.131	0.022
E03	**0.172***	**0.173***	**0.192***	**0.237****	-0.041
F01	0.131	0.101	0.079	0.051	0.071
F02	**0.348****	**0.296****	**0.369****	**0.388****	-0.064
F03	0.095	0	0.012	**0.211****	-0.023
F04	0.08	0.064	0.029	**0.198***	-0.109
G01	**0.441****	**0.414****	**0.435****	**0.253****	**0.182***
G02	**0.259****	**0.281****	**0.310****	0.087	0.076
G03	**0.346****	**0.389****	**0.358****	0.135	**0.248****
* p<0.05 ** p<0.01					

By comparing hundreds of recent studies, Khalilzadeh & Tasci (2017) found that in tourism and hospitality academia, the small, medium, and large effect sizes of Pearson correlation analysis set to 0.10, 0.30, and 0.50 can more effectively reflect the actual situation[170]. Through horizontal comparison of variable sizes in similar studies, it was found that prior brand awareness and positive quality perception are significant drivers of the attitudes with effect size of 0.33-0.56[171]. The influence factor

[170] Khalilzadeh, J., & Tasci, A. D. (2017). Large sample size, significance level, and the effect size: Solutions to perils of using big data for academic research. Tourism Management, 62, 89-96.

[171] Koech, A. K., Buyle, S., & Macário, R. (2023). Airline brand awareness and perceived quality effect on the attitudes towards frequent-flyer programs and airline brand choice-Moderating effect of frequent-flyer programs. Journal of Air Transport Management, 107, 102342.

of the hotel's distance from key facilities (such as beaches and ski resort entrance ropeways) on price is 0.25-0.27[172]. Personal innovativeness has a significant positive effect on intention to use technology in hospitality and tourism research, with an overall effect size (ESr) of 0.38. [173]Therefore, it can be considered that the significant positive correlation between BDIA and booking.com Scores has a medium effect size, and may have a small to medium effect size with other factors. However, due to the small sample size (N= 26) Unable to present significance. Some BDIA may have benign effects on positive rankings, with medium effect sizes.

By comparing the results of 49 studies on biophilic design effect size from 43 articles identified through a meta-analysis by Gaekwad et al., the mean POMS effect size for the relative impact of exposure to natural and urban environments on positive and negative effects was -0.65/-0.6.[174] Therefore, it can be considered that the effect size of BDIA has not reached the mean impact of biophilic design, and the degree of impact is smaller than that of other elements (such as natural light and water views).

When using the random forest algorithm, first include all BDIA and use the following values for calculation: The proportion of the training set is adjusted to 0.9, and the calculation is carried out when the data is normalized. Because the amount of data in the training set is small (N=26*3=78<100), the proportion of the training set is increased from 0.8 to 0.9. The specific value is set as follows:

Table 15. Machine Learning Parameter Model summary table

Name	Parameter names	Parameter settings
Model parameter	Data preprocessing	None
	Training set proportion	0.9

[172] Lixăndroiu, R., & Lupșa-Tătaru, D. (2023). Switzerland? The Best Choice for Accommodation in Europe for Skiing in the 2023 Season. Sustainability, 15(5), 4032.

[173] Ciftci, O., Berezina, K., & Kang, M. (2021, January). Effect of personal innovativeness on technology adoption in hospitality and tourism: meta-analysis. In Information and Communication Technologies in Tourism 2021: Proceedings of the ENTER 2021 eTourism Conference, January 19–22, 2021 (pp. 162-174). Cham: Springer International Publishing.

[174] Gaekwad, J. S., Sal Moslehian, A., Roös, P. B., & Walker, A. (2022). A meta-analysis of emotional evidence for the biophilia hypothesis and implications for biophilic design. Frontiers in Psychology, 13, 2476.

settings	Number of decision trees	100
	Node Split Criteria	squared_error
	Minimum number of samples for node splitting	2
	Minimum number of samples for leaf nodes	1
	tree maximum depth	None
	Maximum number of features	auto
	Whether there is sampling with replacement	Yes
	Whether to perform out-of-bag data testing	Yes

Following this setting, the model training and testing results are matched, and the interpretation rate is high. This shows that the weight distribution results of this model have good explanation for online scoring.

Table 16. Machine Learning Parameter Model evaluation result

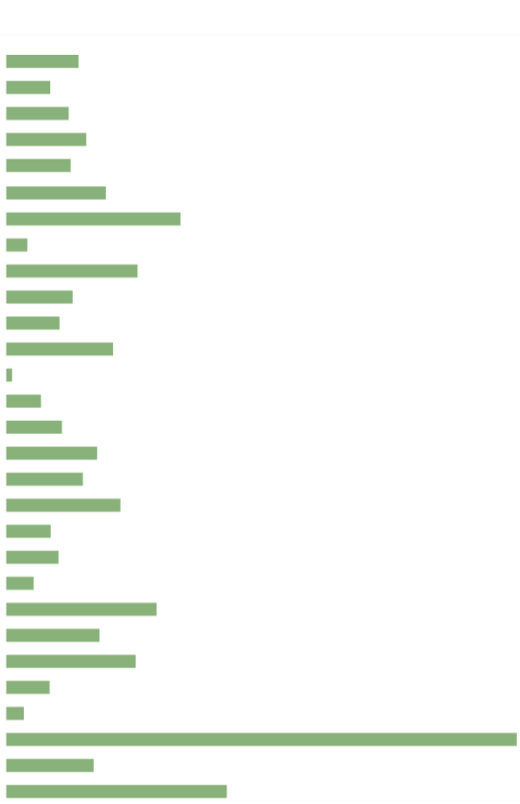
Index	Training	Testing
R squared value	0.961	0.732
Mean absolute error MAE	0.057	0.218
Mean square error MSE	0.007	0.066
root mean square error RMSE	0.085	0.257
Median absolute error MAD	0.033	0.247
Mean Absolute Percent Error MAPE	0.01	0.004
Explained variance score EVS	0.961	0.733
Root mean square log error MSLE	0	0.001

By charting the weight assignments, it can be found that the results of weight analysis are basically consistent with the correlation distribution. That is, the results of machine learning also confirm that some BDIAs have greater weight and higher effectiveness in predicting customer quality perception. Explanatory strength, the impact of some BDIA is smaller. However, the correlation analysis results of some elements are not related to the weight results. For example, B05 (Nature Material Decorations) is significantly related to higher scores, but the explanation strength is smaller. This is because unlike correlation, the level of explanatory weight does not represent whether the element itself is positively related to higher evaluations, but only represents the predictive strength of the element. For example, B05 (Nature

Material Decorations) may appear in most hotels, but only a small number of hotels are not equipped with this element. Therefore, this element cannot be reflected as a unique design feature, so the interpretation is not strong.

Table 17. weight analysis Result

Attributes	Weighted value	
A01	0.026	Nature Image Finishing
A02	0.016	Nature Image Furnishing
A03	0.023	Nature Image Decorations
B01	0.029	Natural materials Walls
B02	0.024	Natural materials Window Frame
B03	0.036	Natural materials Ceiling
B04	0.064	Natural materials Floor
B05	**0.008**	Natural materials Furnitures
B06	0.048	Natural materials Decorations
C01	0.024	Natural colors Walls
C02	0.02	Natural colors Window Frame
C03	0.039	Natural colors Ceiling
C04	**0.002**	Natural colors Floor
C05	**0.013**	Natural colors Furnitures
C06	0.02	Natural colors Decorations
D01	0.033	Adjustable Windows
D02	0.028	Adjustable Sunshades
D03	0.042	Indirect Lighting
D04	0.016	Soft Lighting Lamps
E01	0.019	Naturalistic shapes and forms Furnishing
E02	**0.01**	Naturalistic shapes and forms Finishing
E03	0.055	Naturalistic shapes and forms Decorations
F01	0.034	Nostalgic Decorations
F02	0.047	Nostalgic Craftsmanship
F03	0.016	Recycled Materials
F04	**0.007**	Weathered Materials
G01	0.186	Natural geometries Furnitures
G02	0.032	Natural geometries Finishing
G03	0.081	Natural geometries Furnishing



The current indicators include 29 BDIA scoring indicators. In order to simplify the model and facilitate the application and understanding of the model, BDIAs with an explanation strength less than 0.02 and that do not show significant correlation in the correlation analysis are deleted from the model to simplify the evaluation process. , these indicators include 13 items: A02 (Nature Image Furnishing), B02 (Natural Material Window Frame), B05 (Natural materials Furnitures), C02 (Natural colors Window Frame), C04 (Natural colors Floor), C05 (Natural colors Furnitures) , C06 (Natural colors Decorations), D02 (Adjustable Sunshades), D04 (Soft Lighting Lamps), E01 (Naturalistic shapes and forms Furnishing), and E02 (Naturalistic shapes and forms Walls), F03 (Recycled Materials), F04 (Weathered Materials). The simplified evaluation form will only include the 16 key indicators most important for predictive scoring, simplifying the evaluation process. Machine learning is performed again on the simplified system, and the parameters are set as in the previous step.

Table 18. Simplified BDIA prediction indicator

Attributes	
A01	Nature Image Finishing
A02	Nature Image Furnishing
A03	Nature Image Decorations
B01	Natural materials Walls
B02	Natural materials Window Frame
B03	Natural materials Ceiling
B04	Natural materials Floor
B05	Natural materials Furnitures
B06	Natural materials Decorations
C01	Natural colors Walls
C02	Natural colors Window Frame
C03	Natural colors Ceiling
C04	Natural colors Floor
C05	Natural colors Furnitures
C06	Natural colors Decorations
D01	Adjustable Windows
D02	Adjustable Sunshades
D03	Indirect Lighting
D04	Soft Lighting Lamps
E01	Naturalistic shapes and forms Furnishing
E02	Naturalistic shapes and forms Finishing
E03	Naturalistic shapes and forms Decorations
F01	Nostalgic Decorations
F02	Nostalgic Craftsmanship
F03	Recycled Materials
F04	Weathered Materials
G01	Natural geometries Furnitures
G02	Natural geometries Finishing
G03	Natural geometries Furnishing

Selected Attributes
No Significant Correlation
Weight<0.2

Machine learning was performed on the simplified 16-item BDIA to establish a prediction model for the overall rating of Booking.com by BDIA. The calculation results show that although the interpretation power is smaller than the multi-parameter model, the model training and testing results are matched, and the interpretation rate is high, which can still meet the prediction needs. The simplified calculation results also have better uniformity, and the weight difference between each indicator is smaller.

Table 19. Simplified Machine Learning Parameter Model evaluation result

Index	Training	Testing
R squared value	0.871	-1.116
Mean absolute error MAE	0.118	0.431
Mean square error MSE	0.023	0.296

root mean square error RMSE	0.152	0.544
Median absolute error MAD	0.083	0.411
Mean Absolute Percent Error MAPE	0.01	0.004
Explained variance score EVS	0.871	0.756
Root mean square log error MSLE	0	0.004

Table 20. Simplified weight analysis Result

Attributes		
A01	Nature Image Finishing	0.056
A03	Nature Image Decorations	0.03
B01	Natural materials Walls	0.05
B03	Natural materials Ceiling	0.06
B04	Natural materials Floor	0.054
B06	Natural materials Decorations	0.103
C01	Natural colors Walls	0.045
C03	Natural colors Ceiling	0.052
D01	Adjustable Windows	0.05
D03	Indirect Lighting	0.061
E03	Naturalistic shapes and forms Decorations	0.07
F01	Nostalgic Decorations	0.03
F02	Nostalgic Craftsmanship	0.048
G01	Natural geometries Furnitures	0.18
G02	Natural geometries Finishing	0.068
G03	Natural geometries Furnishing	0.042

Using the same method, machine learning is used to assign weights to the three scenes (guest rooms, lobby, lounge/lounge) in the same hotel. The result is:

Overall = 0.368*Guestroom+0.285*Loundge+0.347*Lobby

This simplified model is used as a prediction model, and the BDIA-Index is proposed as the weighted sum of 16 BDIAs. Its calculation formula is:

BDIA-Scenario = SUM(BDIA*Corresponding weight)

BDIA-Index = 0.368*Guestroom+0.285*Loundge+0.347*Lobby

A correlation analysis was conducted between the calculated BDIA-Index and the hotel's online point rating. It was found that although only the overall rating was used in the calculation of BDIA-Index, the BDIA-Index was related to the hotel's various space-related subdivision ratings (facilities, comfort) are also strongly correlated, and have a moderate correlation with price factors (cost-effectiveness, average price), and have no correlation with control ratings (such as location, service, cleanliness, WIFI). This is consistent with speculation, indicating that BDIA-Index can better reflect and predict customers' quality perceptions of hotel facilities.

Table 21. Simplified correlation analysis Result

	Overall Score	Facilities	Comfort	Value of Money	Average Price	Services	Cleaness	Location	WiFi
BDIA-Index	**0.698****	**0.717****	**0.758****	**0.422****	**0.250***	0.346	0.228	0.483	0.141

* $p<0.05$ ** $p<0.01$

In order to further explore the quantitative correlation between the BDIA-Index and the overall Booking.com rating, a linear regression analysis was performed on the two. Using BDIA-Index as the independent variable and Overall Score as the dependent variable, linear regression analysis is performed. As can be seen from the table above, the model formula is:

Overall Score=7.292 + 1.854*BDIA-Index

and the model R-squared value is 0.488, which means The BDIA-Index can explain 48.8% of the changes in Overall Score. When the F test was performed on the model, it was found that the model passed the F test (F=72.309, p=0.000<0.05), which means that BDIA-Index will definitely have an impact on the Overall Score. The final detailed analysis shows the regression coefficient value of BDIA-Index is 1.854 (t=8.503, p=0.000<0.01), which means that BDIA-Index will have a significant positive impact on Overall Score. The calculation parameters are as follows:

Table 22. Linear regression analysis parameters

Parameter Estimates (n=156)							
	Unstandardized Coefficients		Standardized Coefficients	t	p	Collinearity diagnosis	
	B	Std. Error	Beta			VIF	Tolerance
Constant	7.292	0.136	-	53.757	0.000**	-	-
BDIA-Index	1.854	0.218	0.698	8.503	0.000**	1	1
R 2	0.488						
Adj R 2	0.481						
F	F (1,76)=72.309,p=0.000						
D-W Value	0.623						
Dependent Variable: Overall Score							
* p<0.05 ** p<0.01							

In this section, based on the BDIA scores given by the judges in 4.1, through correlation analysis and the random forest algorithm in machine learning, the 29 BDIAs were reduced to 16 after optimization, simplifying the amount of data required for the prediction model. After that, through linear regression analysis, a prediction

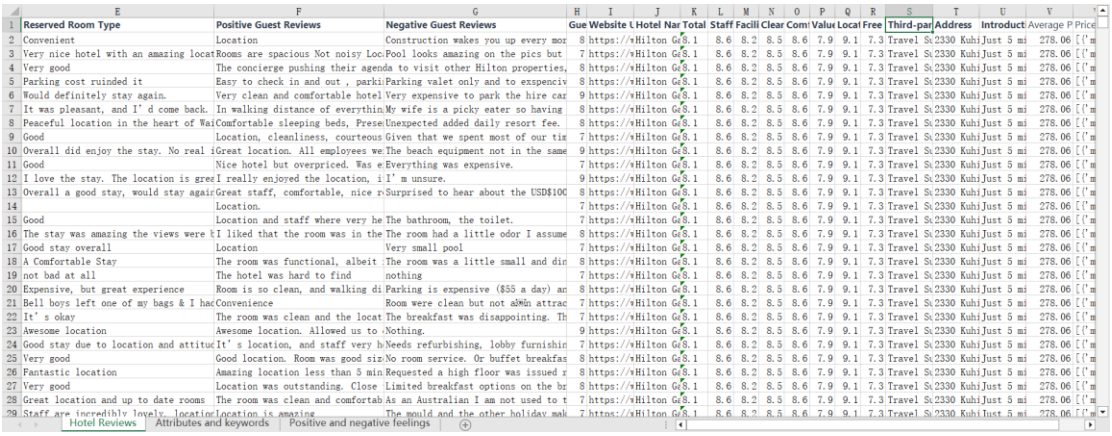
model of BDIA score for various Booking.com scores were established, and the calculation method of BDIA-Index was proposed to quantitatively predict the level of Booking.com's total score. The model's impact on the change of the total score was the explanation power is as high as 48.8%. After determining the quantitative relationship, in the next chapter, the relationship between BDIA and customer perception will be further analyzed through qualitative (NLP) methods.

4.3 Comments Emotional Tendency

Although in 4.2, a quantitative connection between BDIA and customer ratings was established through machine learning, customers' true emotional tendencies and experiences with hotel BDIA still require further qualitative analysis. In order to ensure that the obtained reviews come from real hotel experiences, this study did not use the virtual scene modeling method commonly used in previous studies. Instead, it collected all four-star hotels in Waikiki on Booking.com through crawler big data technology. There are more than 35,000 public customer reviews on the official website, and natural language processing technology (NLP) is used to quantify customers' emotional tendencies and evaluations of BDIA. This process involves four steps of data collection, classification, processing and analysis.

The crawler technology uses python code (see appendix) to initially crawl the data on booking.com and summarize and return each piece of information in the format of a data string. Each data string contains the following two categories of information:

(1) **Basic information:** hotel name, location, star rating, various ratings, average price per night for a standard room, number of reviews.



Reserved Room Type	Positive Guest Reviews	Negative Guest Reviews
Convenient	Location	Construction wakes you up every mor
Very nice hotel with an amazing locat	Rooms are spacious Not noisy Loc	Pool looks amazing on the pics but
Very good	The concierge pushing their agenda	to visit other Hilton properties,
Parking cost ruinded it	Easy to check in and out , parki	Parking valet only and to exspenciv
Would definitely stay again.	Very clean and comfortable hotel	Very expensive to park the hire car
It was pleasant, and I' d come back.	In walking distance of everythin	My wife is a picky eater so having
Peaceful location in the heart of Wai	Comfortable sleeping beds, Prese	Unexpected added daily resort fee.
Good	Location, cleanliness, courteous	Given that we spent most of our tir
Overall did enjoy the stay. No real i	Great location. All employees we	The beach equipment not in the same
Good	Nice hotel but overpriced. Was e	Everything was expensive.
I love the stay. The location is grea	I really enjoyed the location, i	I' m unsure.
Overall a good stay, would stay agair	Great staff, comfortable, nice r	Surprised to hear about the USD$100
	Location.	
Good	Location and staff where very he	The bathroom, the toilet.
The stay was amazing the views were b	I liked that the room was in the	The room had a little odor I assume
Good stay overall	Location	Very small pool
A Comfortable Stay	The room was functional, albeit	The room was a little small and din
not bad at all	The hotel was hard to find	nothing
Expensive, but great experience	Room is so clean, and walking di	Parking is expensive ($55 a day) an
Bell boys left one of my bags & I hac	Convenience	Room were clean but not a□□n attrac
It' s okay	The room was clean and the locat	The breakfast was disappointing. Th
Awesome location	Awesome location. Allowed us to v	Nothing.
Good stay due to location and attituc	It' s location, and staff very h	Needs refurbishing, lobby furnishin
Very good	Good location. Room was good siz	No room service. Or buffet breakfas
Fantastic location	Amazing location less than 5 min	Requested a high floor was issued r
Very good	Location was outstanding. Close	Limited breakfast options on the br
Great location and up to date rooms	The room was clean and comfortab	As an Australian I am not used to t
Staff are incredibly lovely location	Location is amazing	The mould and the other holiday mak

Hotel customer reviews	Attributes and keywords	Positive and negative feelings

Figure 14 Crawler data string indication Source: Author

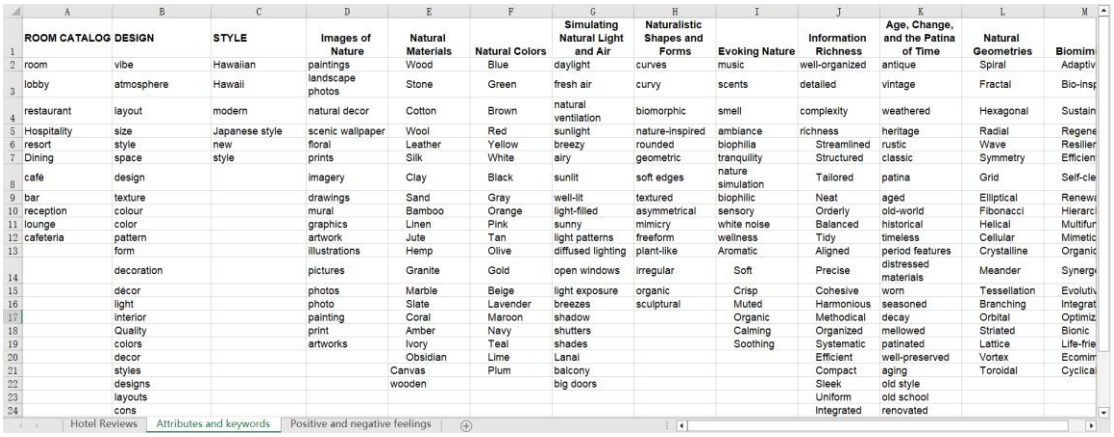
(2) Hotel customer reviews: open information including overall evaluation, positive experience, and negative experience. Private information such as customers' personal information, length of stay, and booked room type will not be collected.

In order to make full use of the initially collected comments, additional analysis tools were added for natural language disassembly and analysis: this study introduced two language packages to perform sentiment analysis and BDIA correlation analysis respectively.

(1) **Emotional Tendency Analysis**: uses the NLTK (Natural Language Toolkit) library under the python platform. NLTK is a powerful Python library for human language data processing, which provides some simple tools for sentiment analysis.

(2) **BDIA correlation analysis:** is based on the author's self-developed BDIA keyword activation tool "Attributes and Keywords", which is linked to ChatGPT and can subdivide various associations in the hotel based on the design elements that need to be analyzed, such as functional rooms, decoration, style, etc. Dozens of design keywords such as functional rooms, decoration, style, and biophilic design elements are extracted from customer reviews. For example, in order to study the impact of "Natural Colors" on customer perception, more than 20 natural color keywords were placed into relevant associations and captured for subsequent emotional tendency

analysis.



ROOM CATALOG	DESIGN	STYLE	Images of Nature	Natural Materials	Natural Colors	Simulating Natural Light and Air	Naturalistic Shapes and Forms	Evoking Nature	Information Richness	Age, Change, and the Patina of Time	Natural Geometries	Biomim
room	vibe	Hawaiian	paintings	Wood	Blue	daylight	curves	music	well-organized	antique	Spiral	Adaptiv
lobby	atmosphere	Hawaii	landscape photos	Stone	Green	fresh air	curvy	scents	detailed	vintage	Fractal	Bio-insp
restaurant	layout	modern	natural decor	Cotton	Brown	natural ventilation	biomorphic	smell	complexity	weathered	Hexagonal	Sustain
Hospitality	size	Japanese style	scenic wallpaper	Wool	Red	sunlight	nature-inspired	ambiance	richness	heritage	Radial	Regene
resort	style	new	floral	Leather	Yellow	breezy	rounded	biophilia	Streamlined	rustic	Wave	Resilier
Dining	space	style	prints	Silk	White	airy	geometric	tranquility	Structured	classic	Symmetry	Efficien
café	design		imagery	Clay	Black	sunlit	soft edges	nature simulation	Tailored	patina	Grid	Self-cle
bar	texture		drawings	Sand	Gray	well-lit	textured	biophilic	Neat	aged	Elliptical	Renew
reception	colour		mural	Bamboo	Orange	light-filled	asymmetrical	sensory	Orderly	old-world	Fibonacci	Hierarc
lounge	color		graphics	Linen	Pink	sunny	mimicry	white noise	Balanced	historical	Helical	Multifur
cafeteria	pattern		artwork	Jute	Tan	light patterns	freeform	wellness	Tidy	timeless	Cellular	Mimetic
	form		illustrations	Hemp	Olive	diffused lighting	plant-like	Aromatic	Aligned	period features	Crystalline	Organic
	decoration		pictures	Granite	Gold	open windows	irregular	Soft	Precise	distressed materials	Meander	Synerg
	décor		photos	Marble	Beige	light exposure	organic	Crisp	Cohesive	worn	Tessellation	Evoluti
	light		photo	Slate	Lavender	breezes	sculptural	Muted	Harmonious	seasoned	Branching	Integrat
	interior		painting	Coral	Maroon	shadow		Organic	Methodical	decay	Orbital	Optimiz
	Quality		print	Amber	Navy	shutters		Calming	Organized	mellowed	Striated	Bionic
	colors		artworks	Ivory	Teal	shades		Soothing	Systematic	patinated	Lattice	Life-frie
	decor			Obsidian	Lime	Lanai			Efficient	well-preserved	Vortex	Ecomim
	styles			Canvas	Plum	balcony			Compact	aging	Toroidal	Cyclical
	designs			wooden		big doors			Sleek	old style		
	layouts								Uniform	old school		
	cons								Integrated	renovated		

Hotel Reviews | Attributes and keywords | Positive and negative feelings

Figure 15 Part of the Attributes and Keywords Source: Author

After improving the analysis tools, we began to classify and summarize the hotel's original data according to the hotel name. Each time a hotel is processed, a separate hotel is analyzed and the results are output to a different subtable named after the hotel. Use the NLTK library and the "Attributes and Keywords" library to make the following judgments on each comment:

If a specific keyword is mentioned in a positive review, record it in the "Number of Positive Feedback" column and count the occurrence of the emotional keyword and its frequency (count). Similarly, if a certain element is mentioned in a negative review, it is recorded in the "Number of Negative Feedback" column and marked with the corresponding emotional keyword. If an element is mentioned in the overall review, it is categorized and recorded based on the positive or negative tendency of the sentiment keywords.

The Result will be outputted in the following format:

(1) For each hotel, a sheet would be created in the form of an Excel table, with the first row including Attributes and Keywords, Positive Feedback Numbers, Positive Characters, Negative Characters, and Negative Feelings.

(2) Each row records the statistical data of a specific element. Rows where all data for an element are zeros or blank should be deleted.

(3) Summarize the results of specific elements under each general category,

presented as a union, and highlight them in bold.

For example, here is the output for OUTRIGGER Reef Waikiki Beach Resort, one of the 26 hotels. It can be seen that corresponding to each keyword, the number of occurrences of positive and negative comments and related emotional words are summarized for subsequent analysis.

Table 23. Output example: OUTRIGGER Reef Waikiki Beach Resort

Attributes and Keywords	Positive Feedback Number	Positive Character	Negative Feedback Number	Negative Character
room	43	Comfy(2), nice(11), wonderful(2), Comfortable(5), Amazing (1), local(3), Great(11), Modern(1), Open(2), Quality(2), Superb(1), lovely(2), Spacious(4), Fantastic(2), beautiful(5), renovated(3), Perfect(1), super(1), Excited(1), Warm(1), stunning(1), private(1), updated(1), Welcoming(1)	29	Inconvenient(1), Bland(1), Noise(6), Outdated(2), Cheap(1), little(2), Mad(1), Noisy(1), hot(1)
vibe	1	renovated(1), welcome(1), beautiful(1), Modern(1), Comfortable(1)	0	
Hawaiian	5	beautiful(2), local(1), Great(1), Classic(1), touching(1)	3	hot(1)
Blue	1	lovely(1), Amazing (1), Happy(1), Great(1), Perfect(1), Welcoming(1)	2	Noise(1)
music	8	Modern(1), Spacious(2), Open(1), local(3), Great(3), lovely(1), Quality(1), nice(2), beautiful(1), Comfortable(1)	3	hot(1)
lobby	1	super(1)	11	little(2), Noisy(1), Noise(2), Mad(1)
atmosphere	2	Great(2), Fantastic(1), Relaxed(1)	0	
Hawaii	5	Superb(1), Amazing (2), Happy(2), Great(2), Open(1), lovely(1), Perfect(1), Welcoming(1)	1	
lounge	11	private(2), nice(4), wonderful(1), Comfortable(2), Great(3), Modern(1), Open(2), Quality(1), Fantastic(1), Fabulous(1), Bright(1), beautiful(1), Perfect(1)	14	Bland(1), Disappointed(1), Unpleasant(1), little(1), Mad(1)
layout	1	nice(1), Great(1), Modern(1), Open(1), Quality(1)	0	
modern	8	nice(2), Great(3), Modern(7), Open(1), Quality(1), Spacious(1), Amazing (1), Comfortable (1), Comfortable(3), renovated(1), welcome(1), beautiful(1)	0	
Hospitality	1	Pleased(1)	0	
size	4	Great(2), Amazing (1), Comfortable (1), Modern(1), Comfortable(1)	1	
ambiance	1		0	

resort	16	Great(5), wonderful(4), stunning(1), beautiful(1), nice(4), lovely(1), local(1), Spacious(1), Quality(1), private(1)	6	
new	5	local(1), Great(2), Perfect(1), renovated(1), Excited(1), nice(1)	1	
Dining	0		1	Disappointed(1), Unpleasant(1)
space	1	nice(1), Great(1), Modern(1), Open(1), Quality(1)	3	little(1)
airy	1	nice(1), Great(1), Modern(1), Open(1), Quality(1)	0	
classic	2	Classic(2)	0	
Efficient	4	wonderful(2), Great(2)	0	
design	1	nice(1), Great(1)	0	
bar	5	Great(3), Spacious(1), Welcoming(1), Open(1), nice(1), beautiful(1), Comfortable(1), Perfect(1)	3	
Sand	1	super(1), Great(1)	1	
reception	1	Great(1), Spacious(1), Welcoming(1)	0	
hotel	49	Great(17), nice(12), wonderful(4), Comfortable(4), Amazing (2), beautiful(8), Luxurious(1), local(2), Modern(4), Open(2), Quality(1), Perfect(4), lovely(3), Fabulous(1), Vibrant(1), renovated(3), super(2), Welcoming(3), Spacious(2), Classic(1), welcome(1), Pleasant(1), touching(1), Fantastic(1)	33	Bland(1), little(2), Noisy(1), hot(2), Outdated(2), Cheap(2), Mad(1), Noise(3)
lounge	5	nice(1), Great(2), Modern(1), Open(1), Quality(1), Superb(1), beautiful(1), private(1)	2	
Tidy	1	Comfortable (1), Great(1), Spacious(1), Comfortable(1), Tidy(1)	0	
photos	0		2	Noise(1)
worn	0		1	little(1)
light	1	nice(1), Great(1), Modern(1), Open(1), Quality(1)	1	hot(1), little(1), Noise(1)
Quality	2	nice(1), Great(1), Modern(1), Open(1), Quality(2), lovely(1), local(1), Spacious(1)	1	Outdated(1), Cheap(1)
shutters	1	nice(1), Comfortable(1)	0	
Calming	1	local(1), Great(1)	0	
decor	1	nice(1), Great(1), Modern(1), Open(1), Quality(1)	0	
balcony	7	Amazing (1), Comfortable (1), Great(2), Modern(1), Comfortable(1), Open(1), local(2), lovely(1), Spacious(2), Quality(1), Perfect(1)	0	
renovated	5	renovated(5), welcome(1), beautiful(1), Modern(1), Comfortable(1), wonderful(1), Great(1), Excited(1), Spacious(1)	0	
furniture	0		2	Outdated(1), Cheap(1), Noise(1)

At this point, after preliminary processing, the comment data can be used for subsequent analysis.

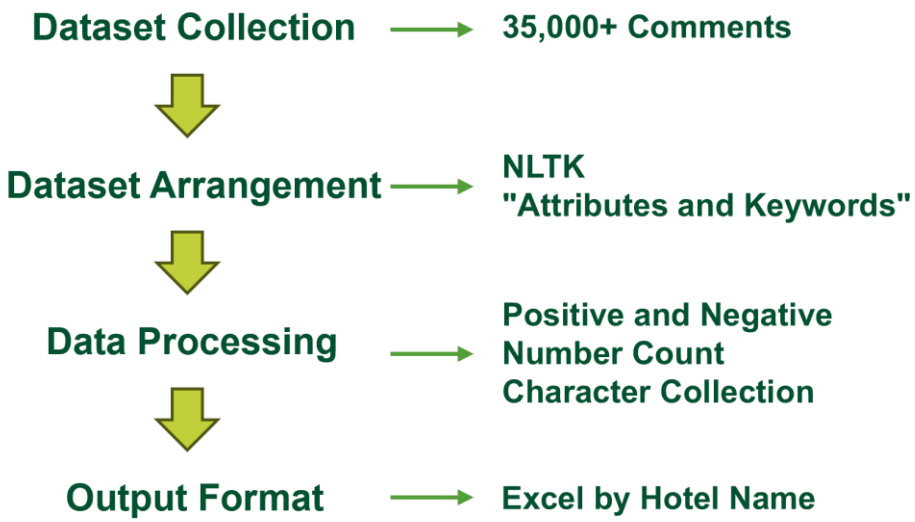


Figure 15 Schematic diagram of data preprocessing process Source: Author

4.3.1 Comments Emotional Tendency by Scenarios

Classify the NLP emotional tendency analysis results according to the hotel scene, and study the characteristics of the designs with the highest and lowest praise rates in different space scenes.

4.3.3.1 Guestroom

Table 27. Room space review emotional tendency ranking list Source: Author

	Hotel Name	Comments	PFN	Positive Character	NFN	Negative Character	Positive Feedback rate	Positive Feedback rate	Feedback Overall rate	Booking	BDIA-Index	BDIA-Scenarios
17	Hilton Grand Vacations Club Grand V	153	20	nice(6), Welcoming(2), Amazing (1), Gr	7	Uncomfortable(1), Crowded(1)	13.07%	4.58%	**8.50%**	8.8	0.71999	0.722
5	Prince Waikiki	705	81	lovely(5), Great(25), Comfortable(9), Fr	24	hot(2), Uncomfortable(1), Mediocre(1),	11.49%	3.40%	**8.09%**	8.8	0.62812	0.673
4	Hyatt Centric Waikiki Beach	321	45	super(3), Spacious(13), beautiful(3), Pe	21	Dated(1), Uncomfortable(1), hot(1), Noi	14.02%	6.54%	**7.48%**	8.6	0.44014	0.324
14	Aston Waikiki Sunset	1380	155	Amazing (7), Great(43), stunning(1), Sp	83	little(7), Cramped(2), Dated(7), Dingy(3	11.23%	6.01%	**5.22%**	7.8	0.30906	0.525
22	Halepuna Waikiki by Halekulani	1519	172	beautiful(21), wonderful(7), Comfortable	96	Noise(6), Noisy(4), Heavy(2), hot(1), litt	11.32%	6.32%	**5.00%**	9	0.8798	0.792
19	Hilton Grand Vacation Club The Gran	106	11	Perfect(1), Great(4), beautiful(3), Spaci	6	Lost(1)	10.38%	5.66%	**4.72%**	8.6	0.83066	0.668
11	The Equus	673	74	nice(12), super(6), Great(15), Fantastic	48	little(3), Noisy(10), Heavy(2), Outdated(	11.00%	7.13%	**3.86%**	7.8	0.39567	0.294
18	Embassy Suites by Hilton Waikiki Be	679	70	beautiful(5), Comfortable (5), Comfortab	45	Noisy(4), Uncomfortable(2), hot(1), Nois	10.31%	6.63%	**3.68%**	8.7	0.51514	0.601
15	Waikiki Beach Marriott Resort & Spa	506	55	Vibrant(1), Great(16), Welcoming(2), su	38	Overlooked(1), Crowded(1), little(2), No	10.87%	7.51%	**3.36%**	8.5	0.47382	0.273
24	OUTRIGGER Reef Waikiki Beach Re	441	43	Comfy(2), nice(11), wonderful(2), Comf	29	Inconvenient(1), Bland(1), Noise(6), Ou	9.75%	6.58%	**3.17%**	8.7	0.6501	0.432
23	The Laylow, Autograph Collection	221	26	Amazing (2), wonderful(2), Great(9), Pe	19	Overlooked(1), Cheap(1), Grim(1), little	11.76%	8.60%	**3.17%**	8.4	0.61282	0.347
21	OUTRIGGER Waikiki Beachcomber	913	94	nice(15), Pleasant(3), Great(31), lovely(	72	little(10), Noisy(12), hot(1), Crowded(2),	10.30%	7.89%	**2.41%**	8.7	0.62785	0.635
3	Sheraton Waikiki Beach Resort	2063	177	nice(21), lovely(6), private(3), Amazing	144	Crowded(5), Noise(3), little(21), Dated(4	8.58%	6.98%	**1.60%**	8.6	0.63036	0.693
1	Hilton Garden Inn Waikiki Beach	2638	242	nice(47), Great(62), Comfortable(37), C	208	Dark(4), little(16), Dingy(2), Dated(6), N	9.17%	7.88%	**1.29%**	8.1	0.46931	0.287
7	The Surfjack Hotel & Swim Club	695	66	Comfortable (8), Great(22), Comfortable	57	Tired(3), little(5), Noise(3), Disorganized	9.50%	8.20%	**1.29%**	8.5	0.65452	0.867
20	OUTRIGGER Waikiki Beach Resort	514	45	super(2), Great(18), Comfortable(9), Pe	41	Noise(4), little(3), Disappointed(1), Date	8.75%	7.98%	**0.78%**	8.7	0.83566	0.639
2	Wayfinder Waikiki	531	59	nice(17), lovely(10), Amazing (6), super	60	Noise(8), Noisy(5), little(6), Rushed(1),	11.11%	11.30%	**-0.19%**	8.5	0.73259	0.68
13	Hyatt Regency Waikiki Beach Resort	1178	97	Comfortable(15), beautiful(6), Fresh(3),	102	Uncomfortable(2), little(5), Disappointed	8.23%	8.66%	**-0.42%**	8.3	0.71007	0.581
9	Hilton Waikiki Beach	679	64	lovely(1), Hospitable(1), nice(13), Amaz	67	Outdated(2), Tired(3), Stale(1), little(8),	9.43%	9.87%	**-0.44%**	8.1	0.43436	0.218
8	Queen Kapiolani Hotel	2941	359	Great(133), beautiful(36), nice(78), Com	378	little(32), Disappointed(10), Ashamed(1	12.21%	12.85%	**-0.65%**	8.4	0.70725	0.542
10	Moana Surfrider, A Westin Resort &	862	78	nice(8), incredible(2), Great(22), Tidy(1)	93	Noisy(10), little(7), Mediocre(2), Dated(5	9.05%	10.79%	**-1.74%**	8.5	0.58194	0.52
16	Hotel La Croix	913	114	nice(22), Amazing (8), Great(32), Comf	135	Humiliated(1), Uncomfortable(7), Rushe	12.49%	14.79%	**-2.30%**	7.5	0.39192	0.303
12	Shoreline Hotel Waikiki	772	50	nice(14), super(6), Perfect(3), Great(9),	76	Noise(5), Uncomfortable(2), little(10), N	6.48%	9.84%	**-3.37%**	8.1	0.34339	0.182
6	Hilton Hawaiian Village Waikiki Beac	2904	254	Amazing (12), beautiful(41), Great(58),	368	little(20), Painful(1), Unpleasant(2), Hea	8.75%	12.67%	**-3.93%**	7.8	0.52465	0.376

Using guest rooms as keywords to extract customer reviews, and sorting

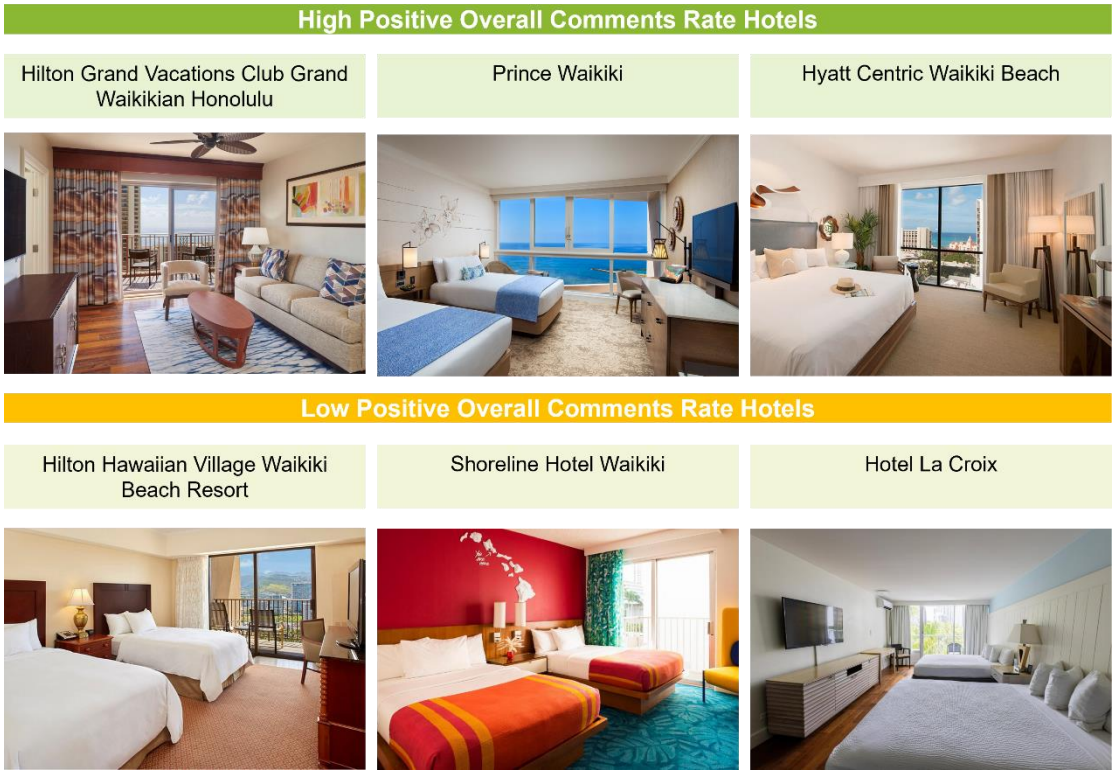


Figure 16 Hotel rooms with the most positive and negative emotional experiences

Source: https://www.booking.com/searchresults.es.html?ss=Waikiki

participating hotels according to the overall positive rate in guest room reviews, it can be found that the top three hotels and their most frequently occurring adjectives are Hilton Grand Vacations Club Grand Waikikian Honolulu (Nice, 25), Prince Waikiki (Great, 25), Hyatt Centric Waikiki Beach (Spacious, 13). Among them, the Hilton Grand Vacations Club Grand Waikikian Honolulu adopts a warm family-style suite layout. The furniture is made of original mahogany color, with wave patterns and natural plant hangings. The lighting uses lamps with a soft lighting effect. Prince Waikiki uses a combination of wooden furniture and off-white furniture and decorations, with large landscape windows and Hawaiian-style wall decorations and wallpapers. Hyatt Centric Waikiki Beach uses simple and soft earth tones with soft-effect lamps and Hawaiian-style decorations. In addition to the simple business style, it adds a natural atmosphere through natural vegetation accessories. The common features of the three hotel rooms are that the decoration style is more casual and business-like, with soft colors and extensive use of wooden furniture. At the same time, the room has a wide view and the density of furniture is appropriate.

The top three hotels include Hilton Hawaiian Village Waikiki Beach Resort (Overcrowded, 4), Shoreline Hotel Waikiki (Noisy, 14), and Hotel La Croix (Uncomfortable, 7). The decoration style of Hilton Hawaiian Village Waikiki Beach Resort is relatively retro, with natural colors, but it lacks Hawaiian patterns or decorations, and the facilities are relatively old. The decoration colors of Shoreline Hotel Waikiki are relatively bright. While receiving positive reviews from some guests, some guests felt that the colors were too bright and caused discomfort. At the same time, the hotel's windows have poor sealing performance, causing noise at night. The rooms at Hotel La Croix are cramped, and the furniture colors are inconsistent. Guests believe the furniture is of poor quality, smelly and uncomfortable. The common feature of the three hotel rooms is that the quality of furnishing or finishing used is poor or outdated, resulting in a deterioration of the customer's experience.

Attributes	Positive Feedback Rate	Negative Feedback Rate	Attribute Feedback Rate
BDIA-Scenarios	0.088	-0.356	0.327
C04	0.484*	-0.123	0.341
C06	0.497*	-0.338	0.518**
E03	0.346	-0.469*	0.545**
G02	-0.416*	-0.349	0.068

Table 28. Correlation Analysis Result by Design Scenarios - Guestroom Source: Author

The correlation analysis between the overall favorable rating of the guest room scene and BDIA-Scenarios did not show a significant correlation. This may indicate that whether the guest room scene received favorable reviews has little correlation with the total score of BDIA-Scenarios. However, a correlation analysis between the 29 items of BDIA and the overall praise rate of the guest room found that the customer praise rate was significantly positively correlated with C04 (Natural colors Floor), C06 (Natural colors Decorations), and E03 (Naturalistic shapes and forms Decorations). Relevance proves that having these elements can improve customer praise rate; G02 (Natural geometries Finishing) will cause a significant reduction in positive review rate. This is consistent with the previous analysis. Using simple and comfortable hard furnishings and avoiding overly complex geometric hard furnishings or large-area jumping colors in the room can effectively improve customers'

perception of the quality of the guest rooms.

4.3.3.2 Lobby

Table 29. Lobby space review emotional tendency ranking list Source: Author

	Hotel Name	Comments	PFN	Positive Character	NFN	Negative Character	Positive Feedback rate	Positive Feedback rate	Feedback Overall rate	Booking	BDIA-Scenarios
12	Shoreline Hotel Waikiki	772	36	lovely(1), Great(9), Quaint(1), nice(5), C	7	Nauseated(1), Mad(1)	4.66%	0.91%	**3.76%**	8.1	0.481
8	Queen Kapiolani Hotel	2941	121	nice(24), Great(44), Fun(8), Comfortabl	33	little(3), Disappointed(1), Ashamed(1), D	4.11%	1.12%	**2.99%**	8.4	0.81
7	The Surfjack Hotel & Swim Club	695	16	Great(9), Open(2), nice(3), Sparkling(1),	3	Disorganized(1), Disappointed(1)	2.30%	0.43%	**1.87%**	8.5	0.498
5	Prince Waikiki	705	14	Great(7), incredible(1), Fresh(2), Tidy(1)	3		1.99%	0.43%	**1.56%**	8.8	0.651
15	Waikiki Beach Marriott Resort & Spa	506	10	Welcoming(2), super(1), wonderful(2), F	3		1.98%	0.59%	**1.38%**	8.5	0.665
2	Wayfinder Waikiki	531	9	lovely(2), Fun(1), Great(3), Open(3), nic	3	little(2), Noise(2), Restless(1), Uncomfo	1.69%	0.56%	**1.13%**	8.5	0.772
19	Hilton Grand Vacation Club The Grand Islander	106	2	wonderful(2)	1		1.89%	0.94%	**0.94%**	8.6	0.846
1	Hilton Garden Inn Waikiki Beach	2638	33	Comfortable (3), Great(11), Comfortable	12	Uncomfortable(2), Dated(1), Noise(3), N	1.25%	0.45%	**0.80%**	8.1	0.756
9	Hilton Waikiki Beach	679	6	lovely(1), Spacious(1), beautiful(2), Ope	1	hot(1)	0.88%	0.15%	**0.74%**	8.1	0.772
23	The Laylow, Autograph Collection	221	2	Welcoming(1)	1		0.90%	0.45%	**0.45%**	8.4	0.761
21	OUTRIGGER Waikiki Beachcomber Hotel	913	8	nice(3), Amazing (1), beautiful(1), wond	4	Cheap(1), Frustrated(1), Uncomfortable	0.88%	0.44%	**0.44%**	8.7	0.873
10	Moana Surfrider, A Westin Resort & Spa, Waik	862	12	beautiful(5), Great(7), Upscale(1), Fun(1	9	Mediocre(1), Noise(2), Empty(1), Lost(1	1.39%	1.04%	**0.35%**	8.5	0.644
4	Hyatt Centric Waikiki Beach	321	4	beautiful(1), Great(1), Modern(1), nice(1	3		1.25%	0.93%	**0.31%**	8.6	0.451
18	Embassy Suites by Hilton Waikiki Beach Walk	679	2	Amazing (1), Happy(1), Premium(1), Pe	1		0.29%	0.15%	**0.15%**	8.7	0.544
13	Hyatt Regency Waikiki Beach Resort & Spa	1178	4	lovely(1), Amazing (1), beautiful(1), Fant	3	Rundown(1)	0.34%	0.25%	**0.08%**	8.3	0.795
11	The Equus	673	7	Fantastic(1), Comfortable(3), beautiful(1	7	Heavy(2), Dark(1)	1.04%	1.04%	**0.00%**	7.8	0.492
22	Halepuna Waikiki by Halekulani	1519	12	stunning(1), beautiful(2), wonderful(2), n	12	little(1), Unpleasant(1), Empty(1)	0.79%	0.79%	**0.00%**	9	0.941
3	Sheraton Waikiki Beach Resort	2063	3	private(1), Comfortable (1), Great(1), Cc	7	Crowded(1)	0.15%	0.34%	**-0.19%**	8.6	0.668
14	Aston Waikiki Sunset	1380	3	nice(1), Spacious(1)	8	Noisy(1), little(1), Tired(1), Uncomfortabl	0.22%	0.58%	**-0.36%**	7.8	0.305
16	Hotel La Croix	913	14	nice(4), Great(3), Amazing (1), Spaciou	18	Dated(1), little(1), Noise(2), Empty(1), In	1.53%	1.97%	**-0.44%**	7.5	0.241
6	Hilton Hawaiian Village Waikiki Beach Resort	2904	2	Great(1)	17	Heavy(2), little(1), Crowded(2), Bored(1)	0.07%	0.59%	**-0.52%**	7.8	0.63
20	OUTRIGGER Waikiki Beach Resort	514	1		4	Dated(1), Tired(1), Dark(1)	0.19%	0.78%	**-0.58%**	8.7	0.899
17	Hilton Grand Vacations Club Grand Waikikian H	153	1	Comfortable (1), Great(1), Luxurious(1),	2	Inconvenient(1), Disappointed(1), Uncon	0.65%	1.31%	**-0.65%**	8.8	0.841
24	OUTRIGGER Reef Waikiki Beach Resort	441	1	super(1)	11	little(2), Noisy(1), Noise(2), Mad(1)	0.23%	2.49%	**-2.27%**	8.7	0.821

Using lobby as the keyword to extract customer reviews, and sorting participating hotels according to the overall positive rate in room reviews, it can be found that the top three hotels and their most frequently occurring adjectives are Shoreline Hotel Waikiki (Great, 9), Queen Kapiolani Hotel (Nice, 44), The Surfjack Hotel & Swim Club (Great, 9). Among them, Shoreline Hotel Waikiki ranks lower in the guest room evaluation, but ranks higher in the lobby scene. Related evaluation adjectives include Colorful(2), Happy(1), Fun(2), Bright(1), incredible(2), renovated(2), Unique(1), Vibrant(1), Fresh(1), Exquisite(1), Heavenly(1), which proves that the use of color and Hawaiian-style decoration in the lobby can effectively enhance customer loyalty. Evaluate rates, but be cautious in the room. Queen Kapiolani Hotel uses soft wood-colored furniture and finishing to create a resting space for guests of different group sizes. It is also paired with a large number of natural vegetation, murals and decorations, giving it a strong Hawaiian style. The Surfjack Hotel & Swim Club uses waves as its theme, using native Hawaiian rattan decorations and background walls transformed from surfboards to create a relaxed and enjoyable Hawaiian resort style.

The Booking scores of the top three hotels are all above 8.6, which means that the overall evaluation of the hotel is high, and only the lobby space has a high rate of

negative reviews. Including two hotels owned by OUTRIGGER and the Hilton Grand Vacations Club Grand Waikikian Honolulu. Among them, OUTRIGGER Waikiki Beach Resort (Dated, Tired, Dark) and OUTRIGGER Reef Waikiki Beach Resort (Noisy, 3) may have improper layout problems, causing the customer resting and waiting areas to be overcrowded or empty, making customers feel uncomfortable. Hilton Grand Vacations Club Grand Waikikian Honolulu lacks seating and uses too many hard furnishings, resulting in poor acoustics. Therefore, in the design of the lobby, more Native Hawaiian style elements, rich colors, well-organized and

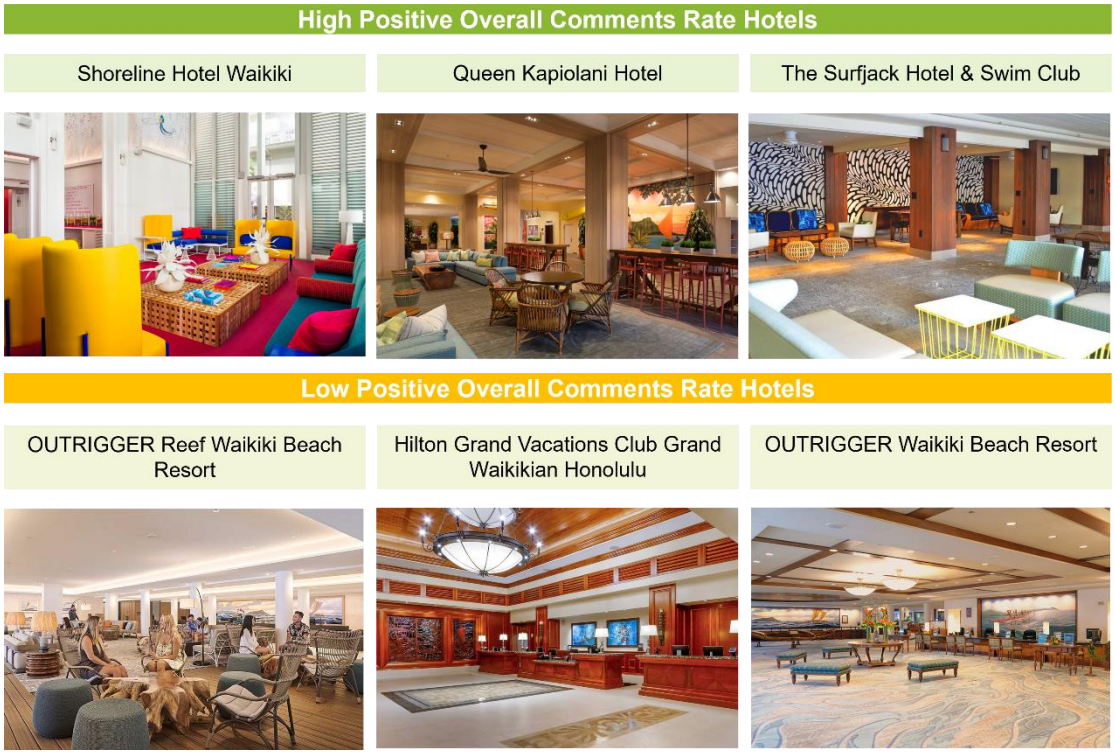


Figure 17 Hotel Lobbies with the most positive and negative emotional experiences

Source: https://www.booking.com/searchresults.es.html?ss=Waikiki

comfortable seating and waiting areas, while providing adequate lighting and soft surfaces for sound attenuation are key to improving customer perception.

Table 30. Correlation Analysis Result by Design Scenarios - Lobby Source: Author

Attributes	Positive Feedback Rate	Negative Feedback Rate	Attribute Feedback Rate
BDIA-Scenarios	-0.142	-0.104	-0.087
B01	-0.437*	-0.255	-0.298
B04	-0.368	0.133	-0.405*
C01	-0.437*	-0.255	-0.298
C05	-0.046	-0.461*	0.161
C06	-0.618**	-0.05	-0.559**
F04	0.558**	-0.103	0.570**

The correlation analysis between the overall favorable rating of the lobby scene and BDIA-Scenarios did not show a significant correlation. This may indicate that whether the guest room scene received favorable reviews has little correlation with the total score of BDIA-Scenarios. However, a correlation analysis between the 29 BDIA items and the overall positive rating of the lobby found that the positive correlation between the customer's positive rating and C05 (Natural colors Furnitures) and F04 (Weathered Materials) proves that having these elements can improve customer satisfaction. Positive review rate; B01 (Natural materials Walls), B04 (Natural colors Floor), C01 (Natural colors Walls), C06 (Natural colors Decorations) will cause a significant decrease in the positive review rate. B04 Contrary to the guest room results, Zhenxing's use of natural-colored flooring in guest rooms improves quality perceptions, while in the lobby it decreases customer experience. Adopting some Hawaiian-style, colorful decoration styles in the lobby, using wall paintings and retro styles will help improve customer reviews.

4.3.3.3 Lounge

Table 31. Lounge space review emotional tendency ranking list Source: Author

	Hotel Name	Comments	PFN	Positive Character	NFN	Negative Character	Positive Feedback rate	Negative Feedback rate	Feedback Overall rate	Booking	BDIA-Scenarios
2	Wayfinder Waikiki	531	84	nice(9), lovely(3), Amazing (2), super	23	Noisy(7), Inadequate(1),little(2), Spen	15.82%	4.33%	**11.49%**	8.5	0.772
23	The Laylow, Autograph Collection	221	35	Comfortable (2), Comfortable(2), Ama	11	Noise(6), Overwhelming(1), Congeste	15.84%	4.98%	**10.86%**	8.4	0.761
7	The Surfjack Hotel & Swim Club	695	92	local(2), Great(20), Perfect(3), nice(9	19	Tired(2), Disorganized(1), Dingy(1), D	13.24%	2.73%	**10.50%**	8.5	0.498
11	The Equus	673	55	local(1), wonderful(1), Happy(1), Com	9	Noise(1), Noisy(1), Dark(1)	8.17%	1.34%	**6.84%**	7.8	0.492
12	Shoreline Hotel Waikiki	772	52	Satisfied(1), Fantastic(1), lovely(1), G	8	Rough(2), Disgusted(1), Outdated(1),	6.74%	1.04%	**5.70%**	8.1	0.481
8	Queen Kapiolani Hotel	2941	290	nice(72), Great(108), Spacious(4), cc	153	Disappointed(2), Overcrowded(1), No	9.86%	5.20%	**4.66%**	8.4	0.81
9	Hilton Waikiki Beach	679	42	Quality(1), Great(10), nice(8), Perfect	18	Tired(1), hot(1), Cheap(1), Uncomfort	6.19%	2.65%	**3.53%**	8.1	0.772
20	OUTRIGGER Waikiki Beach Resort	514	28	beautiful(2), wonderful(2), Great(7), C	15	Overwhelming(1), Dated(1), Outdated(	5.45%	2.92%	**2.53%**	8.7	0.899
21	OUTRIGGER Waikiki Beachcomber Hotel	913	34	nice(4), Great(7), Open(1), Enjoyable	18	Noise(4), Crowded(2)	3.72%	1.97%	**1.75%**	8.7	0.873
16	Hotel La Croix	913	21	Great(1), Perfect(1), nice(1), wonderf	7		2.30%	0.77%	**1.53%**	7.5	0.241
18	Embassy Suites by Hilton Waikiki Beach Wall	679	15		8		2.21%	1.18%	**1.03%**	8.7	0.544
10	Moana Surfrider, A Westin Resort & Spa, Wa	862	48	Great(22), nice(4), wonderful(5), Perf	43	Lost(2), Empty(1), Noise(1), Noisy(1)	5.57%	4.99%	**0.58%**	8.5	0.644
1	Hilton Garden Inn Waikiki Beach	2638	46	Great(8), beautiful(1), nice(5), Comfo	34	Dated(1), Noisy(3), Uncomfortable(5),	1.74%	1.29%	**0.45%**	8.1	0.756
15	Waikiki Beach Marriott Resort & Spa	506	11	Great(3), Comfortable (1), Comfortabl	9	hot(1), Sad(1), Uncomfortable(1), Tas	2.17%	1.78%	**0.40%**	8.5	0.665
24	OUTRIGGER Reef Waikiki Beach Resort	441	21	private(2), nice(4), wonderful(1), Com	20	Bland(1), Disappointed(1), Unpleasan	4.76%	4.54%	**0.23%**	8.7	0.821
14	Aston Waikiki Sunset	1380	10	Great(2), Spacious(1), Comfortable(1	10	Unpleasant(1), Cramped(1), Dingy(1)	0.72%	0.72%	**0.00%**	7.8	0.305
19	Hilton Grand Vacation Club The Grand Islande	106	1	Comfortable(1)	1		0.94%	0.94%	**0.00%**	8.6	0.846
6	Hilton Hawaiian Village Waikiki Beach Resort	2904	64	nice(3), super(3), Great(5), Welcomir	80	Disappointed(2), little(2), Empty(1), A	2.20%	2.75%	**-0.55%**	7.8	0.63
5	Prince Waikiki	705	17	Great(4), nice(4), Amazing (1), Qualit	21	Dated(1), small(1), Noise(1)	2.41%	2.98%	**-0.57%**	8.8	0.651
13	Hyatt Regency Waikiki Beach Resort & Spa	1178	12	Amazing (2), nice(1), Great(2), Comf	19	little(1), Spent(1), Inconvenient(1), D	1.02%	1.61%	**-0.59%**	8.3	0.795
3	Sheraton Waikiki Beach Resort	2063	34	lovely(1), private(1), Amazing (2), Gre	63	little(1), Frustrated(1), Tired(1), Noise	1.65%	3.05%	**-1.41%**	8.6	0.668
17	Hilton Grand Vacations Club Grand Waikikian	153	0		5	hot(1), little(1)	0.00%	3.27%	**-3.27%**	8.8	0.841
4	Hyatt Centric Waikiki Beach	321	4	Great(2), Modern(1)	15	little(3), hot(1)	1.25%	4.67%	**-3.43%**	8.6	0.451
22	Halepuna Waikiki by Halekulani	1519	36	nice(4), Amazing (2), Modern(3), upd	104	little(2), Sterile(1), Empty(1), Noisy(2)	2.37%	6.85%	**-4.48%**	9	0.941

Using lounge as the keyword to extract customer reviews, and sorting the participating hotels according to the overall favorable rating in the reviews, it can be found that the top three hotels and their most frequently occurring adjectives are Wayfinder Waikiki (Nice, 9), The Laylow, Autograph Collection (Comfortable, 4), The Surfjack Hotel & Swim Club (Great, 20). The Booking scores of the top three hotels are all above 8.6, which means that the overall evaluation of the hotel is high, and only the lobby space has a high rate of negative reviews. Including Halepuna Waikiki by Halekulani (Bland, 3), Hyatt Centric Waikiki Beach (Little, 3), Hilton Grand Vacations Club Grand Waikikian Honolulu (hot, 1). From the comparison, it can be seen that semi-open spaces that use a variety of colors, Hawaiian-style furniture and decorations, and bamboo weaving craftsmanship are more likely to be praised by customers, while lounge spaces with a business minimalist style or lack of color are likely to receive lower evaluations.

A correlation analysis was performed between the overall positive rating of the lounge scene and BDIA-Scenarios. The results showed that BDIA-Scenarios was significantly positively correlated with the negative rating rate. That is, in the lounge scene, a higher BDIA would actually cause a decrease in the customer positive rating. A correlation analysis between the 29 BDIA items and the overall favorable rating of the lobby found that B04 (Natural colors Floor), D01 (Adjustable Windows), D04 (Soft Lighting Lamps), and E01 (Naturalistic shapes and forms Furnishing) all will cause negative effects. Significantly lower comment rate to BDIA. It should be avoided wherever possible in the design of lounges and lounges. The results show that easy-to-clean terrazzo or polished concrete floors should be used in the lounge design, while natural light sources and regular-shaped furniture should be used as

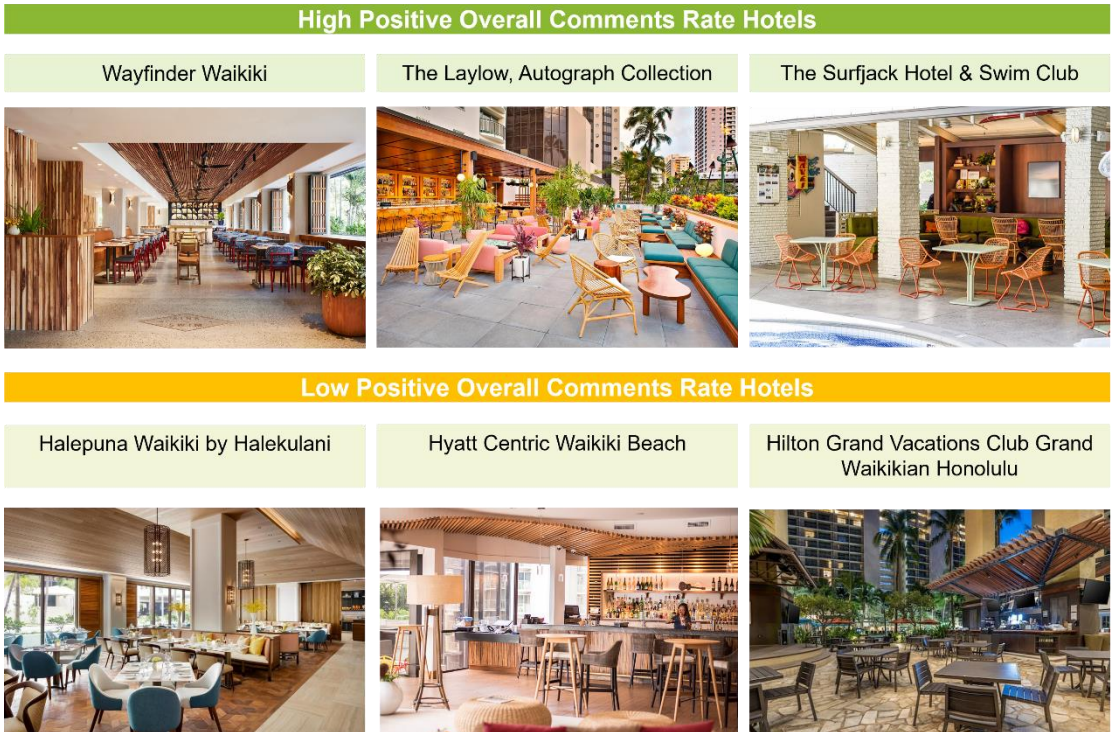


Figure 18 Hotel Loundges with the most positive and negative emotional experiences

Source: https://www.booking.com/searchresults.es.html?ss=Waikiki

much as possible to facilitate dining.

Table 32. Correlation Analysis Result by Design Scenarios - Lounge Source: Author

4.3.3.4 Overall

Lounges are similar to lobbies. Semi-open spaces that use a variety of colors, Hawaiian-style furniture and decorations, and bamboo weaving craftsmanship are

Attributes	Positive Feedback Rate	Negative Feedback Rate	Attribute Feedback Rate
BDIA-Scenarios	0.07	0.459*	-0.103
B04	-0.511*	0.137	-0.598**
D01	0.151	0.463*	-0.02
D04	-0.035	0.442*	-0.209
E01	-0.201	0.528**	-0.420*
G01	0.438*	0.426*	0.302

more likely to be praised by customers, while spaces with business simplicity or a lack of color are likely to receive lower evaluations. In the guest room results, interior spaces that adopt a simple business style with a small amount of Hawaiian style decoration are the most popular. Too many colors or Hawaiian style will not easily obtain the highest positive evaluation rate. Therefore, when designing, it can be

configured according to space type, and the design strategies used in public spaces can be different from those in guest rooms.

4.3.2 Comments Emotional Tendency by Keywords

In this analysis, evaluative keywords frequently encountered in interior design evaluation, such as "Design," "Decor," and "Vibe," are extracted from the comments. The sentiment associated with these keywords, in relation to various Building Design and Interior Architecture (BDIA) aspects, is assessed as either positive or negative. This evaluation aims to determine the impact of each BDIA element on the achievement of favorable feedback for these design keywords, discerning whether the effect is positive or negative. For example, extract the comments related to "Design" from all customer reviews, aggregate the extraction results of all hotels, and calculate the review rate through the following formula:

Positive Feedback Rate= Number of positive feedback/total Comments Number

Negative Feedback Rate= Number of Negative feedback/total Comments Number

Attribute Feedback Rate= (Number of positive feedback-Number of Negative feedback)/total Comments Number

Both positive and negative comments are included to determine whether the emotional tendency caused by BDIA is bidirectional or unidirectional (for example, if it causes an increase in positive comments but not a decrease in negative comments, it is unidirectional). After that, Pearson correlation analysis was used to determine the correlation between BDIA and the emotional tendency of the review, and the significant results were summarized.

Table 33. Output example: “Design” as Keyword

Hotel Number	Hotel Name	Comment Number	Booking score prediction	BDIA-Overall	Average Price	Positive Feedback Number	Negative Feedback Number	Attribute Feedback rate
1	Hilton Garden Inn Waikiki Beach	2638	8.20084759	0.469308	278.06	21	39	-68.23351024
2	Wayfinder Waikiki	531	8.70398713	0.732594	224.31	124	7	2203.389831
3	Sheraton Waikiki Beach Resort	2063	8.50862178	0.630362	896.37	12	11	4.847309743
4	Hyatt Centric Waikiki Beach	321	8.1450999	0.440136	355.67	8	6	62.30529595
5	Prince Waikiki	705	8.50434114	0.628122	445.71	26	6	283.6879433
6	Hilton Hawaiian Village Waikiki Beach Resort	2904	8.3065966	0.524645	536.32	41	67	-89.53168044
7	The Surfjack Hotel & Swim Club	695	8.55478963	0.654521	197	147	22	1798.561151
8	Queen Kapiolani Hotel	2941	8.65555093	0.707248	349.95	197	62	459.0275417
9	Hilton Waikiki Beach	679	8.1340696	0.434364	311.83	12	12	0
10	Moana Surfrider, A Westin Resort & Spa, Waikiki Beach	862	8.41608925	0.581941	656.42	35	21	162.412993
11	The Equus	673	8.06011582	0.395665	117.67	53	25	416.0475483
12	Shoreline Hotel Waikiki	772	7.96021447	0.343388	227	107	23	1088.082902
13	Hyatt Regency Waikiki Beach Resort & Spa	1178	8.66093804	0.710067	524.52	12	20	-67.91171477
14	Aston Waikiki Sunset	1380	7.89460793	0.309057	281.37	35	46	-79.71014493
15	Waikiki Beach Marriott Resort & Spa	506	8.20946429	0.473817	348.52	8	6	39.5256917
16	Hotel La Croix	913	8.0529553	0.391918	249	34	26	87.62322015
17	Hilton Grand Vacation Club The Grand Islander Waikiki Honolulu	153	8.67989707	0.719988	470.38	8	4	261.4379085
18	Embassy Suites by Hilton Waikiki Beach Walk	679	8.28842299	0.515135	599.03	20	9	162.0029455
19	Hilton Grand Vacations Club Grand Waikikian Honolulu	106	8.8913989	0.830664	388.5	5	2	283.0188679
20	OUTRIGGER Waikiki Beach Resort	514	8.90094053	0.835657	896.29	9	19	-194.5525292
21	OUTRIGGER Waikiki Beachcomber Hotel	913	8.50382708	0.627853	470.93	38	9	317.6341731
22	Halepuna Waikiki by Halekulani	1519	8.98529207	0.879797	425	76	16	394.9967084
23	The Laylow, Autograph Collection	221	8.47510093	0.612821	474.86	29	3	1176.470588
24	OUTRIGGER Reef Waikiki Beach Resort	441	8.54634874	0.650104	578.06	10	7	68.02721088

The analysis results show that among different design evaluation keywords, some elements are repeatedly related to negative evaluations. For example, higher B04 (Natural materials Floor) is related to lower design, atmosphere, decoration, and style evaluations. Related Adjectives include Dated(10), Outdated(6), Overcrowded(5), which represent natural material floors that may be associated with outdated styles or physical experiences. On the contrary, some BDIAs, such as A01 (Nature Image Finishing) and C01 (Natural colors Walls), will significantly increase the frequency of positive evaluations, while F (Age, change, and the patina of time) will be the same as the whole group. Related to high positive emotional experience.

Some BDIAs, such as B05 (Natural materials Furnitures), have more complicated functions. They improve some feelings (Decoration) while lowering the evaluation of others (Furniture).

The correlation between some BDIA and design evaluation is weak. For example, D01 (Adjustable Windows) has no relevance to any design keyword.

Table 34. Correlation Analysis Result by Design Keywords

Attributes		Positive Feedback Rate	Negative Feedback Rate	Attribute Feedback Rate
Design	B04	-0.388	-0.094	-0.505*
	F02	0.22	-0.292	0.435*
	F03	0.322	-0.214	0.548**
	F04	0.583**	-0.058	0.685**
Atmosphere	B03	0.261	0.371**	0.229
	B04	-0.279	0.111	-0.307*
	F01	0.343*	0.106	0.347*
	F02	0.340*	0.104	0.344*
	F03	0.422**	0.159	0.423**
	F04	0.342*	-0.047	0.365**
Decoration	B01	-0.387*	-0.232	-0.353*
	B04	-0.549**	-0.221	-0.514**
	B05	0.149	-0.379*	0.195
	B06	-0.018	-0.519**	0.048
	C01	-0.151	-0.419*	-0.096
	C04	-0.399*	-0.157	-0.373*
	F02	0.318	-0.252	0.345*
	F04	0.447**	-0.105	0.454**
	G01	0.111	-0.359*	0.155
	G03	-0.207	-0.376*	-0.156
Form	A01	0.119	-0.542**	0.488*
	A03	0.282	-0.371	0.426*
	C02	-0.11	-0.441*	0.308
	D02	0.205	-0.576**	0.553**
	E01	-0.33	0.377	-0.461*
	F03	0.562**	-0.151	0.383
	G02	0.016	-0.527**	0.433*
Furniture	B05	-0.504*	-0.326	0.174
	D04	-0.441*	-0.459*	0.328
	E02	-0.442*	0.295	-0.436*
	G02	-0.422*	-0.432*	0.306
	G03	-0.282	-0.409*	0.326
Lighting	A01	-0.179	-0.522**	0.415*
	G01	-0.281	-0.771**	0.605**
Space	A02	0.301	-0.128	0.346*
	B02	-0.013	0.359*	-0.217
	B03	0.278	-0.158	0.342*
Style	B04	-0.367	0.149	-0.424*
	E01	0.534**	0.28	0.481*
	F04	0.555**	0.078	0.558**

4.3.3 Comments Emotional Tendency by Attributes

Furthermore, this analysis involves extracting comments related to BDIA and evaluating whether the sentiment associated with these comments in relation to the corresponding BDIA element is positive or negative. This process is instrumental in discerning the impact of BDIA elements that elicit customer comments, whether positive or negative, and identifying the reasons behind the positive or negative perceptions of specific BDIA aspects. In this part of the analysis, according to the 7 elements of indirect natural experience of biophilic design, the data of 26 hotels were summarized as a whole, and their related keywords were summarized, along with related adjectives and their mention frequencies, to observe the key points in customer reviews. What elements are mentioned and why they are focused.

Table 35. Comments Attributes Emotional Tendency Result by BDIA

Attributes	Keywords	PFN	Positive Character	NFN	Negative Character	Overall
Local	Hawaiian	96	Great(6), Happy(1), Comfy(1), stunning(1), wonderful(6), renovated(1), nice(13), beautiful(13), lovely(5), updated(1), Comfortable(1), welcome(4), local(4), super(6), private(1), Open(1), Grand(2), insulated(1), blinds(1)	25	hot(1)	71
	Hawaii	222	Great(14), Happy(1), Comfy(1), nice(30), stunning(1), beautiful(21), super(20), wonderful(18), renovated(2), private(4), lovely(13), updated(1), Comfortable(2), welcome(5), local(6), Open(1), reasonable(1), incredible(1), Perfect(1), Grand(2), insulated(1), blinds(1)	139	Dated(1), hot(4), Sad(1), Spent(1)	83
Images of Nature	paintings	2	local(1), beautiful(1)	1		1
	floral	1	lovely(1)	1		0
	prints	2		3		-1
	mural	1	nice(2), super(1)	0		1
	artwork	7	super(1), local(2), beautiful(2), Great(2)	1		6
	pictures	28	lovely(1), nice(5), super(5), private(1), incredible(1), cute(2), renovated(1), beautiful(1)	73	Bad(2), Sad(1)	-45
	photos	23	renovated(1), nice(6), super(2), Great(3), interesting(2), lovely(1), reasonable(1)	51		-28
	photo	32	nice(11), Great(4), renovated(1), super(2), local(1), Welcoming(1), beautiful(2), interesting(2), lovely(2), reasonable(1)	61		-29
	painting	7	local(1), beautiful(1), touching(1)	10		-3

	print	7	secure(1), nice(1), local(1)	10	Awful(1)	-3
Natural Materials	Wood	0		1	hot(1)	-1
	Sand	8	Perfect(1), lovely(1), Great(1), nice(3)	1		7
	Bamboo	1	lovely(1), nice(1), interesting(1)	0		1
	wooden	0		9		-9
Natural Colors	Blue	2		1		1
	Green	1	Great(1)	0		1
	Red	7	Fresh(1), Cool(1), Cozy(1), Great(1), nice(1), Comfy(1)	2		5
	White	1		0		1
	Black	2	nice(1), beautiful(1)	3		-1
	Pink	1		0		1
	Tan	2		0		2
	Olive	1	wonderful(2)	0		1
	Gold	3	nice(1)	0		3
Simulating Natural Light and Air	daylight	2	nice(1)	2		0
	fresh air	10	beautiful(1), lovely(1), Great(1), super(1), nice(1)	3		7
	sunlight	1	wonderful(3)	3		-2
	breezy	2	nice(2)	0		2
	airy	9	private(1), local(1), cute(1), nice(1)	2		7
	well-lit	1		1		0
	sunny	1		4		-3
	open windows	0		1		-1
	breezes	2	Great(1), nice(1)	1		1
	shadow	0		5		-5
	shutters	1	nice(1)	1		0
	shades	7	nice(4), super(3), beautiful(1)	5		2
	Lanai	3		5	hot(1)	-2
	balcony	439	nice(92), Great(56), beautiful(33), super(17), private(4), Fresh(1), Cool(1), Cozy(1), lovely(15), wonderful(10), Ideal(2), Comfy(1), incredible(4), Spacious(3), cosy(1), blinds(2), Welcoming(2), cute(5), stunning(3), Perfect(3), Classic(1), Comfortable(10), local(5), Fantastic(5), renovated(1), updated(1), welcome(2), reasonable(1), Happy(2)	288	hot(14), Disappointed(1), Noisy(2), Exhausted(1), Shabby(1)	151
	big doors	1		0		1
Natural Forms	rounded	15	private(1), lovely(1), nice(5), wonderful(1), welcome(1), local(1), incredible(1), super(2), blinds(1), interesting(1), updated(1), cute(1)	7	hot(1)	8
	organic	2	Great(1)	0		2
Evoking Nature	music	147	Comfy(1), super(6), Great(18), Spacious(1), nice(21), private(2), Perfect(2), local(5), beautiful(7), Comfortable(2), lovely(4), wonderful(3), Classic(1), welcome(3), Fun(1), insulated(1), blinds(1)	100	Crowded(1), hot(6), Noisy(1)	47
	scents	1		1		0

	smell	64	nice(6), Spacious(1), updated(2), super(2), lovely(2), Perfect(1), incredible(3), wonderful(2), Great(2), beautiful(1)	252		-188
	ambiance	33	Fantastic(1), Comfortable(1), lovely(1), Great(1), wonderful(1)	5		28
	tranquility	1	incredible(1), beautiful(1)	0		1
	white noise	0		3		-3
	Soft	1	Sunny(1)	1		0
Information Richness	detailed	2	aged(1), welcome(1)	7		-5
	Neat	1	Great(1)	0		1
	Tidy	1	Tidy(1)	0		1
Age, change, and the patina of time	vintage	11	super(1), nice(3), Great(3), beautiful(2)	1		10
	rustic	0		1		-1
	classic	8	super(2), local(1), beautiful(1), Great(1)	2		6
	aged	25	nice(4), aged(5), Fantastic(1), Great(2), super(1), wonderful(1), secure(1), lovely(1), beautiful(1)	43	hot(1)	-18
	historical	11	charming(1), stunning(1), Great(1), beautiful(1)	0		11
	worn	9	private(1), nice(1)	55	Shabby(1), hot(1)	-46
	seasoned	2	beautiful(1), lovely(1)	2		0
	decay	1		0		1
	aging	5	Great(1)	9	hot(2)	-4
	old style	0		2		-2
	old school	2	beautiful(1), nice(1)	1		1
	renovated	79	renovated(56), wonderful(4), Stylish(1), Great(8), lovely(4), nice(11), super(3), stunning(1), welcome(3), incredible(2), beautiful(2), Fantastic(1), cute(2), Fresh(1), Happy(1)	42	hot(1)	37

The analysis results reflect some customers' environmental emotional preferences:

（1） Adopting Hawaiian style is generally an excellent strategy and will make customers think that the design is beautiful(34), super(26), wonderful(24), but it may also increase negative comments, especially with Dated(1), hot (5) Contact. Therefore, while adopting Hawaiian style is an effective strategy to enhance the experience, care should also be taken in the design to avoid the problem of lack of thermal environment regulation for the sake of creating style. There are more than 800 comments related to Hawaiian local style, which means that Hawaiian style is the focus of customer experience.

(2) Nature paintings and artwork were associated with positive reviews, while nature photos, prints, and pictures were more common in negative reviews. However, further analysis of the reviews revealed that the negative reviews for “Photo” were more related to customers taking photos than to photos of interior decoration. There are 526 comments related to Images of Nature, most of which are neutral. This means that customers will pay attention to Images of Nature, but not to a high degree.

(3) Likewise, there are a limited number of reviews on natural materials. Most of the material is not mentioned. Wood was associated with negative reviews, while bamboo and sand were associated with positive reviews, however, it is still necessary to exclude the possibility of sand as an external beach evaluation term.

(4) Almost all colors received more positive reviews, with the exception of black, which is associated with discomfort and darkness. Although natural colors are relevant to the experience, they are not a focus in customer reviews.

(5) The air quality inside the room has received more mention and attention. Higher quality of Simulating Natural Light and Air is associated with a better experience. At the same time, there are more positive than negative reviews about balconies, but it can also be related to thermal discomfort; letting in breeze and fresh air has a positive effect, and direct sunlight can cause overheating.

(6) Organic shapes are associated with positive reviews, and circles are particularly popular; richness of information is not reflected more in reviews.

(7) Playing music to enhance the atmosphere, scents help increase positive comments, and adding a good tactile experience also has a slightly positive effect. However, white noise and bad smells can have negative effects. Overall, it is an effective design strategy to use the five senses to assist in increasing the experience, especially using hearing and smell. There are

more than 700 reviews about Evoking Nature, which can be considered as an element that customers value more.

（8） A customer's emotional disposition toward age, change, and the patina of time depends on the perception of the object. If it is considered to be a retro and nostalgic style, it will increase its positive evaluation; if it is thought to be caused by overuse, it will increase its negative evaluation. There are more than 270 comments about Age, change, and the patina of time, which can be considered as factors that customers will pay attention to.

4.4 Discussion and Conclusion

In order to establish correlation between multiple analyses, the results of correlation analysis, machine learning weight analysis, and keyword analysis were juxtaposed, and sorted according to the effectiveness of BDIA in multiple analyses. The results showed that F02 (Nostalgic Craftsmanship), F04 (Weathered Materials), and G03 (Natural geometries Furnishing) showed significant correlation with positive results in each analysis and were strong predictors. At the same time, in the current design There are fewer occurrences (Mean=0.34), which is not a mainstream design style. It is a new design potential opportunity and can be more actively adopted in future designs.

Some BDIAs, such as A (Nature Image) and G (Natural geometries), have shown significant correlation with positive results in multiple analyses, and are strong predictors. At the same time, they appear more often in the current design (Mean > 0.5), has become a potential design standard for local hotels and can continue to be used in future designs and can refer to existing design plans.

Some design elements appear to be less clearly associated with more positive design outcomes. Most of these elements are smaller in size or occupy a smaller footprint and have little impact on the overall spatial perception. Such as natural material window frames and furniture (B02, B05), natural color furniture and decoration (C05, C06).

Some BDIA results show a negative correlation, such as floors with natural

colors and materials (B04, C04). The NLP Attributes analysis results show that this may be because customers associate this type of flooring with a lower grade or outdated According to the association, hotels with higher scores commonly use materials such as industrial carpets or slate. E02 (Naturalistic shapes and forms Finishing) is either associated with negative outcome scores or not associated with results in all analyses, and should be avoided in design as much as possible.

Table 36. Analysis results summary by BDIA

Attributes		Correlation	Machine	Guestroom	Lobby	Lounge	Design	Atmosphere	Decoration	Form	Furniture	Space	Style
F02	Nostalgic Craftsmanship	●	●	◎	◎	◎	●	●	●	◎	◎	◎	◎
F04	Weathered Materials	●	○	◎	●	◎	●	●	●	◎	◎	◎	●
G03	Natural geometries Furnishing	●	●	◎	◎	◎	◎	◎	●	◎	●	◎	◎
A01	Nature Image Finishing	●	●	◎	◎	◎	◎	◎	◎	●	◎	◎	◎
A03	Nature Image Decorations	●	●	◎	◎	◎	◎	◎	◎	●	◎	◎	◎
B06	Natural materials Decorations	●	●	◎	◎	◎	◎	◎	●	◎	◎	◎	◎
E03	Naturalistic shapes and forms Decorations	●	●	●	◎	◎	◎	◎	◎	◎	◎	◎	◎
F01	Nostalgic Decorations	●	●	◎	◎	◎	◎	●	◎	◎	◎	◎	◎
F03	Recycled Materials	●	○	◎	◎	◎	●	●	◎	●	◎	◎	◎
G01	Natural geometries Furnitures	●	●	◎	◎	○	◎	◎	●	◎	◎	◎	◎
B03	Natural materials Ceiling	●	●	◎	◎	◎	◎	○	◎	◎	◎	●	◎
C03	Natural colors Ceiling	●	●	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎
C01	Natural colors Walls	●	●	◎	○	◎	◎	◎	●	◎	◎	◎	◎
D01	Adjustable Windows	●	●	◎	◎	○	◎	◎	◎	◎	◎	◎	◎
G02	Natural geometries Finishing	●	●	○	◎	◎	◎	◎	◎	●	○	◎	◎
A02	Nature Image Furnishing	●	○	◎	◎	◎	◎	◎	◎	◎	◎	●	◎
C05	Natural colors Furnitures	●	○	◎	●	◎	◎	◎	◎	◎	◎	◎	◎
C02	Natural colors Window Frame	●	○	◎	◎	◎	◎	◎	◎	●	◎	◎	◎
D03	Indirect Lighting	●	●	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎
B02	Natural materials Window Frame	◎	○	◎	◎	◎	◎	◎	◎	◎	◎	●	◎
B05	Natural materials Furnitures	●	○	◎	◎	◎	◎	◎	●	◎	○	◎	◎
B01	Natural materials Walls	●	●	◎	○	◎	◎	◎	○	◎	◎	◎	◎
C06	Natural colors Decorations	●	○	●	○	◎	◎	◎	◎	◎	◎	◎	◎
D02	Adjustable Sunshades	◎	○	◎	◎	◎	◎	◎	◎	●	◎	◎	◎
E01	Naturalistic shapes and forms Furnishing	●	○	◎	◎	○	◎	◎	◎	○	◎	◎	●
D04	Soft Lighting Lamps	●	○	◎	◎	◎	◎	◎	◎	◎	○	◎	◎
C04	Natural colors Floor	●	○	●	◎	◎	◎	◎	○	◎	◎	◎	◎
B04	Natural materials Floor	●	●	◎	○	○	○	○	○	◎	◎	◎	○

E02	Naturalistic shapes and forms Finishing	◎	○	◎	◎	◎	◎	◎	◎	◎	○	◎	◎

○ Negatively correlated with the analysis results
◎ No correlation with the analysis results
● Positively correlated with the analysis results

■ Very highly related to positive emotions/higher rankings
■ Highly related to positive emotions/higher rankings
■ Moderately related to positive emotions/higher rankings
■ Likely related to positive emotions/higher rankings
■ May related to positive emotions/higher rankings
Not related to positive emotions/higher rankings
May negatively related to positive emotions/higher rankings
■ Negatively related to positive emotions/higher rankings

Through the research in this chapter, the relationship between 29 BDIAs, hotel online rating scores and customer emotional tendencies has been initially explored and quantified. The predictive validity of each BDIA has also been verified, which can provide suggestions and references for the design of future Waikiki hotels. However, in order to further verify the universality of this result, further experimental analysis is still needed.

CHAPTER 5. MODEL TESTING BY CUSTOMER ONLINE SURVEY

In order to further verify the accuracy of the big data analysis results in Chapter 4, and to exclude factors that have nothing to do with the perception of interior design quality such as hotel online advertising, service quality, and operation strategies, the study uses an online questionnaire to further review and verify the relationship between BDIA and The relationship between customer quality and perceived quality. In order to study customers' environmental perception and booking intention of Waikiki four-star hotels, the 6 hotels selected in 3.2 were selected as samples, and 120 online anonymous questionnaires were issued to each hotel to obtain a pure hotel interior design evaluation. . The questionnaire will include all interior pictures of the hotel, but will not include the hotel's exterior, swimming pool, aerial view, surrounding environment and other content that has nothing to do with interior design. At the same time, all other information about the hotel, such as name, star rating, reputation and other factors, are also hidden to prevent customer evaluations from being affected by these factors.

5.1 Survey Design

In order to more comprehensively assess the participants' environmental perception, the following review and evaluation questions were included in the questionnaire design to obtain more and more comprehensive data information while verifying the accuracy and authenticity of the answers. Questions include the following:

（1） Pre-qualification: In order to ensure that customers are familiar with hotel check-in and online booking, 5 pre-qualification questions are designed to screen participants who have online booking experience and habits and are familiar with the resort hotel booking process.

（2） Demographic information: In order to increase the demographic diversity of participants, the experiment included a wider range of participant types where possible and ensured that each hotel scenario was evaluated by people from different genders, ages, education levels, income levels,

marriages Status and number of children were provided by test takers, and demographics were similar between groups.

（3） Environmental Perception: In order to measure participants' intuitive reactions to hotel design and space, we referred to the S-O-R model and ART model theory mentioned in the literature review and the questionnaire design in previous studies, and included factors related to pleasure and arousal, approach and avoidance. 10 questions.

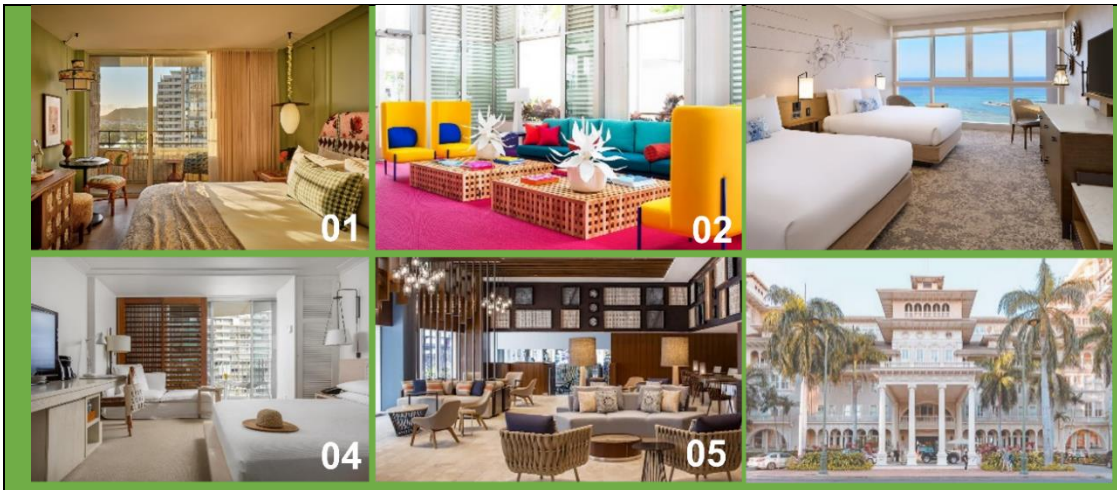
（4） Local Identity: In the research in Chapter 4, it was found that “local style” is an important positive factor in the perception of public space quality. In order to verify whether this difference in style can also be clearly perceived by consumers and affect their quality judgments, a Local Identity section was included in the questionnaire to measure the degree of "natural" and "Hawaiian" in interior design. This part will be measured with 10 questions about how they feel about the design. The design of the questionnaire refers to previous research questionnaires on place perception and biophilic design.

（5） Servicecape Evaluation: In order to measure whether customers' perception of the quality of the environment and Local Identity perception will affect their consumption intention, this part of the questionnaire follows the service landscape theory and uses 9 questions to measure customers' preference, quality perception and satisfaction with hotel interior design. .

（6） Quantitative decision-making and behavior: In order to measure whether customers' perception of environmental quality and local identity will affect their final consumption decision, this part uses 8 questions commonly used in other consumption intention studies to measure people's reservation intention and WTPP/payment premium will.

Participants were recruited through online data platforms and snowballing methods. Participants were asked to answer the following questions in the form of an online anonymous questionnaire:

Table 37. Customer Online Questionnaire Design



Theory	Attributes	Questions	Scale
Demographic characteristics	Prequalification	Are you a US consumer?	YES/NO
		Have you ever booked a hotel online?	YES/NO
		What is the name/brand of the hotel you recently booked online?	
		In a nutshell, what are you looking for in a resort hotel?	
		Date of birth verification (the start and end of the question must be consistent, and you must be an adult)	
	Demographic	Gender	
		Age	
		Education level	
		Income level	
		Travel mode	
		Traveling frequency	
(Show the hotel image) "Imagine that you are planning a vacation trip to Hawaii and when searching for a hotel to book, the following hotel room designs come up. Please look carefully at the picture presented, as the following questions will gauge your opinion of this hotel."			
S-O-R Model	Pleasure and Arousal	When I saw this hotel, I had a feeling of excitement.	Five-point Likert scale
		When I saw this hotel, I had a pleasant feeling.	Five-point Likert scale
	Approaching	I like this environment.	Five-point Likert scale
		I would love to vacation in a hotel like this.	Five-point Likert scale
		I would spend more time in hotel like this.	Five-point Likert scale
		I might end up spending more than I originally planned to spend at this hotel.	Five-point Likert scale
		I would be more friendly and chatty with other travelers and staff at this hotel.	Five-point Likert scale
	Avoidance	I will avoid returning to this hotel.	Five-point Likert scale
		I want to avoid exploring this environment.	Five-point Likert scale

		I would probably try to avoid talking to other people in this hotel.	Five-point Likert scale
Local identity	Natural Attachment	I think the hotel is very natural.	Five-point Likert scale
		I think the hotel reflects a feeling of nature.	Five-point Likert scale
		I feel like spaces like this are a part of me.	Five-point Likert scale
	Local Attachment	I think the room is very Hawaiian-style.	Five-point Likert scale
		I think the room reflects the culture of Hawaii.	Five-point Likert scale
		I think the room reflects the ecology and climate of Hawaii.	Five-point Likert scale
Servicescape Evaluation	Servicescape Preference	I absolutely love this hotel design.	Five-point Likert scale
		I prefer this hotel design to any other.	Five-point Likert scale
	Servicescape Quality Perception	How would you describe the environmental quality of this hotel design?	poor-excellent
		How attractive you would describe this hotel design?	unfavorable-favorable
		What do you think the environment quality of this hotel should be in the top percent of the local price?	0-100%
Decision And Behavior	Predetermined intention	After looking at this hotel, the likelihood of booking this hotel is high.	Five-point Likert scale
		The probability of me booking this hotel is high.	Five-point Likert scale
		I have high intention to book this hotel.	Five-point Likert scale
		There is a high chance that I will recommend my friends to book this hotel.	Five-point Likert scale
		If the actual space matches the pictures, I would love to come back.	Five-point Likert scale
	WTPP/willingness to pay premium	I would pay more to stay longer at a hotel like this.	Five-point Likert scale
		I would be willing to pay this extra percentage on my hotel bill to support the hotel like this.	0%, 1–5%, 6–10%, 11–15%, 16–20%, and more than 20%.
		What do you think the price of this hotel should be in the top percent of the local price?	0-100%

Based on the relevant theories in the literature review and the analysis results in Chapter 4, hypotheses are put forward for the results of the questionnaire survey. Based on the basic theory and previous research data, the results of the questionnaire should show the following results:

（1） S-O-R Model: Biophilic design elements are theorized to increase residents' arousal while increasing approach behavior. Therefore, in the results, high arousal, Approching and Pleasure should be positively

correlated with high BDIA.

（2） Local Identity: According to the results of the study in Chapter 4, positive ratings were significantly associated with high local identity. Especially in public spaces, more Hawaiian style elements were significantly associated with higher perceived quality ratings. Therefore, in the questionnaire results, Local Identity may be related to higher Booking scores, higher Quantitative Feeling And Evaluation, and more positive consumption decisions. It may also be related to the presence or absence of some BDIA.

（3） Servicescape Evaluation: According to the results of the correlation analysis, participants' quality evaluation of the hotel space may be positively correlated with the existence of some BDIAs. At the same time, the results of machine learning show that the ranking of the score results of this evaluation should be able to be ranked by the BDIA-Index Partially predicted, the prediction level is 50.4%.

（4） Decision And Behavior: The previous analysis shows that the relationship between BDIA and average price is not significant, and BDIA-Index has no significant positive correlation with price. Therefore, the explanatory power of BDIA-Index to consumption intention may not be significant, but there may be a positive correlation.

These hypotheses will be tested in subsequent analysis of results.

5.2 Results

5.2.1 Descriptive Statistics

The online questionnaire was distributed and recovered in January 2024. 130 questionnaires were distributed, and a total of 115 valid questionnaires that passed the pre-qualification questions were recovered, with a recovery rate of 88.4%. On average, 18 participants per hotel submitted valid answers. Before analyzing the data, chi-square cross analysis was first used to verify that there were no significant differences in the demographic information of the participants between different hotel groups, and that there were gradient differences among participants within the groups to ensure the objectivity and universality of the answers.

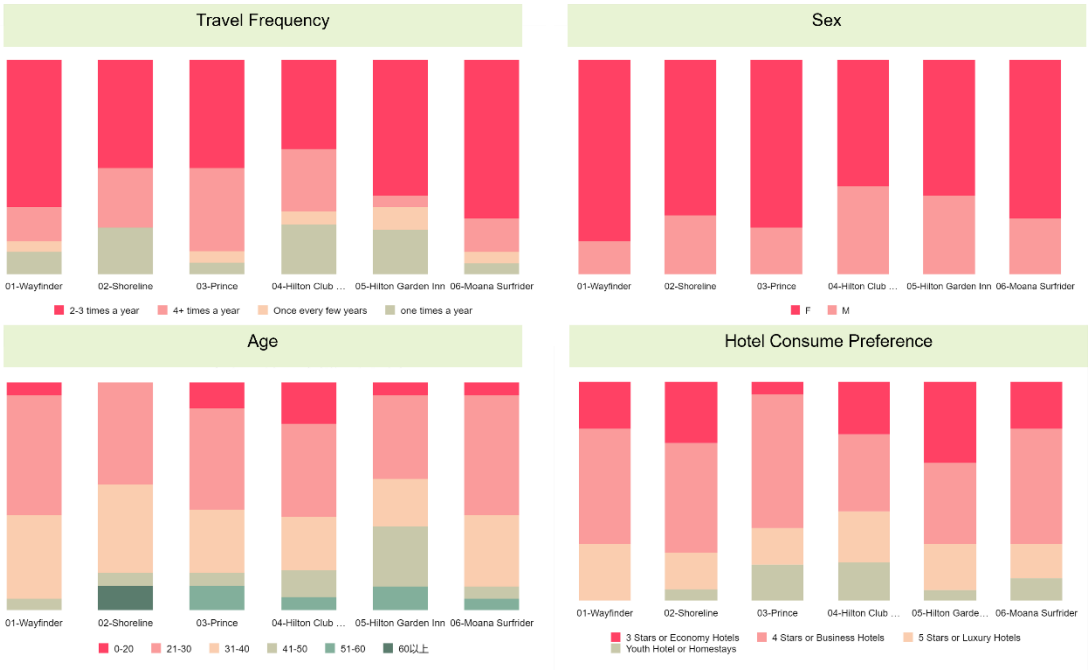


Figure 19 Demographic cross-analysis results Source: Author

The results of the chi-square analysis showed that the eight demographic questions among different hotel groups showed consistency and no difference; the results within the group showed stratified differences. The data meets the requirements and can be used for subsequent analysis and research.

5.2.2 Chi-square analysis

Table 38. Perception and Evaluation Chi-square analysis results

	01-Wayfinder	02-Shoreline	03-Prince	04-Hilton Club modern	05-Hilton Garden Inn	06-Moana Surfrider	f	
Arousal Overall	4.05±1.13	3.61±2.28	4.89±0.76	2.59±2.69	3.26±1.41	3.74±1.76	3.331	**0.008 ** **
Approching Overall	8.89±2.96	7.67±6.11	10.61±2.93	7.94±4.96	8.32±2.67	8.21±5.74	1.033	0.402
Aviodance Overall	-5.79±3.65	-5.83±3.28	-5.83±3.00	-5.88±2.26	-4.21±4.33	-4.16±4.56	1.030	0.404
Nature Attachment	5.11±2.45	3.56±4.91	5.78±2.80	4.82±3.28	5.00±2.38	3.74±3.23	1.253	0.290
Local Attachment	6.32±1.53	5.39±4.02	5.06±4.36	3.71±4.84	3.47±3.81	3.37±4.27	1.779	0.124
Servicecape Likeness	15.16±3.99	10.83±12.17	17.83±5.96	11.47±11.14	14.58±3.96	13.74±6.75	1.878	0.104
Servicecape Evaluation	12.11±1.05	11.39±2.73	12.61±1.14	11.53±1.59	11.53±1.65	11.74±1.56	1.338	0.254
WTPP by Percentage	2.21±1.13	2.44±1.29	2.44±1.20	1.88±1.22	1.89±1.15	2.42±1.26	0.892	0.489
Price Perception	66.00±16.22	67.72±16.38	75.39±11.92	64.53±15.03	65.21±11.99	72.32±16.08	1.598	0.167
WTPP by Amount	120.37±114.71	155.39±118.20	231.39±447.27	127.35±55.79	219.74±228.35	160.00±107.82	0.813	0.543

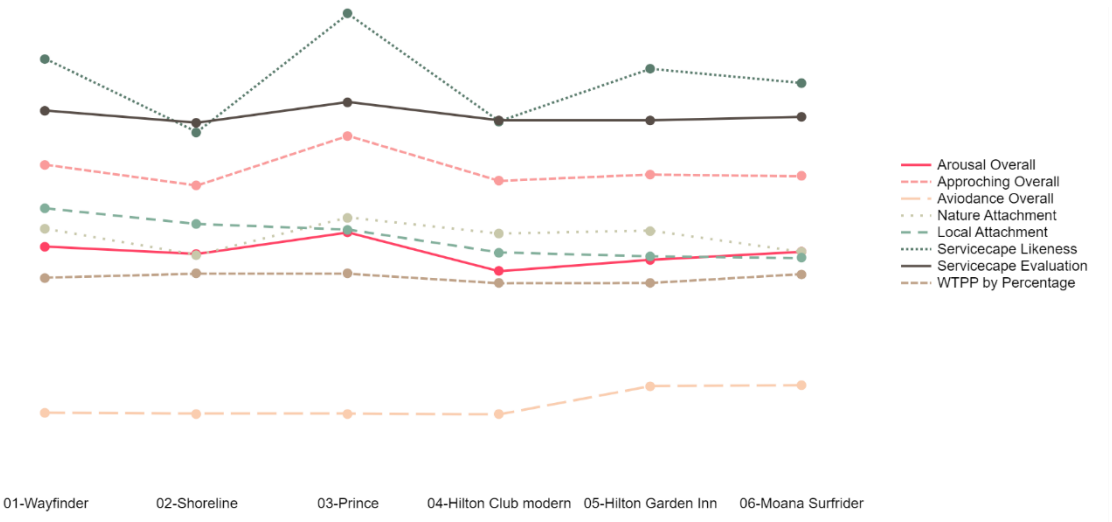


Figure 20 Perception and Evaluation Chi-square analysis results Source: Author

The results of variance analysis on the main Evaluation and Perception analysis factors show that Prince Waikiki's S-O-M Model performs best, with the highest Arousal Overall, Approching Overall and the lowest Aviodance Overall, representing the highest overall arousal, followed by Wayfinder Waikiki, which performs the best The worst group is 04-Hilton Club modern. Participants believed that 01-Wayfinder Waikiki was the most Hawaiian-style hotel, followed by 02-Shoreline, and 06-Manoa Surfrider was considered the least Hawaiian-style hotel. The most popular hotel among consumers is 03-Prince, followed by 01-Wayfinder, and the least popular

hotel is 02-Shoreline. Consumers believe that 03-Prince is the most expensive hotel and are willing to pay the most extra money for 03-Prince. Except for Arousal Overall, none showed significant differences between groups. This may be due to the small sample size and the inability to produce significant results for small effects.

5.2.3 Correlation Analysis

5.2.3.1 S-O-M Model Correlation Analysis

Table 39. S-O-M Model Correlation Analysis results

Correlation analysis was conducted between the three evaluation parameters of the S-O-M model and BDIA-Index. The results showed that BDIA-Index was significantly positively correlated with the degree of Arousal, that is, higher BDIA was significantly related to higher arousal, but it did not cause closeness. or significant changes in avoidance levels. Further correlation analysis including 29 BDIA elements found that all BDIA would not cause significant changes in the degree of approach or avoidance, but some BDIA would cause a significant increase in the degree of

Attributes	Nature Attachment	Local Attachment
BDIA-Index	0.09	0.125
C03	0.184	**0.214***
E03	**0.194***	0.179
F02	0.101	**0.220***
F03	-0.068	**0.234***
F04	0.013	**0.274****

arousal. However, comparing with the comprehensive analysis results in Chapter 4, it is found that the BDIA that will cause an improvement in Arousal is not completely consistent with the BDIA that may cause an improvement in customer

Attributes	Arousal Overall	Approching Overall	Aviodance Overall
BDIA-Index	**0.285****	0.137	0.026
A01	**0.296****	0.114	0.004
C03	**0.193***	0.145	-0.149
C05	**0.255****	0.065	0.073
C06	**0.217***	0.142	0.070
E01	**0.226***	0.160	-0.092
E02	**0.200***	0.047	0.003
E03	**0.252****	0.185	-0.137
F02	**0.279****	0.127	-0.053
G01	**0.245****	0.080	0.051
G02	**0.203***	0.047	0.104

perception in the big data analysis results, and further research is still needed.

5.2.3.2 Local Identity Correlation Analysis

Table 40. Local Identity Correlation Analysis results

Correlation analysis was conducted between the two evaluation parameters of the Local Identity model and BDIA-Index. The results showed that there was no obvious correlation between BDIA-Index and the two parameters, that is, a higher BDIA would not cause a perception of the degree of "Hawaiian" style. Increase. Further correlation analysis by incorporating 29 BDIA elements found that some BDIAs were significantly related to participants' perceptions of the "Hawaiian" degree of the space. This includes C03 (Natural colors Ceiling), E03 (Naturalistic shapes and forms Decorations), and F (Age and time) series BDIA. Therefore, although BDIA-Index does not show a significant correlation with Local Identity in general, the application of BDIA above can still be added in the design to increase the "Hawaiian" feel of the space. According to the NLP analysis results in 4.3, this will be helpful in improving the lobby. The quality perception and praise rate of public spaces such as , lounges and so on play an effective role.

5.2.3.3 Servicecape Perception Correlation Analysis

Table 41. Servicecape Evaluation Correlation Analysis results

The correlation analysis between the two evaluation parameters of the Servicecape Evaluation model and BDIA-Index shows that BDIA-Index has a significant positive correlation with Servicecape Likeness. That is to say, the more BDIA, the more customers like and are satisfied with Servicecape. The higher it is. Although BDIA-Index shows a positive correlation with Servicecape Evaluation, its correlation is not significant. This may be due to the small sample size of each hotel (N<20), which cannot show significance below 0.2. Some BDIAs are positively related to positive Servicecape Perception. However, comparison with the comprehensive analysis results in Chapter 4 shows that the BDIAs that will cause an

Attributes	Servicecape Likeness	Servicecape Evaluation
BDIA-Index	**0.227***	0.181
B01	**0.188***	0.076
C06	**0.240***	0.179
E01	**0.189***	**0.214***
E03	**0.199***	**0.200***
F02	**0.189***	0.171
G03	**0.232***	0.143

increase in Servicecape Perception are not completely consistent with the BDIAs that may cause an increase in customer perception in the big data analysis results. Still need to be further research.

5.2.3.4 WTPP Correlation Analysis

Table 42. WTPP Correlation Analysis results

The correlation analysis between the three evaluation parameters of the WTPP model and the BDIA-Index shows that there is no significant correlation between the BDIA-Index and the three parameters, that is, a higher BDIA will not increase the customer's willingness to pay additional economic costs. However, some BDIA elements were associated with higher price perceptions, which would make participants perceive the space as "more luxurious", including A01 (Nature Image Finishing) and E02 (Naturalistic shapes and forms Finishing), but this perception was not apparent. This increases participants' willingness to pay higher prices.

5.2.3. Correlation Analysis Comparison

By comparing hundreds of recent studies, Khalilzadeh & Tasci (2017) found that in tourism and hospitality academia, the small, medium, and large effect sizes of Pearson correlation analysis set to 0.10, 0.30, and 0.50 can more effectively reflect

Attributes	WTPP by Percentage	Price Perception	WTPP by Amount
BDIA-Index	0.082	0.118	0.025
A01	0.184	**0.245****	0.035
E02	0.182	**0.220***	0.057

the actual situation.[175] Therefore, it can be considered that the significant positive correlation between BDIA-Index and Arousal Overall and Servicecape Likeness has a medium effect size, and may have a small to medium effect size with other factors. However, due to the small sample size (N= 26) Unable to present significance. Some BDIA may have benign effects on positive perceptions, with medium effect sizes.

5.2.4 Path Analysis

In order to verify whether the mechanism of BDIA's impact on customer

[175] Khalilzadeh, J., & Tasci, A. D. (2017). Large sample size, significance level, and the effect size: Solutions to perils of using big data for academic research. Tourism Management, 62, 89-96.

psychology is as hypothesized above, a path analysis was conducted on the questionnaire results. Path Analysis is a statistical technique used to study complex networks of direct and indirect relationships between variables. It is part of structural equation modeling (SEM) and is commonly used in social sciences, biostatistics, market research, and other fields. The main purpose of path analysis is to determine the strength and direction of causal relationships between different variables. In this study, through literature analysis, a theoretical model is constructed that describes the hypothesized relationships between variables, which can be direct or indirect through one or more mediating variables. Afterwards, SPSS software was used to

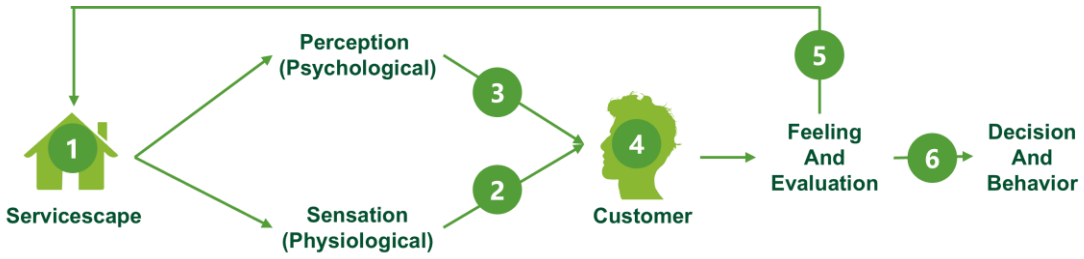


Figure 21 Theoretical model of path analysis Source: Author

estimate the coefficients of each path in the model. These coefficients reflect the strength of the relationship between variables. By analyzing these path coefficients, we can understand the direct and indirect effects between variables, as well as the causal links between them.

This path analysis investigates the relationships among multiple variables. The BDIA-Index demonstrates a positive effect on Arousal Overall, a relationship also observed in Nature Attachment (z=0.942, p=0.346). However, it does not show significant effects on Aviodance Overall (z=0.553, p=0.581) and Approching Overall (z=0.841, p=0.400), suggesting that the BDIA-Index may not be a crucial predictive variable in these specific relationships. Therefore, it is plausible that the BDIA could indirectly influence Servicecape Perception by enhancing Nature Attachment while simultaneously increasing Arousal Overall.

Table 43. Path analysis model regression coefficient summary

X	→	Y	unstandardized path coefficient	*SE*	*z* (CR Value)	*p*	standardized path coefficient (t)
Servicecape Evaluation	→	WTPP by Amount	23.756	12.592	1.887	0.059	0.228
Servicecape Likeness	→	WTPP by Amount	1.265	2.67	0.474	0.636	0.057
Nature Attachment	→	Local Attachment	0.793	0.088	9.049	0	0.653
Servicecape Evaluation	→	Price Perception	3.743	1.011	3.703	0	0.404
Servicecape Likeness	→	Price Perception	0.243	0.214	1.131	0.258	0.123
BDIA-Index	→	Nature Attachment	0.97	0.099	0.942	0.046	0.489
Local Attachment	→	Aviodance Overall	-0.184	0.085	-2.171	0.03	-0.203
BDIA-Index	→	Aviodance Overall	1.14	2.062	0.553	0.581	0.052
Local Attachment	→	Approching Overall	0.638	0.086	7.404	0	0.575
BDIA-Index	→	Approching Overall	1.761	2.093	0.841	0.4	0.065
Servicecape Evaluation	→	WTPP by Percentage	0.327	0.09	3.613	0	0.422
Servicecape Likeness	→	WTPP by Percentage	-0.017	0.019	-0.887	0.375	-0.103
Local Attachment	→	Arousal Overall	0.278	0.034	8.255	0	0.6
BDIA-Index	→	Arousal Overall	2.392	0.82	2.919	0.004	0.212
Aviodance Overall	→	Servicecape Evaluation	-0.072	0.025	-2.833	0.005	-0.169
Approching Overall	→	Servicecape Evaluation	0.199	0.022	8.932	0	0.57
Arousal Overall	→	Servicecape Evaluation	0.261	0.053	4.898	0	0.312
Aviodance Overall	→	Servicecape Likeness	-0.048	0.102	-0.47	0.639	-0.024
Approching Overall	→	Servicecape Likeness	0.898	0.089	10.082	0	0.546
Arousal Overall	→	Servicecape Likeness	1.875	0.213	8.804	0	0.477

Moreover, a significant positive relationship exists between Nature Attachment and Local Attachment ($z=9.049$, $p<0.01$), indicating that Nature Attachment strengthens feelings towards Local Attachment. This positive correlation is also reflected in the impact of Local Attachment on Approching Overall and Arousal Overall, where Local Attachment exhibits a significant positive effect on both variables (Approching Overall: $z=7.404$, $p<0.01$; Arousal Overall: $z=8.255$, $p<0.01$). However, Local Attachment shows a significant negative effect on Aviodance Overall ($z=-2.171$, $p<0.05$), suggesting that individuals with a strong attachment to a place are inclined to reduce avoidance behaviors.

The analysis indicates varied effects of Servicecape Evaluation on WTPP by Amount and Price Perception. Specifically, while the impact of Servicecape

Evaluation on WTPP by Amount does not reach significance (z=1.887, p=0.059), its influence on Price Perception is significant and positive (z=3.703, p<0.01). This finding may imply that Servicecape Evaluation plays a role in affecting customers' price perception, but has a limited impact on the amount they are willing to pay.

On the other hand, Servicecape Likeness does not show significant effects in any of the examined pathways. Whether it is regarding WTPP by Amount (z=0.474, p=0.636), Price Perception (z=1.131, p=0.258), or WTPP by Percentage (z=-0.887, p=0.375), the influence of Servicecape Likeness is not significant. This suggests that Servicecape Likeness may not be a key factor in determining these outcome variables.

Finally, the effects of Aviodance Overall, Approching Overall, and Arousal Overall on Servicecape Evaluation and Servicecape Likeness reveal complex relationships. Aviodance Overall has a significant negative effect on Servicecape Evaluation (z=-2.833, p<0.01), but no significant effect on Servicecape Likeness (z=-0.470, p=0.639). Conversely, both Approching Overall and Arousal Overall show significant positive effects on Servicecape Likeness (Approching Overall: z=10.082, p<0.01; Arousal Overall: z=8.804, p<0.01) and also on Servicecape Evaluation (Approching Overall: z=8.932, p<0.01; Arousal Overall: z=4.898, p<0.01). This might suggest that in service environments, customers' approach and arousal responses significantly influence their evaluation of the service landscape.

Based on the above results, the model is verified and significant influence paths are marked to revise the hypothesized model. The revised results are basically in line with the previous mechanism hypothesis, and show how BDIA-Index and Local Identity affect participants' Servicecape Preference and Decision And Behavior through the intermediary effect of S-O-M Model.

Through horizontal comparison of the Standardized Estimate, t-Statistic and p-

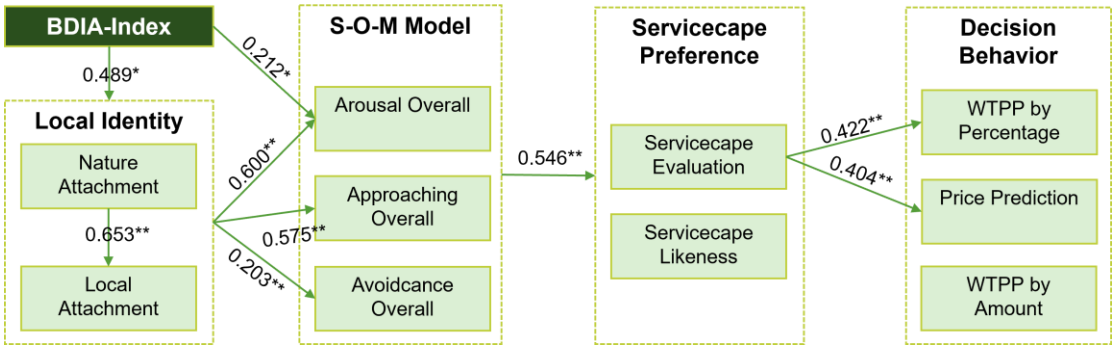


Figure 22 Modified impact path model Source: Author

Value results of other research path analysis, it was found that the effect size of BDIA-Index on improving Local Identity and other design perception factors was between medium and large. However, the overall hotel environment plays a limited role in the success and development of the hotel, so the improvement of BDIA has no obvious effect on the improvement of WTPP, hotel cost-effectiveness perception[176].

Previous research on how the S-O-R model affects hotel perception has proven that the S-O-R model can explain how the hotel's atmosphere elements affect customer perception. For example, the impact of Room design factor on Customer satisfaction is 0.311-0.489, and the impact of Customer satisfaction on Willingness to suggest is not significant, [177][178]these are consistent with the path analysis conclusions of the study. Combined with the goodness-of-fit statistics of the model showed that the structural model fit the data very well, it can be considered that this model better reflects the impact mechanism of BDIA on real customer decision-making and Impact size.

[176] Yadegaridehkordi, E., Nilashi, M., Nasir, M. H. N. B. M., & Ibrahim, O. (2018). Predicting determinants of hotel success and development using Structural Equation Modelling (SEM)-ANFIS method. Tourism Management, 66, 364-386.

[177] Choi, H., & Kandampully, J. (2019). The effect of atmosphere on customer engagement in upscale hotels: An application of SOR paradigm. International Journal of Hospitality Management, 77, 40-50.

[178] Jana, A., & Chandra, B. (2016). Mediating role of customer satisfaction in the mid-market hotels: An empirical analysis. Indian Journal of Science and Technology, 9(1), 1-16.

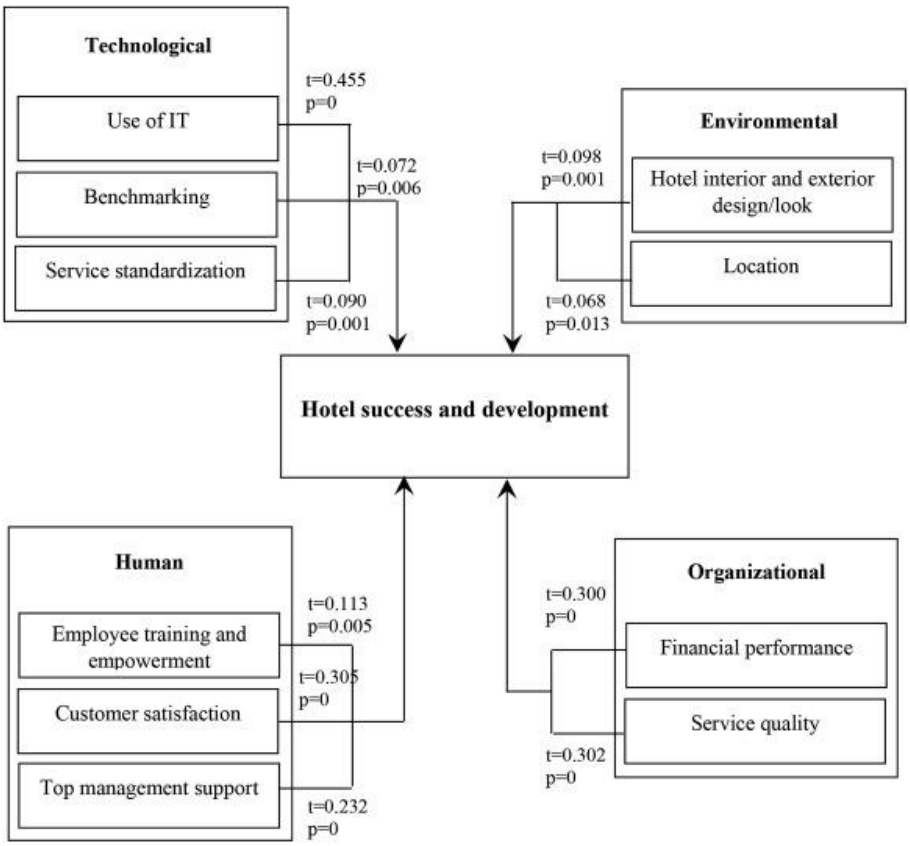
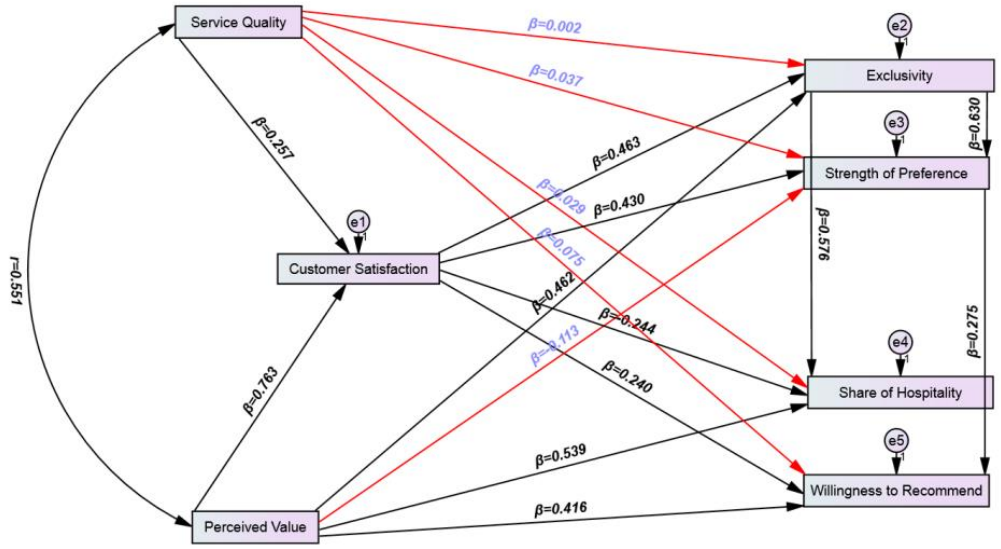


Figure 23 Structural model results from Literature Review Source: Yadegaridehkordi, E., Nilashi, M., Nasir, M. H. N. B. M., & Ibrahim, O. (2018). Predicting determinants of hotel success and development using Structural Equation Modelling (SEM)-ANFIS method. *Tourism Management*, *66*, 364-386.



Note:
β= Rejected Path
β= Accepted Path
r= Correlation coefficient, β= Standardized regression weight, e1= Error term of Customer Satisfaction,
e2= Error term of Exclusivity, e3= Error term of Strength of Preference, e4= Error term of Share of Hospitality and e5= Error term of Willingness to Recommend.

Figure 24 Re-specified Loyalty Outcome Model Revealing the Mediating Role of Customer Satisfaction in the Mid-Market Source: Jana, A., & Chandra, B. (2016). Mediating role of customer satisfaction in the mid-market hotels: An empirical analysis. *Indian Journal of Science and Technology*, *9*(1), 1-16.

5.3 Discussion

5.3.1 Hypothesis Verifying

When designing the questionnaire, the researcher proposed four hypotheses and discussed them in the analysis. Here, by comparing the results of the questionnaire analysis with the hypothesis, you can verify whether it is true.

Hypothesis 1: S-O-R Model: According to theory, biophilic design elements will increase residents' arousal and increase approach behavior. Therefore, in the results, high arousal, Approching and Pleasure should be positively correlated with high BDIA. The results of data analysis show that the increase in BDIA-Index is indeed related to higher arousal levels, but Approching and Avoidance are not significantly related to BDIA. relation. This may be due to the limited amount of data, or it may be because BDIA does not directly cause changes in avoidance or approach intentions. Therefore, part of the hypothesis holds true.

Hypothesis 2: Local Identity may be related to higher Booking scores, higher Quantitative Feeling And Evaluation, and more positive consumption decisions. It may also be related to the existence of some BDIA. The results indeed show that Local Identity and these factors are significantly positive and are strong predictors. Therefore, hypothesis 2 holds.

Hypothesis 3: Participants' quality assessment of hotel space may be positively correlated with the existence of some BDIA, and the ranking of the score results of this assessment should be partially predicted by the BDIA-Index. The correlation analysis results show that A01 (Nature Image Finishing) and E (Naturalistic shapes and forms) are related to higher willingness to pay and price prediction; BDIA-Index can be used as a predictor of Servicecape Likeness, but it does not reflect the relationship with Servicecape Evaluation. The significant correlation may be due to

the small sample size. Therefore, hypothesis three is established.

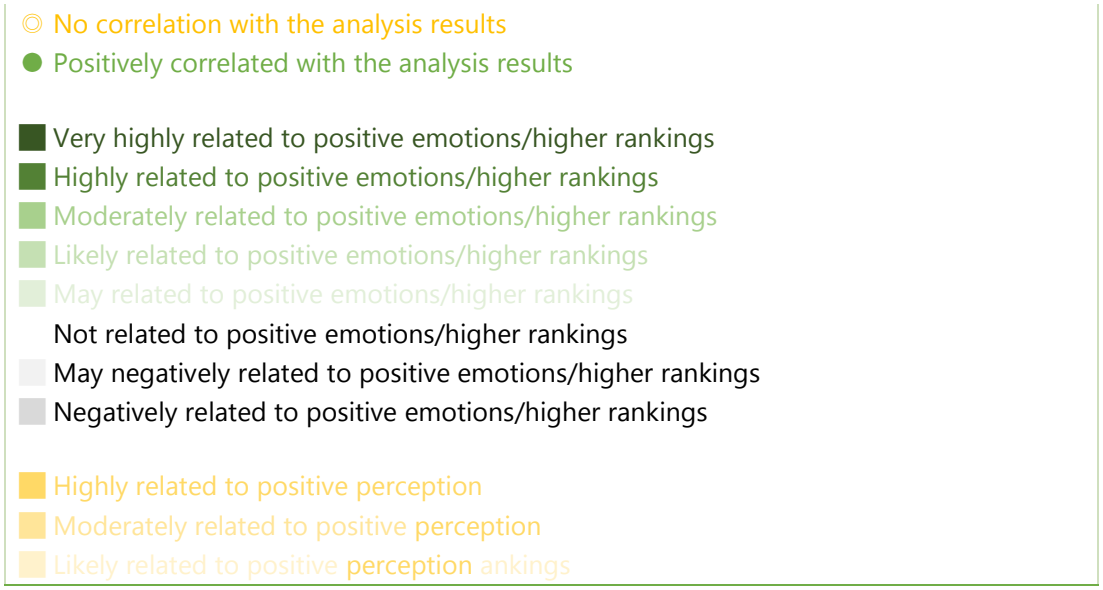
Hypothesis 4: The explanatory power of BDIA-Index to consumption intention may not be significant, but there may be a positive correlation. Although the correlation analysis results show a positive correlation, they do not show a significant correlation with WTPP. Therefore, hypothesis 4 holds.

5.3.2 Comparison of correlation analysis results

Table 44. Analysis results summary for comparison

Attributes	Correlation	Machine	Guestroom	Lobby	Lounge	Design	Atmosphere	Decoration	Form	Furniture	Space	Style	S-O-Model	Local Identity	Service scape	WTPP
F02	●	●	◎	◎	◎	●	●	●	◎	◎	◎	◎	●	●	●	◎
F04	●	○	◎	●	◎	●	●	●	◎	◎	◎	●	◎	●	◎	◎
G03	●	●	◎	◎	◎	◎	◎	●	◎	●	◎	◎	◎	◎	●	◎
A01	●	●	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	●	◎	◎	●
A03	●	●	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎
B06	●	●	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎	◎
E03	●	●	●	◎	◎	◎	◎	◎	◎	◎	◎	◎	●	●	●	◎
F01	●	●	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎	◎	◎
F03	●	○	◎	◎	◎	●	●	◎	●	◎	◎	◎	◎	●	◎	◎
G01	●	●	◎	◎	○	◎	◎	●	◎	◎	◎	◎	●	◎	◎	◎
B03	●	●	◎	◎	◎	◎	○	◎	◎	◎	●	◎	◎	◎	◎	◎
C03	●	●	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	●	●	◎	◎
C01	●	●	◎	○	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎	◎
D01	●	●	◎	◎	○	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎
G02	●	●	○	◎	◎	◎	◎	◎	●	○	◎	◎	●	◎	◎	◎
A02	●	○	◎	◎	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎
C05	●	○	◎	●	◎	◎	◎	◎	◎	◎	◎	◎	●	◎	◎	◎
C02	●	○	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎
D03	●	●	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎
B02	◎	○	◎	◎	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎
B05	●	○	◎	◎	◎	◎	◎	●	◎	○	◎	◎	◎	◎	◎	◎
B01	●	●	◎	○	◎	◎	◎	○	◎	◎	◎	◎	◎	◎	●	◎
C06	●	○	●	○	◎	◎	◎	◎	◎	◎	◎	◎	●	◎	●	◎
D02	◎	○	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎
E01	●	○	◎	◎	○	◎	◎	◎	○	◎	◎	●	●	◎	●	◎
D04	●	○	◎	◎	◎	◎	◎	◎	◎	○	◎	◎	◎	◎	◎	◎
C04	●	○	●	◎	◎	◎	◎	○	◎	◎	◎	◎	◎	◎	◎	◎
B04	●	●	◎	○	○	○	○	○	◎	◎	◎	○	◎	◎	◎	◎
E02	◎	○	◎	◎	◎	◎	◎	◎	◎	○	◎	◎	●	◎	◎	●

○ Negatively correlated with the analysis results



Juxtaposing the results of the analysis in Chapter 4 with the results of the questionnaire survey and conducting correlation analysis, it was found that there is no strong correlation between the two results. Some are related to higher online rating scores and higher praise rates under big data. The related BDIA does not show the same effectiveness in double-blind testing; and vice versa. However, some elements maintained stable predictive validity across all tests, including F02 (Nostalgic Craftsmanship), F04 (Weathered Materials), G03 (Natural geometries Furnishing), A01 (Nature Image Finishing), E03 (Naturalistic shapes and forms Decorations), F03 (Recycled Materials), G01 (Natural geometries Furnitures), C03 (Natural colors Ceiling), and they are all BDIA related to Local Identity.

Some BDIA have different performances in different tests. For example, A03 (Nature Image Decorations) has been proven to be related to higher experience or evaluation in big data analysis, but this correlation was not reflected in the questionnaire survey.

Some BDIAs show opposite results in two tests. For example, E01 (Naturalistic shapes and forms Furnishing) and E02 (Naturalistic shapes and forms Finishing) are considered to be results that will lead to lower evaluations in big data analysis, but the results of the questionnaire survey prove that these two items are associated with higher arousal levels. and higher WTPP correlation.

Some BDIAs have been proven to have no correlation or only negative

correlation with positive environmental perception in all tests, including B02 (Natural materials Window Frame), B05 (Natural materials Furnitures), D02 (Adjustable Sunshades), D04 (Soft Lighting Lamps), C04 (Natural colors Floor), B04 (Natural materials Floor).

Overall, the two tests showed some similar and different results. What is clear is that in both real-life situations and controlled experimental groups, the use of Native Hawaiian-inspired designs and vintage-style craftsmanship is associated with higher ratings or perceptions of quality, as well as with lighting patterns and architectural details such as window frames. Color)-related BDIA cannot have a significant impact on quality perception.

5.3.3 Possible impact mechanism of BDIA

The results of the path analysis verified and revised the hypothesis model established based on the literature research results, and established a mechanism model of the impact of BDIA on participants' consumption decisions. BDIA directly affects participants' arousal levels and indirectly affects participants' approach or avoidance responses by affecting Local Identity. The S-O-M Model will affect participants' quality perception and satisfaction with the space. Quality perception will affect consumers' decision-making and behavior, and they will accept prices higher than the budget or market price. This proves that the impact of BDIA on consumer decision-making is indirect and needs to be achieved through intermediary effects.

5.4 Conclusion

In this chapter, through questionnaire analysis, the study evaluated four hypotheses, elucidating the interplay between BDIA and guest responses.The analysis partially corroborated the Stimulus-Organism-Response (S-O-R) Model, indicating that biophilic design enhances arousal levels, though its impact on approach or avoidance behaviors was inconclusive. This suggests a nuanced influence of biophilic elements on guest psychology, extending beyond straightforward behavioral responses.

Further, the study confirmed the positive correlation between Local Identity and

higher booking scores, qualitative evaluations, and consumption decisions, highlighting the significance of integrating local identity elements in biophilic design to enrich guest experience. Regarding guest quality assessments and willingness to pay, specific biophilic features, notably Nature Image Finishing and Naturalistic shapes and forms, were associated with enhanced perceptions and valuation of hotel spaces. However, the BDIA-Index's predictive capacity for these outcomes was limited, indicating a complex relationship between biophilic design and consumer valuation metrics. While a positive correlation between the BDIA-Index and consumption intention was observed, the explanatory power of BDIA-Index on consumption intentions was not significant, pointing to an indirect influence of biophilic design on guest decision-making processes.

Comparative analysis across different evaluative methods revealed both consistencies and discrepancies in the effectiveness of BDIA elements, with certain aspects consistently enhancing guest perceptions, particularly those tied to Local Identity. This underscores the variable impact of biophilic design across diverse assessment frameworks.

In summary, the study establishes that the impact of BDIA on consumer decision-making is mediated through complex psychological responses, including arousal and perceptions of local identity. These findings suggest that the application of biophilic design in hospitality environments influences guest experiences and decisions in multifaceted ways, necessitating further investigation into the underlying mechanisms and optimizing BDIA's implementation for enhanced guest satisfaction.

CHAPTER 6. MODEL TESTING BY OTHER HAWAII HOTELS

In the previous chapter, the study verified the results of NLP big data research through online anonymous questionnaires. In this chapter, the BDIA-Index Prediction Model established by machine learning is verified by incorporating a larger number of hotel data in Hawaii. When establishing the BDIA-Index, in order to reduce the impact of location, consumption level, hotel supporting facilities, etc. on space evaluation, the sample only included 26 four-star hotels in Waikiki. When testing the model, for the sake of increasing diversity, we considered including hotels with different geographical locations and different star ratings to test the prediction of Booking.com scores by BDIA-Index after incorporating environmental factors unrelated to interior design quality.

6.1 Hotel Selection and Comparisons

In order to test the predictive power of BDIA-Index on online ratings in a complex actual environment, the experiment used random sampling method to randomly select 5 hotels from all four-star hotels in Hawaii and different star-rated hotels in downtown Honolulu. hotels, and score a total of 10 hotels in two groups according to the BDIA standards established in Chapter 3.

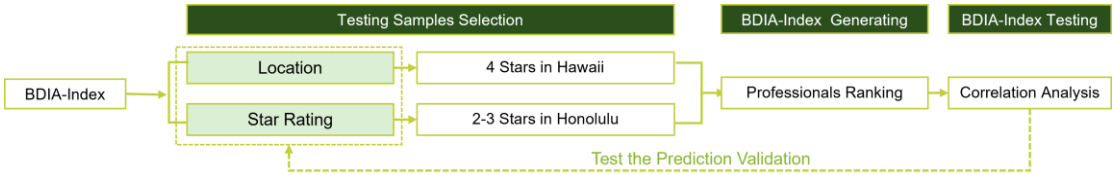


Figure 25 Roadmap for sampling to verify the validity of BDIA-Index

During the extraction process, crawler technology is first used to obtain a list of all hotel information that meets the required categories. In order to ensure the accuracy of the extracted hotel ratings and ensure that some hotels in non-popular tourist areas can be included in the test, only 20 or more reviews are retained. hotels, and then select hotel samples of corresponding category groups through random numbers. To prevent raters from being affected by hotel operations, services, and

customer reviews, only photos related to the hotel's BDIA rating were extracted and other information was hidden. Each hotel was rated by three professionals according to the standard, and the final result was taken as the majority.

The hotels that were finally included in the ratings and their corresponding rating information are shown in the table below:

Table 45. Selected Hotels for BDIA-Index Validation Testing

No.	Type	Name	Google	Expedia	Booking	Services	Facilities	Cleanness	Comfort	Value of Money	Location	WiFi	Average Price
27	Location	Marriott's Kaua'i Beach Club	4.5	8	7.6	8.4	8.4	8.1	8.6	5.9	9.2	7.5	539
28	Location	Volcano Inn	4.4	8.4	7.5	7.9	7.9	8.3	8.2	7.4	9	9.2	179
29	Location	The Westin Maui Resort & Spa, Ka'anapali	4.3	8.6	8.7	9.1	9.2	9.3	9.1	7.4	7.4	9.4	553
30	Location	OUTRIGGER Kona Resort & Spa	4.1	8.2	8	8.7	8.7	8.6	8.4	7.4	8.5	7.5	327
31	Location	Hana-Maui Resort - Destination by Hyatt	4.3	9	8.5	9.2	8.5	8.9	8.9	7.1	9.4	7.9	398
32	Ratings	Waikiki Monarch Hotel	3.4	5.6	5.8	7.1	6.1	6.4	6.4	6.1	7.5		129
33	Ratings	Royal Grove Waikiki	3.7	7.2	6.8	8.5	8.5	7	7	7.3	9.1	8.6	162
34	Ratings	Royal Kona Resort	4.1	8.2	7.7	8.4	7.7	8.1	8.1	8.1	9.2	8.3	280
35	Ratings	Kula Lodge	4.2		6.7	7.2	6.8	7.4	7.3	5.8	8.4	7.9	444
36	Ratings	Tip Top Motel, Cafe, And Bakery	4.4	7.4	6.8	7.6	6.4	7.3	7.1	6.7	8	7.7	162

To help readers understand the basic information of the selected hotels, the introduction of the 12 hotels finally selected is as follows. In order to avoid affecting the results, these text introductions and online scoring information will not be presented to the judges during the BDIA scoring stage, only the photos will be displayed on the big screen.

Table 46. Selected hotels for BDIA-Testing introduction

Hotel Name	Description
Marriott's Kaua'i Beach Club	**Group:** Other 4-Stars in Hawaii **Brief Introduction:** Located near Kalapaki Beach, the vacation ownership resort in Lihue provides oceanfront accommodations. The resort features a range of guest

  	rooms, parlors, and condo-style villa vacation rentals, all equipped with resort amenities, a gourmet center for meal preparation, and views of the ocean or gardens. Guests have access to a 26,000-square-foot outdoor pool, several whirlpool spas, and a children's pool. Additional facilities include a fitness center, offering a variety of activities and recreation options across Kauai such as scuba diving, surfing, and snorkeling. For relaxation, a day spa is available on-site, along with beachfront dining options. The resort is co-located with The Royal Sonesta Kaua'i Resort, enhancing the variety of services and experiences available to guests. [179] **Location:** Kalapaki Beach, Marriott Vacation Club, 3610 Rice St, Lihue, HI 96766
Volcano Inn   	**Group:** Other 4-Stars in Hawaii **Brief Introduction:** Situated in Volcano Village, the lodge presents spacious rooms featuring high ceilings and large windows, providing panoramic views of the surrounding rainforest. It is designed to cater to guests seeking either relaxation or adventure. The lodge, known as Lotus Garden Cottages, prioritizes guest experience, with a focus on comfort and quality service. The staff is committed to ensuring a memorable and hassle-free stay. The accommodations combine comfort with the natural beauty of its location, making it a suitable option for travelers planning their next vacation in Hawaii. [180] **Location:** 19-3820 Old Volcano Rd, Volcano, HI 96785
The Westin Maui Resort & Spa, Ka'anapali	**Group:** Other 4-Stars in Hawaii **Brief Introduction:** Located in Maui, this hotel is positioned close to the Pacific Ocean and near volcanic mountains, offering a scenic setting that is characteristic of Hawaii's natural beauty. The hotel provides access to a

[179] https://www.marriott.com/en-us/hotels/lihka-marriotts-kauai-beach-club/overview/

[180] https://www.volcanoinnhawaii.com/

  	variety of activities that cater to both adventure seekers and those looking to relax. Guests can enjoy local cuisine at any of the five dining venues available on-site, several of which offer outdoor seating with picturesque views. The hotel also offers a shuttle service for easy access to attractions in West Maui. Additionally, guests can explore Hokupa'a, a luxury offering within the Westin Maui, which includes access to the exclusive Lanai. This area is dedicated to providing experiences that connect guests with Hawaiian culture and promote wellbeing.[181] **Location:** 2365 Kaanapali Pkwy, Lahaina, HI 96761
OUTRIGGER Kona Resort & Spa   	**Group:** Other 4-Stars in Hawaii **Brief Introduction:** The resort features a multi-level fantasy pool, equipped with a 200-foot water slide and areas for manta-ray viewing, alongside panoramic views of the Pacific Ocean. It offers a variety of guest activities such as yoga, lei making, hula and ukulele lessons, as well as cultural tours of the historic fishing village located within the resort's grounds. **Location:** 78-128 Ehukai Street, Kailua-Kona, Hawaii 96740

[181] https://www.marriott.com/en-us/hotels/hnmwi-the-westin-maui-resort-and-spa-kaanapali/overview/?gclid=Cj0KCQiAn-2tBhDVARIsAGmStVlpt0EqJ4YNXWf5ic5MZZD7K4PZ4VYQW9DKepVFlmwCoYWwjJxTX0QaAieCEALw_wcB&gclsrc=aw.ds&cid=PAI_GLB0004YXD_GLE000BIM5_GLF000OETA

Hana-Maui Resort - Destination by Hyatt   	**Group:** Other 4-Stars in Hawaii **Brief Introduction:** Located in Hana town along the Road to Hana, a 64-mile scenic route winding along Maui's northern coast, the Hana-Maui Resort offers a secluded setting on Hawaii's eastern coast. The resort is situated close to renowned beaches, tropical forests, waterfalls, and the Pacific Ocean, providing guests with access to natural beauty and coastal attractions.[182] **Location:** 5031 Hana Hwy, Hana, HI 96713
Waikiki Monarch Hotel   	**Group:** Other Honolulu Hotels in different star ratings **Brief Introduction:** The Waikiki Monarch Hotel, situated at the gateway to Waikiki and a short distance from the Pacific Ocean, positions guests conveniently for exploring Oahu's attractions. It is well-located for those attending events at the Hawaii Convention Center, shopping at the Ala Moana Center, and visiting some of the world's most renowned beaches. On-site dining includes the Cream Pot Lounge, with additional dining options of various price ranges nearby. The hotel is within walking distance to Waikiki Beach, often cited as the birthplace of surfing. Guests of the hotel benefit from preferred access to island tours and attractions, thanks to its affiliation with Roberts Hawaii.[183] **Location:** 444 Niu St, Honolulu, HI 96815
Royal Grove Waikiki	**Group:** Other Honolulu Hotels in different star ratings **Brief Introduction:** Since its establishment in 1953 by a local family, Royal Grove Waikiki has been a notable

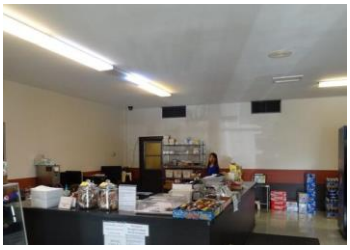
[182] https://www.hyatt.com/en-US/hotel/hawaii/hana-maui-resort/

[183] https://waikikimonarchhotel.com/hotel-specials/

 	presence in Waikiki and holds the distinction of being the state's first hotel owned and operated by a woman. The hotel, recognized for its iconic coral color, offers guests the opportunity to experience authentic local culture, from enjoying live Hawaiian music to exploring the downtown area or observing the sunset.[184] **Location:** 151 Uluniu Ave, Honolulu, HI 96815
Royal Kona Resort   	**Group:** Other Honolulu Hotels in different star ratings **Brief Introduction:** The Royal Kona Resort is set against a backdrop where ocean waves meet lava rock shores, surrounded by gardens filled with tropical flora. The location offers a blend of sea breezes and sunlight. The resort features an oceanfront lounge and bar known for its cuisine, alongside amenities such as a pool with cabana service. Cultural experiences include the Voyagers of the Pacific Luau, showcasing Hawaiian traditions. Guests have opportunities for outdoor adventures like volcano exploration and scuba diving with manta rays. Nearby, the seaside village of Kailua-Kona provides dining and shopping options. The resort also houses a spa, catering to those seeking relaxation. This setting aims to provide a comprehensive island experience on the Big Island of Hawaii.[185] **Location:** 75-5852 Ali'i Dr, Kailua-Kona, HI 96740
Kula Lodge 	**Group:** Other Honolulu Hotels in different star ratings **Brief Introduction:** Kula Lodge, situated at a 3,200-ft elevation in the mountains, provides Upcountry hospitality. Originally built as a private home in 1948 and known for hosting celebrities, it was transformed into a lodge with a restaurant in the 1950s, adding five guest lodges for tourists exploring Haleakala's slopes. Under

[184]https://www.royalgrovewaikiki.com/specials?gad_source=1&gclid=EAIaIQobChMIzPfmqLWLhAMVyB-tBh0XHAIoEAAYASAAEgI8PPD_BwE

[185] https://www.royalkona.com/

	the ownership of Fred Romanchak since 1984, the lodge has been enhanced with a terrace, an artisanal pizza oven, and gardens featuring local and exotic plants. The lounge serves breakfast and fine dining with a wide wine selection, including local varieties. The lodge's bar, the only one in Kula, offers signature cocktails with local ingredients. Accommodations include five lodges, and the on-site Kula Marketplace features gifts from over 200 local artists and artisans.[186] **Location:** 15200 Haleakala Hwy, Kula, HI 96790
Tip Top Motel, Cafe, And Bakery   	**Group:** Other Honolulu Hotels in different star ratings **Brief Introduction:** Founded in November 1916 by Denjiro Ota, the Tip Top Cafe & Bakery was established at the suggestion of the Lihue store manager to provide the local community with a place for breakfast and coffee. Initially starting as a cafe, it expanded to include a bakery that delivered bread across Kauai. In 1925, Denjiro's son, Mitchell Ota, took over and expanded the product range to include pastries, pies, and other baked goods, notably introducing a pancake recipe and the first macadamia nut cookie made in Hawaii.[187] **Location:** 3173 Akahi St, Lihue, HI 96766

By analyzing the description and geographical location of the hotel, it can be found that the reliability and validity test included samples from different geographical locations, different economic levels and average prices. It has good dispersion and can comprehensively cover the common resort hotels in Hawaii's tourism industry. type. Therefore, accept this sampling result and proceed with further scoring and analysis.

[186] https://kulalodge.com/

[187] https://www.tiptop-motel.com/about.html

6.2 Results

Based on the expert rating results and the BDIA-Index calculation formula, the BDIA-Scenarios, BDIA-Index, and Booking Prediction Score of the sample hotels were calculated for subsequent analysis in 4.2.

6.2.1 Descriptive Results

The mean ± standard deviation results of BDIA-Scenarios are classified and summarized according to the scenario group and the hotel group, and the results are presented in the form of a box plot. As can be seen from the picture, the BDIA of other four-star hotels in Hawaii is higher than that of ordinary hotels, both by scene and overall. In four-star hotels, the lobby is the space with the highest BDIA and uses the most indirect biophilic design elements; in ordinary hotels, the lounge scene uses the most.

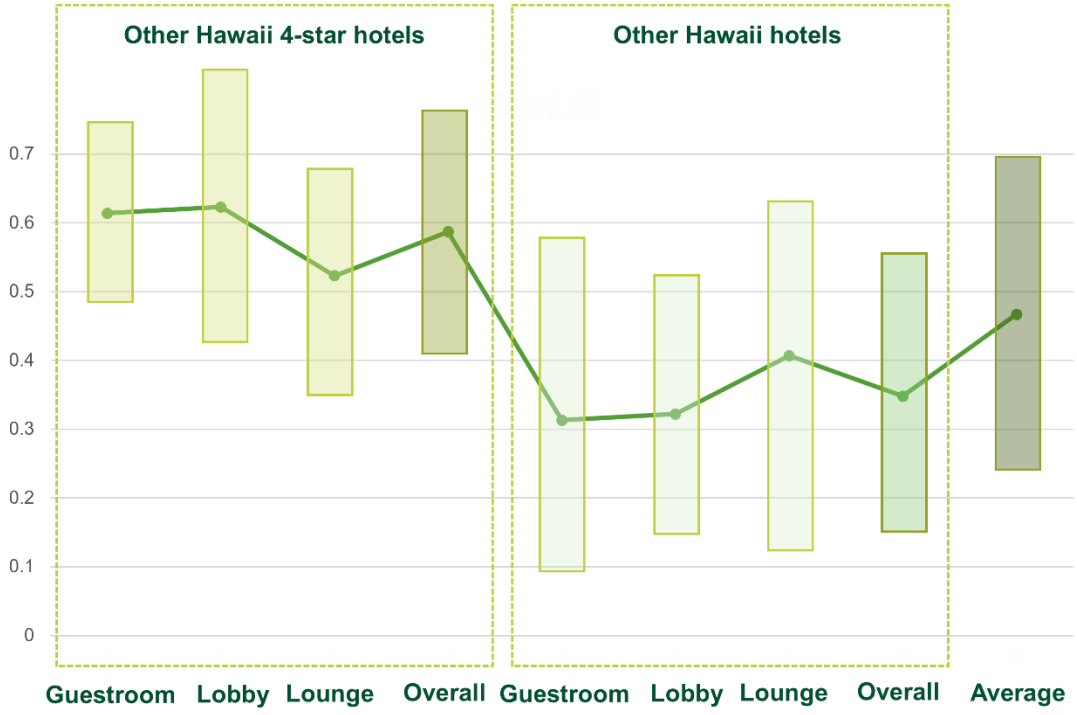


Figure 26 BDIA-Scenario Mean ± Standard deviation Result by Group

The average results of BDIA-Index are classified and summarized according to hotel numbers. The results are shown in the figure below. Among them, hotels

numbered 27-31 are four-star hotels, and 32-36 are ordinary hotels. It can be seen that there is not much difference in the BDIA scores within the four-star hotel group and the average value is basically stable at around 0.6, which is consistent with the test group. The gap between groups in the ordinary hotel group is larger, and the BDIA scores of different spaces within the same hotel group are also unstable. Although the BDIA-Index of most general hotel groups is low, some samples show higher BDIA, which may be due to their location in a nature reserve. Overall, four-star hotels showed higher factor consistency and higher BDIA levels, while ordinary hotels showed lower and more unstable levels.

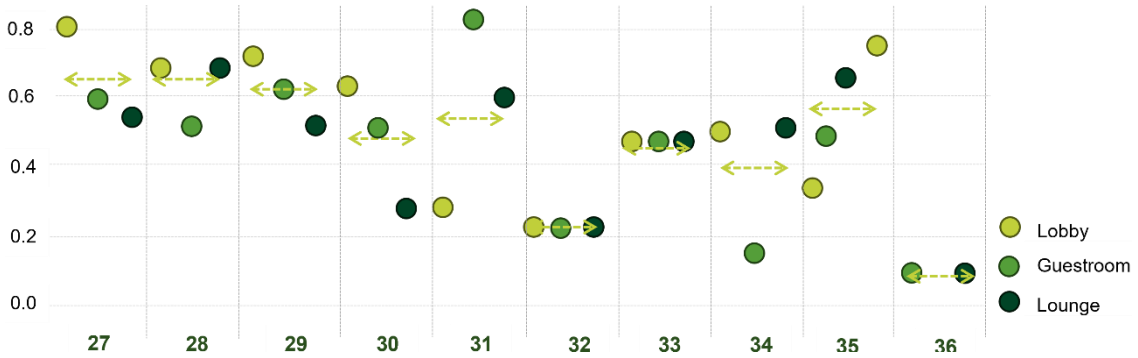


Figure 26 BDIA-Index and Scenarios Mean Result by Hotels Source: Author

The average of the total scores of the 29 BDIA items in the three scenarios were calculated for four-star hotels, ordinary hotels and waikiki four-star hotels in other areas of Hawaii. The results show that in almost all BDIA, four-star hotels score higher than ordinary hotels, except for A03 (Nature Image Decorations) and B05 (Natural materials Furnitures), which may be due to the fact that wooden floors and natural decorative hanging paintings are economical to implement. Higher decorative elements are therefore widely used in budget hotels, while in four-star hotels more expensive decorative elements such as decorative carpets or wood carvings can be used. Among all four-star hotels, for A (Nature Image) and D (SIMULATING NATURAL LIGHT AND AIR) scores, other regional groups are higher than the Waikiki group; E (Naturalistic shapes and forms), the Waikiki group is higher than other regional groups. Groups B (Natural materials), C (Natural colors), F (AGE, CHANGE,

AND THE PATINA OF TIME), and G (Natural geometries) have little difference or depend on specific factors. This shows that four-star hotels use almost all BDIA elements more than ordinary hotels, except for a small number of cost-effective decorative elements. Four-star hotels in the Waikiki area use significantly more natural forms of decoration and less use of decorative images, natural ventilation and other climate opening measures than hotels of the same level in other areas.

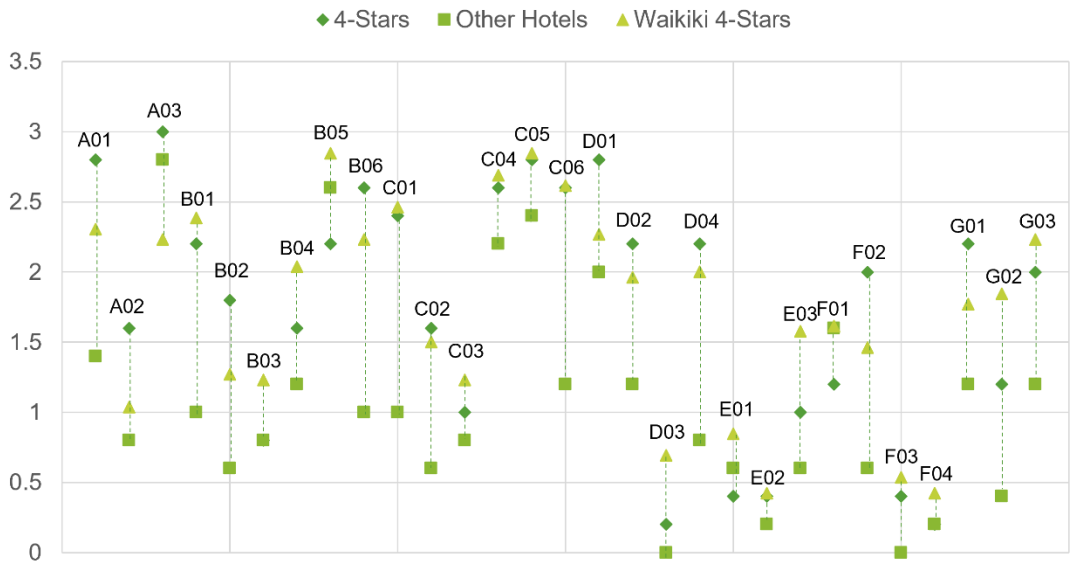


Figure 27 Average BDIA Scores by Hotel Type and Location

6.2.2 Correlation Analysis Results

Table 47. BDIA-Index Correlation Analysis Results for BDIA-Testing Hotels

	BOOKING RANKING PREDICTION	BOOKING SCORE PREDICTION	BDIA-INDEX
GOOGLE SCORE	-0.566	0.416	0.416
EXPEDIA SCORE	-0.583	0.659	0.659
OVERALL SCORE RANKING	**0.685***	**-0.679***	**-0.679***
OVERALL SCORE	-0.5	0.582	0.582
FACILITIES	-0.572	**0.706***	**0.706***
COMFORT	**-0.659***	**0.687***	**0.687***
VALUE OF MONEY	0.153	0.059	0.059
AVERAGE PRICE	**-0.689***	**0.673***	**0.673***

A correlation analysis was conducted between the BDIA-Index of the test hotel

and its various online rating results. It was found that the BDIA-Index showed a significant correlation with all scores except cost-effectiveness. Some results do not show significant correlation in the data because the included test sample size is small (n=10) and cannot achieve correlation significance below 0.6, but it can be inferred based on the significance that has been presented. There is a positive correlation between BDIA-Index and ratings on Google, Expedia, Booking and other platforms, which means that higher BDIA will significantly improve the ratings and customer reviews of online platforms, while also significantly improving the average price, comfort, and quality of facilities of the hotel. Perception. However, customers' perception of cost-effectiveness is not significantly related to BDIA. There is a significant negative correlation between BDIA-Index and hotel rankings, which means that the higher the BDIA, the higher the hotel ranking. BDIA-index can also be used to predict the rating ranking between hotels, and the resulting deviation value is within 3 places. This proves that BDIA-index is an efficient and significant predictor of perceived quality for hotels of different levels and geographical locations in Hawaii.

6.3 Discussion

6.3.1 BDIA in Hawaii Hotel Interior Design

The investigation into the deployment of the BDIA (Big Data Interior Analysis) within Hawaii's hotel sector, particularly focusing on four-star establishments in Waikiki, has illuminated significant trends in interior design quality and element consistency. Notably, four-star hotels exhibited a higher degree of element consistency and BDIA levels compared to more economically priced accommodations. This disparity underscores a broader trend in the hospitality industry, where higher-tier hotels leverage a more comprehensive array of BDIA elements to enhance guest experience, barring a few cost-effective decorative elements.

Interestingly, despite the overall trend for luxury accommodations to adopt a wide range of BDIA components, elements such as Nature Image Decorations (A03) and Natural Materials Furniture (B05) were also prevalent in economically priced

hotels. This suggests that these elements, due to their cost-effectiveness and the ambience they provide (e.g., wood flooring and natural décor paintings), have a universal appeal across different hotel categories. Additionally, four-star hotels in the Waikiki area were observed to utilize natural form decorations more extensively than their counterparts in other regions, indicating a regional preference that possibly aligns with local cultural or environmental factors. Conversely, there was a noted reduction in the use of decorative images, natural ventilation, and other climate-open measures, highlighting a potential area for further exploration in relation to guest preferences and sustainability considerations.

6.3.2 BDIA-Index Validation

The validation of the BDIA-Index through its correlation with online platform ratings from Google, Expedia, and Booking.com reveals its efficacy as a predictive marker for hotel quality perception. The positive correlation between BDIA scores and higher ratings on these platforms not only indicates that a greater use of BDIA elements can significantly improve online scores and customer reviews but also suggests an increase in perceived value, comfort, and facility quality. This correlation underscores the BDIA-Index's relevance across varying hotel categories and geographical locations within Hawaii, establishing it as a robust and significant predictor of quality perception.

However, it is important to note that the perception of value for money did not show a significant correlation with BDIA scores. This finding suggests that while customers may appreciate higher quality and comfort levels, their perceptions of value are influenced by additional factors beyond interior design quality. Moreover, the observation that hotels with higher BDIA scores tend to rank higher in listings further validates the index as a useful tool for enhancing hotel competitiveness. By demonstrating the predictive validity of the BDIA-Index for hotel spatial quality perception, this study not only contributes to the academic discourse on interior design and hospitality management but also offers practical insights for hoteliers aiming to improve guest experience and market positioning through strategic interior

design investments.

6.4 Conclusion

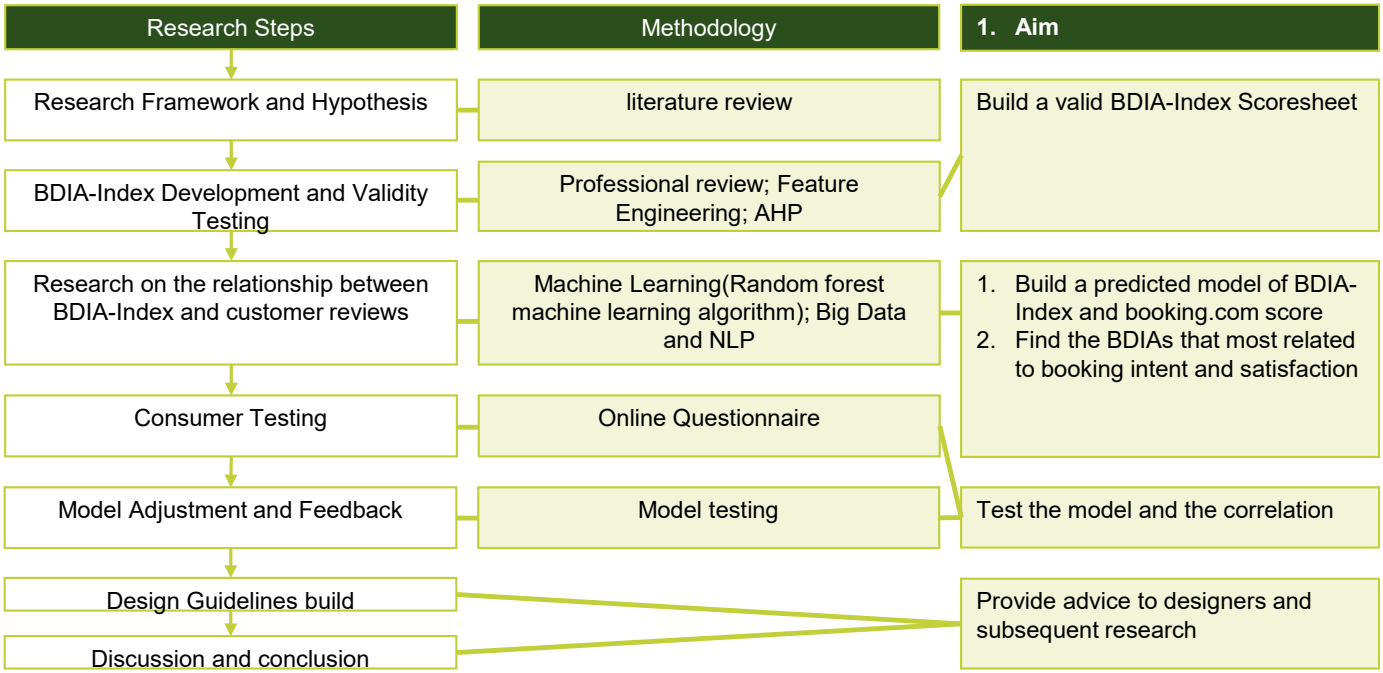
In this chapter, the research validated the BDIA-Index prediction model established through machine learning and expanded the dataset to reduce the impact of factors such as location, consumption levels, and hotel amenities on spatial evaluation. By randomly sampling and scoring BDIA for 10 hotels from different locations and star ratings in Hawaii, the study identified the application characteristics of BDIA in Hawaii hotel interior design and verified the effectiveness of the BDIA-Index as a predictor of hotel space quality perception. This indicates that by improving BDIA levels, hotels can significantly enhance their online ratings, customer reviews, and perceptions of average price and facility quality. By validating the predictive effectiveness of the BDIA-Index, this research provides hotel operators with a practical tool, assisting designers in improving guest experience and market positioning through strategic interior design investments, which will be discussed in depth in the next chapter. The establishment of the BDIA-Index offers a new quantitative reference model for biophilic design principles in the field of interior design and hotel management, providing hotel operators with a practical tool to enhance guest experience and market positioning through strategic interior design investments, contributing to the competitiveness and customer satisfaction of hotels through interior design.

CHAPTER 7. APPLICATION HANDBOOK FOR HOTEL DESIGNERS

What is BDIA?

The Biophilic Design Indirect Attributes (BDIA) model is a quantitative assessment tool for evaluating the use of indirect biophilic attributes in hotel interiors, using a binary scoring system to quantify the indirect experience of nature. Developed through NLP analysis from more than 35,000 comments in Waikiki 4-star hotel, it integrates principles from the Design Quality Indicator (DQI) and the Biophilic Interior Design Matrix (BID-M) to offer a simple approach for predicting hotel evaluation scores on booking platform.

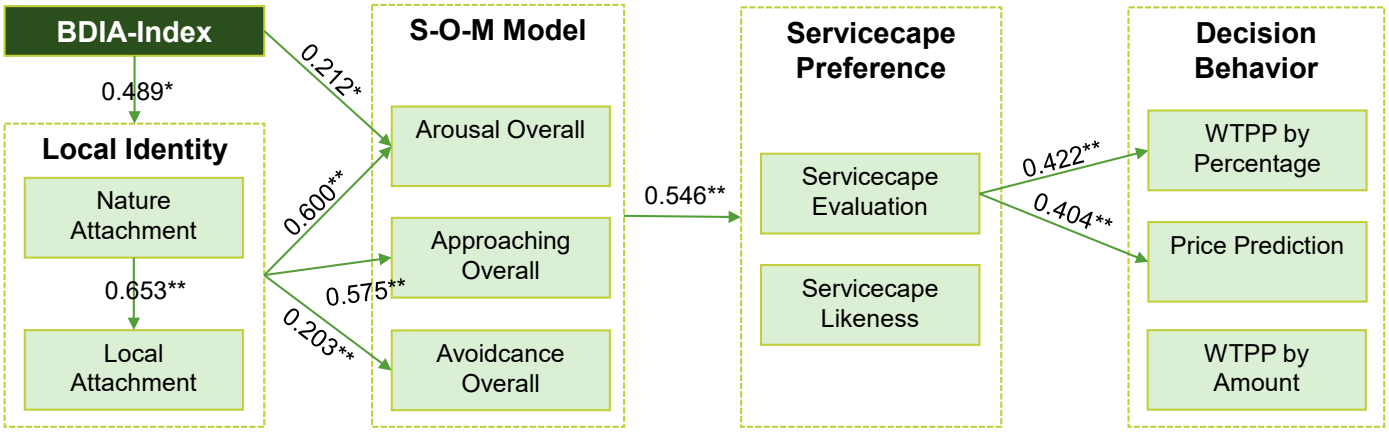
Fig. BDIA development process



How BDIA influence the customer's perception and behavior?

High BDIA directly increases customers' perception and sense of belonging to the "Hawaiian" degree of the design, as well as the degree to which they are aroused by the environment, and indirectly increases customers' willingness to approach the facilities and their love and evaluation of the facilities, thus affecting their willingness to pay.

Fig. BDIA's mechanism of action



Who can use this tool?

Architectural designers, interior designers, hotel managers, and owners can all predict the future performance of the project by referring to the design details.

What is included in BDIA? How can I apply?

BDIA contains 7 categories of measurement indicators from A to G. Each category is used to measure the proportion of an indirect biophilic design quality in the interior and consists of 3-6 subdivision indicators. These categories include Images of Nature, Natural materials, Natural colors, Simulating natural light and air, Naturalistic shapes and forms, Age/change/the patina of time, and Natural geometries. For a detailed description and usage introduction of each category and its subdivision indicators, please refer to the following article.

How to use BDIA in predicting my hotel's Booking.com ranking?

BDIA-Index is a tool that converts BDIA into a predicted Booking.com Overall Score. It contains 16 detailed questions about biophilic design, which you can find in the appendix. For each question, you need to evaluate your hotel in three scenarios (standard room, lobby, lounge) and answer yes (1) / not present (0). For definitions and detailed descriptions of elements. The scores from this table will be converted into scores using the following formula, with an explanatory power of 50.4%:

BDIA-Scenarios = SUM(Attribute*Weight)

For example,

BDIA-Guestroom=0.056*A01+0.03*A03+0.05*B01+0.06*B03+0.054*B04+0.103*B06+0.045*C01+0.052*C03+0.05*D01+0.061*D03+0.07*E03+0.03*F01+0.048*F02+0.18*G01+0.068*G02+0.042*G03

BDIA-Index = 0.368*Guestroom+0.285*Loundge+0.347*Lobby

Booking.com Overall Score Prediction =7.292 + 1.854*BDIA-Index

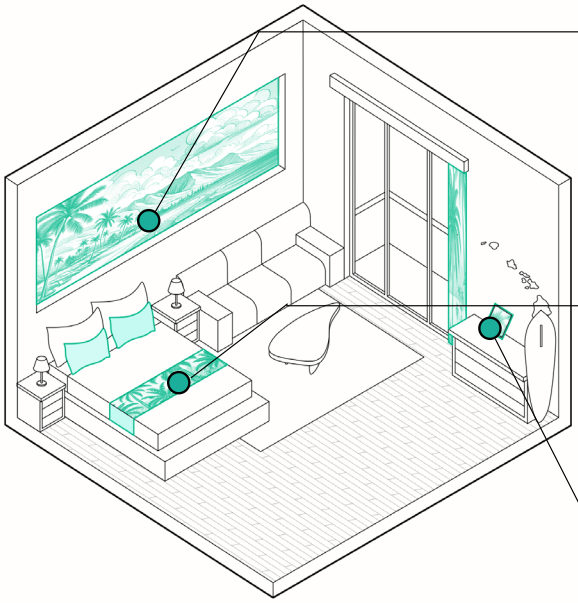
Glossary of terms

- **Recommand Level:** The recommended usage level is obtained by comprehensive analysis of customer questionnaires, machine learning, and big data analysis.
- **Frequency:** Proportion used in existing hotels.
- **↑/↓:** It will cause the corresponding indicator to rise(↑) or fall(↓).
- **Design Quality Perception:** Customers' perception of the quality of a certain design quality discovered through big data analysis. For example, "Form Design Quality Perception" refers to the quality perception of the design quality of "Form". The higher the perception, the more pleasant the hotel is perceived by customers, and is associated with more positive emotions and positive reviews.
- **Arousal Level:** The degree to which customers are psychologically aroused by certain factors. The higher the level of arousal, the more likely it is to produce positive environmental perceptions and experiences.
- **Local Identity Level:** The degree that customers think the hotel design style represents "Hawaii". The higher the perception, the more positive the customer's evaluation.
- **Servicecape Perception Level:** Customers' comprehensive perception of space satisfaction, luxury, and comfort.
- **WTPP Level:** The willingness of customers to pay an additional premium for this design.



A. Images of Nature

The image and representation of nature in the built environment—plants, animals, landscapes, water, geological features—can be both emotionally and intellectually satisfying. These images can occur through the use of photographs, paintings, sculptures, murals, video, computer simulations, and other representational means. Single or isolated images of nature typically exert little impact. Representational expressions of nature should be repeated, thematic, and abundant.



- Strong positive predictors
- Apotential design standard for high-end hotels
- Can continue to be used in future designs
- Can refer to existing design projects

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



Wall paintings or decorative paintings contain direct natural scenes (such as landscape photos).

Frequency
↑ Form Design Quality Perception
↑ Arousal Level
↑ Willing to Pay Premium



Carpets, pillows, curtains and other fabrics contain direct natural scenes (such as cartoon cat pillows)

Frequency
↑ Space Design Quality Perception



Use decorations with natural forms (such as ceramic rockeries).

Frequency
↑ Form Design Quality Perception

Examples



Aston Waikiki Beach Tower / 9.4
1. Using natural images in rooms is associated with positive perceptions of design quality
2. Natural images are the most commonly used BDIA by low-cost hotels



Queen Kapiolani Hotel/ 8.4
Multiple natural images have stronger effects than single locations

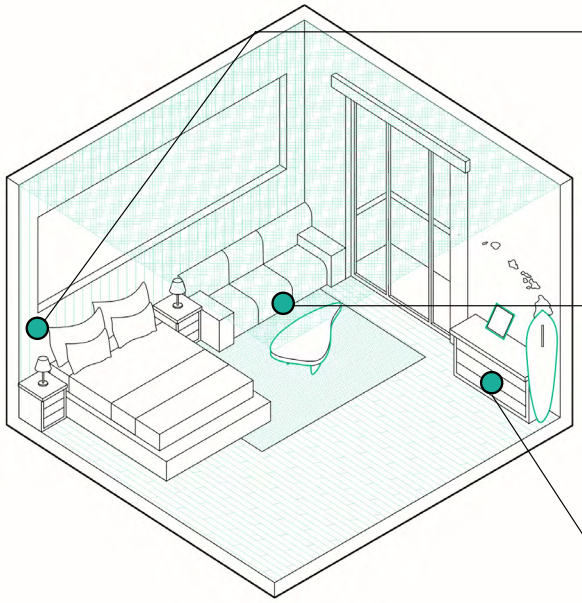


Sheraton Waikiki Beach Resort / 8.6
Paintings/artwork are associated with positive perceptions, prints/prints may be associated with negative perceptions



B. Natural materials

The transformation of materials from nature frequently elicits positive visual and tactile responses, which few artificial materials can duplicate. Prominent natural building and decorative materials include wood, stone, wool, cotton, and leather, used in a wide array of products, furnishings, fabrics, and other interior and exterior designs.



- may negatively correlated to quality perception
- associate with a lower level or outdated
- Widly used in all level hotels

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



Background walls, columns and other hard wall decorations are made of natural materials.

Frequency
↓ Decoration Design Quality Perception
↑ Servicecape Level



The ceiling is made of natural materials.

Frequency
↑ Space Design Quality Perception



Furnitures are made of natural materials.

Frequency
↑ Decoration Design Quality Perception

Examples



Hana-Maui Resort - Destination by Hyatt / 8.5
Natural material flooring (such as wooden flooring) will cause a decrease in quality perception and is associated with low cost and obsolescence. Carpet and slate floors will cause an increase in quality perception.



Halepuna Waikiki by Halekulani/ 9.0
Natural material ceilings are significantly associated with positive quality perceptions

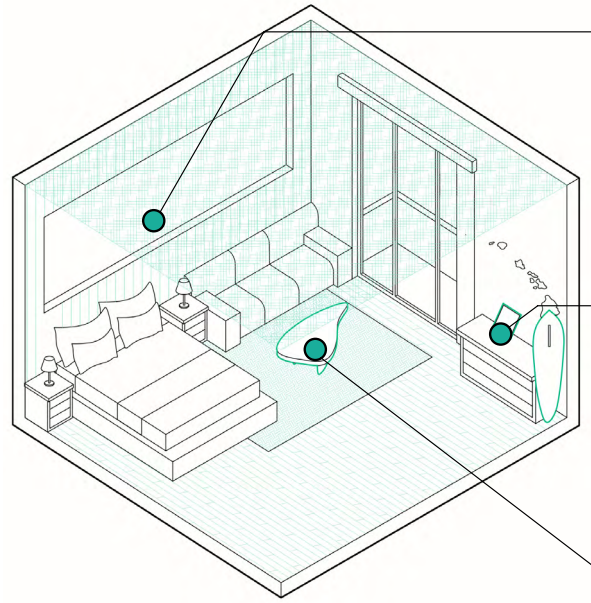


OUTRIGGER Waikiki Beach Resort / 8.7
Almost all Hawaii hotels (92.3%) use wooden furniture and decorations, which has become a design standard



C. Natural colors

The effective biophilic application of color should generally favor muted “earth” tones characteristic of soil, rock, and plants. The use of bright colors should be cautiously applied, and emphasize such appealing environmental forms as flowers, sunsets and sunups, rainbows, and certain plants and animals. The occurrence of highly artificial, contrasting, and “vibrating” colors should be avoided.



- may negatively correlated to quality perception
- associate with a lower level or outdated
- Widly used in all level hotels

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



The ceiling is made of natural colors.

Frequency
↑ Arousal Level
↑ Local Identity Level



Decorations are made of natural colors.

Frequency
↑ Arousal Level



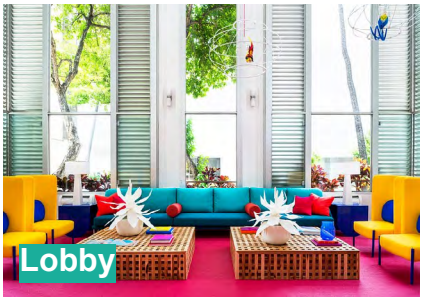
Furnitures are made of natural colors.

Frequency
↑ Arousal Level
↑Servicecape Perception Level

Examples



Prince Waikiki / 8.8
1. Rooms rated best in soft beige or Morandi colors
2. Intense colors are associated with lower price perceptions and higher discomfort perceptions



Shoreline Hotel Waikiki / 8.1
1. The use of strong colors in the lobby is associated with positive “Vibe” and “Atmosphere” ratings
2. The use of strong colors in public spaces increases perceived spatial quality and WTPP

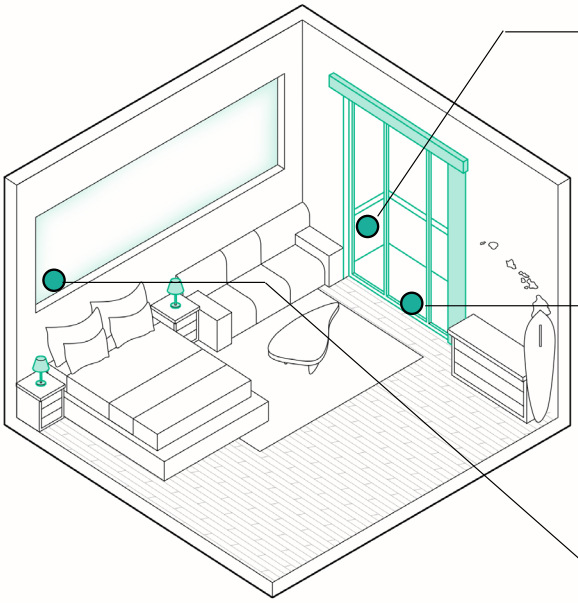


Marriott's Kaua'i Beach Club / 8.4
Using local color matching can increase the perception of "Local identity", thereby enhancing quality perception and booking intention



D. Simulating natural light and air

Indoor lighting and processed air have been made possible by advances in building technology and construction. The trade-of has often been the occurrence of static conditions that can be physically and psychologically debilitating. Artifcial light can be designed to mimic the spectral and dynamic qualities of natural light. Processed air can also simulate qualities of natural ventilation through variations in airfow, temperature, humidity and barometric pressure.



- may not directly correlated to quality perception
- Widly used in high-end hotels
- Balcony is postively related to comments

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



Adjustable, openable sash windows allow for natural ventilation.

Frequency

↑ Form Design Quality Perception



Adjustable interior or exterior sun visors or blinds, or balconies.

Frequency



Use indirect ambient lighting.

Frequency

Examples



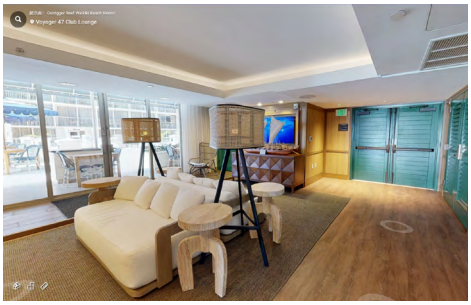
Sheraton Waikiki Beach Resort / 8.6

1. Balcony is a postive element, but it may also be related to thermal discomfort.
2. The direction of the light source is irrelevant to the evaluation.

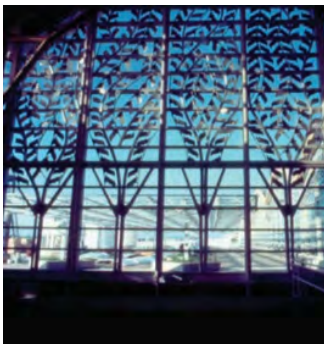


Halepuna Waikiki by Halekulani/ 9.0

Whether public spaces can be ventilated is irrelevant to the evaluation.

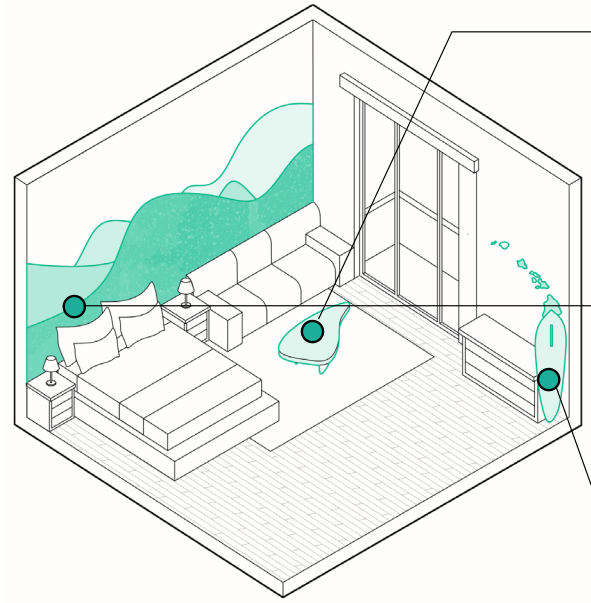


OUTRIGGER Reef Waikiki Beach Resort / 8.7



E. Naturalistic shapes and forms

The experience of shapes and forms characteristic of the natural world can be especially appealing. These naturalistic forms can be extraordinarily diverse from the leaf-like patterns found on columns, the shapes of plants on building facades, to animal facsimiles woven into fabrics and coverings. The occurrence of naturalistic shapes and forms can transform a static space into one that possesses the dynamic and ambient qualities of a living system.



- Indirectly correlated to quality perception
- Rarely used in hotel design

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



The furniture adopts irregular natural forms.

Frequency

↓ Form Design Quality Perception
↑ Style Design Quality Perception
↑ Arousal Level
↑ Serviceape Perception



Hard decoration (walls, ceilings, etc.) adopts irregular natural forms.

Frequency

↑ Arousal Level
↑ Willing to Pay Premium



The decorations adopt irregular natural forms.

Frequency

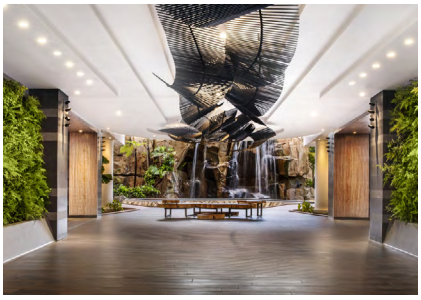
↑ Arousal Level
↑ Local Identity Level
↑ Serviceape Perception

Examples



The Westin Maui Resort & Spa, Ka'anapali / 9.4

Naturalistic shapes and forms in guestroom is either associated with negative outcome scores or not associated with results in all analyses, and should be avoided in design as much as possible.



The Westin Maui Resort & Spa, Ka'anapali / 9.2

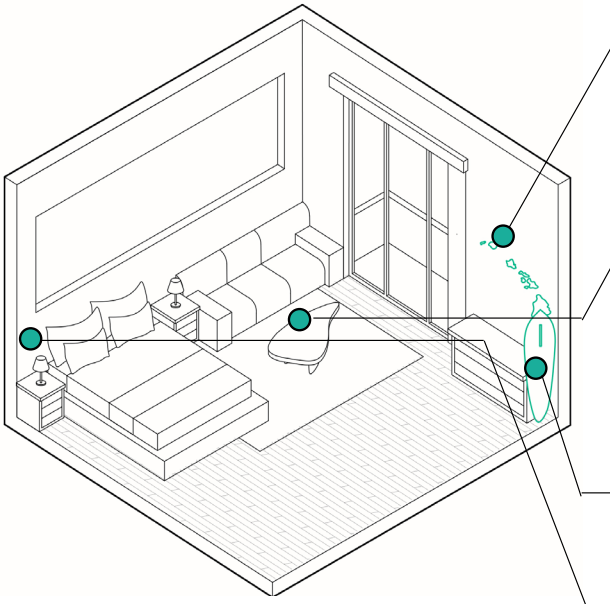


Hyatt Centric Waikiki Beach / 8.6



F. Age, change, and the patina of time

People respond positively to these dynamic forces and the associated patina of time, revealing nature's capacity to respond adaptively to ever-changing conditions. These dynamic tendencies are often most satisfying when balanced by the complementary qualities of unity and stability. Change and a patina of time can be achieved through such design strategies as naturally aging materials, weathering, a sense of the passage of time, and in other ways.



- significantly correlation with positive rankings
- a new design potential opportunity
- can be more actively adopted in future designs

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



There are nostalgic elements in the decoration style (such as 90s posters).
Frequency
↑ Atmosphere Design Quality Perception



The decorative style includes nostalgic craftsmanship (such as bamboo weaving and hand binding).
Frequency
↑ Design/Atmosphere/Decoration Design Quality Perception
↑ Arousal Level
↑ Local Identity Level
↑ Serviceape Perception



Use recycled materials, furniture or decor.
Frequency
↑ Design/Atmosphere/Form Design Quality Perception
↑ Local Identity Level



Use materials that will show natural weathering or aging (such as copper).
Frequency
↑ Design/Atmosphere/Decoration/Style Design Quality Perception
↑ Local Identity Level

Examples



Wayfinder Waikiki / 8.5
If the design is considered local, retro or nostalgic style, it will increase its positive evaluation. But if the customer think it's outdated or worn, their perception will drop.



Wayfinder Waikiki / 8.5
Local Identity is related to higher Booking scores, higher Quantitative Feeling And Evaluation, and more positive consumption decisions.

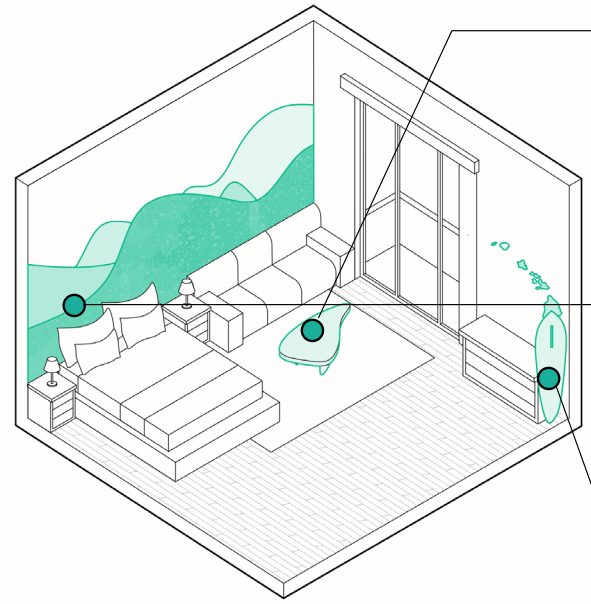


Wayfinder Waikiki / 8.5
Age, change, and the patina of time are the strongest predictors of Postive Comments Emotional Tendency. It may improve overall perception and evaluation by enhancing customers' local identity.



G.Natural geometries

The satisfying experience of nature can also be revealed through imaginative and fantastic depictions. For example, the “wings” of the Sydney Opera House suggest the qualities of a bird; Notre Dame’s stained glass windows, a rose-like fower; while, the skyline of some cities mimic the vertical heterogeneity of a forest. None of these designs actually occurs in nature, but they all draw from design principles and characteristics of the natural world.



- Strong positive predictors
- Apotential design standard for high-end hotels
- Can continue to be used in future designs
- Can refer to existing design projects

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



The furniture adopts abstract and geometric patterns that imitate nature.

Frequency
↑ Decoration Design Quality Perception
↑ Arousal Level



Hard decoration (walls, ceilings, etc.) adopts abstract and geometric patterns that imitate nature.

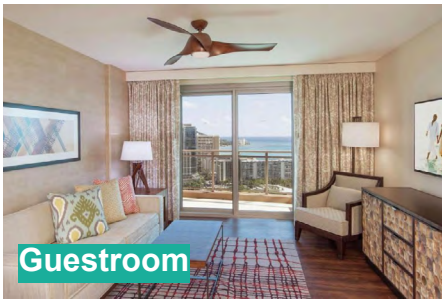
Frequency
↑Form Design Quality Perception
↑ Arousal Level



Decorations and soft furnishings adopt abstract and geometric patterns that imitate nature.

Frequency
↑Form Design Quality Perception
↑ Serviceape Perception

Examples



Hilton Grand Vacations Club Grand Waikikian Honolulu / 8.8
Using room decorations with natural textures can significantly improve perceived design quality and environmental satisfaction



Aston Waikiki Beach Tower / 9.4
Can be harmoniously combined with BDIA: natural images



The Surfjack Hotel & Swim Club / 8.7

Appendix: BDIA Scoresheet

You can access the automatic calculation form by following the link. Please copy it into your Excel for use.

https://docs.google.com/spreadsheets/d/137NjsYtomJauL6hnKsDD50ldK0LIHkyMszzRJh8lZE8/edit?usp=sharing

BDIA		Weight	Guestroom	Lobby	Lounge
Images of nature					
A01	Wall paintings or decorative paintings contain direct natural scenes (such as landscape photos).	0.056			
A03	Use decorations with natural forms (such as ceramic rockeries).	0.03			
Natural materials					
B01	Background walls, columns and other hard wall decorations are made of natural materials.	0.05			
B03	The ceiling is made of natural materials.	0.06			
B04	The floors are made of natural materials.	0.054			
B06	Decorations are made of natural materials.	0.103			
Natural colors					
C01	The hard decoration of the background wall, pillars and other walls are in natural colors.	0.045			
C03	The ceilings are in natural colours.	0.052			
Simulating natural light and air					
D01	Adjustable sash windows allow for natural ventilation.	0.05			
D03	Use indirect ambient light.	0.061			
Naturalistic shapes and forms					
E03	The decorations adopt irregular natural forms.	0.07			
Age, change, and the patina of time					
F01	There are nostalgic elements in the decoration style (such as 90s posters).	0.03			
F02	The decorative style includes nostalgic craftsmanship (such as bamboo weaving and hand binding).	0.048			
Natural geometries					
G01	The furniture adopts abstract and geometric patterns that imitate nature.	0.18			
G02	Hard decoration (walls, ceilings, etc.) adopts abstract and geometric patterns that imitate nature.	0.068			
G03	Decorations and soft furnishings adopt abstract and geometric patterns that imitate nature.	0.042			
BDIA-Scenarios					
BDIA-Index					
Booking.com Overall Score Prediction					

BDIA-Scenarios = SUM(Attribute*Weight)

Example: BDIA-Guestroom=0.056*A01+0.03*A03+0.05*B01+0.06*B03+0.054*B04+0.103*B06+0.045*C01+0.052*C03+0.05*D01+0.061*D03+0.07*E03+0.03*F01+0.048*F02+0.18*G01+0.068*G02+0.042*G03

BDIA-Index = 0.368*Guestroom+0.285*Loundge+0.347*Lobby

Booking.com Overall Score Prediction =7.292 + 1.854*BDIA-Index

Appendix: BDIA Checklist

This checklist helps you quickly check how each BDIA relates to perceived design quality. For example, you can examine which BDIA contributes to the perceived quality of a specific space (e.g. lobby, guest room).

Attributes	Correlation	Guestroom	Lobby	Lounge	Design	Atmosphere	Decoration	Form	Furniture	Space	Style	S-O-M Model	Local Identity	Servicecape	WTPP
F02	●	◎	◎	◎	●	●	●	◎	◎	◎	◎	●	●	●	◎
F04	●	◎	●	◎	●	●	●	◎	◎	◎	●	◎	●	◎	◎
G03	●	◎	◎	◎	◎	◎	●	◎	●	◎	◎	◎	◎	●	◎
A01	●	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	●	◎	◎	●
A03	●	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎
B06	●	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎	◎
E03	●	●	◎	◎	◎	◎	◎	◎	◎	◎	◎	●	●	●	◎
F01	●	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎	◎	◎
F03	●	◎	◎	◎	●	●	◎	●	◎	◎	◎	◎	●	◎	◎
G01	●	◎	◎	○	◎	◎	●	◎	◎	◎	◎	●	◎	◎	◎
B03	●	◎	◎	◎	◎	○	◎	◎	◎	●	◎	◎	◎	◎	◎
C03	●	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	●	●	◎	◎
C01	●	◎	○	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎	◎
D01	●	◎	◎	○	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎
G02	●	○	◎	◎	◎	◎	◎	●	○	◎	◎	●	◎	◎	◎
A02	●	◎	◎	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎
C05	●	◎	●	◎	◎	◎	◎	◎	◎	◎	◎	●	◎	◎	◎
C02	●	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎
D03	●	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎
B02	◎	◎	◎	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎
B05	●	◎	◎	◎	◎	◎	●	◎	○	◎	◎	◎	◎	◎	◎
B01	●	◎	○	◎	◎	◎	○	◎	◎	◎	◎	◎	◎	●	◎
C06	●	●	○	◎	◎	◎	◎	◎	◎	◎	◎	●	◎	●	◎
D02	◎	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎
E01	●	◎	◎	○	◎	◎	◎	○	◎	◎	●	●	◎	●	◎
D04	●	◎	◎	◎	◎	◎	◎	◎	○	◎	◎	◎	◎	◎	◎
C04	●	●	◎	◎	◎	◎	○	◎	◎	◎	◎	◎	◎	◎	◎
B04	●	◎	○	○	○	○	○	◎	◎	◎	○	◎	◎	◎	◎
E02	◎	◎	◎	◎	◎	◎	◎	◎	○	◎	◎	●	◎	◎	●

- **Scenarios (Geustroom | Lobby | Lounge):** The correlation between customer positive evaluation and specific scenarios.
- **Quality Perception (Design | Atmosphere | Decoration | Form | Furniture | Space | Style):** Customers' perception of the quality of a certain design quality discovered through big data analysis. For example, "Form" refers to the quality perception of the design quality of Form.
- **S-O-M Model:** The degree to which customers are psychologically aroused by certain factors. The higher the level of arousal, the more likely it is to produce positive environmental perceptions and experiences.
- **Local Identity:** The degree that customers think the hotel design style represents "Hawaii".
- **Servicecape:** Customers' comprehensive perception of space satisfaction, luxury, and comfort.
- **WTPP:** The willingness of customers to pay an additional premium for this design.

○ Negatively correlated with the analysis results
◎ No correlation with the analysis results
● Positively correlated with the analysis results

CHAPTER 8. CONCLUSION

8.1 BDIA Results

8.1.1 BDIA in Hotel Evaluation

Local Identity is related to higher Booking scores, higher Quantitative Feeling And Evaluation, and more positive consumption decisions. In both real-life situations and controlled experimental groups, the use of Native Hawaiian-inspired designs and vintage-style craftsmanship is associated with higher ratings or perceptions of quality, as well as with lighting patterns and architectural details such as window frames. Color related BDIA cannot have a significant impact on quality perception.

F02 (Nostalgic Craftsmanship), F04 (Weathered Materials), and G03 (Natural geometries Furnishing) showed significant correlation with positive results in each analysis and were strong predictors. At the same time, in the current design There are fewer occurrences (Mean=0.34), which is not a mainstream design style. It is a new design potential opportunity and can be more actively adopted in future designs.

Some BDIAs, such as A (Nature Image) and G (Natural geometries), have shown significant correlation with positive results in multiple analyses, and are strong predictors. At the same time, they appear more often in the current design (Mean > 0.5), has become a potential design standard for local hotels and can continue to be used in future designs and can refer to existing design plans.

Some design elements appear to be less clearly associated with more positive design outcomes. Most of these elements are smaller in size or occupy a smaller footprint and have little impact on the overall spatial perception. Such as natural material window frames and furniture (B02, B05), natural color furniture and decoration (C05, C06).

Some BDIA results show a negative correlation, such as floors with natural colors and materials (B04, C04). The NLP Attributes analysis results show that this may be because customers associate this type of flooring with a lower grade or outdated According to the association, hotels with higher scores commonly use materials such as industrial carpets or slate. E02 (Naturalistic shapes and forms

Finishing) is either associated with negative outcome scores or not associated with results in all analyses, and should be avoided in design as much as possible.

However, the explanatory power of BDIA-Index to consumption intention is not significant in all tests. BDIA-Index may not be a direct factor that can influence WTPP and Price Perception.

8.1.2 Possible impact mechanism of BDIA

The results of the path analysis verified and revised the hypothesis model established based on the literature research results, and established a mechanism model of the impact of BDIA on participants' consumption decisions. BDIA directly affects participants' arousal levels and indirectly affects participants' approach or avoidance responses by affecting Local Identity. The S-O-M Model will affect participants' quality perception and satisfaction with the space. Quality perception will affect consumers' decision-making and behavior, and they will accept prices higher than the budget or market price. This proves that the impact of BDIA on consumer decision-making is indirect and needs to be achieved through intermediary effects.

8.1.3 BDIA in Hawaii Hotel Interior Design

The investigation into the deployment of the BDIA within Hawaii's hotel sector, particularly focusing on four-star establishments in Waikiki, has illuminated significant trends in interior design quality and element consistency. Notably, four-star hotels exhibited a higher degree of element consistency and BDIA levels compared to more economically priced accommodations. This disparity underscores a broader trend in the hospitality industry, where higher-tier hotels leverage a more comprehensive array of BDIA elements to enhance guest experience, barring a few cost-effective decorative elements.

Interestingly, despite the overall trend for luxury accommodations to adopt a wide range of BDIA components, elements such as Nature Image Decorations (A03) and Natural Materials Furniture (B05) were also prevalent in economically priced hotels. This suggests that these elements, due to their cost-effectiveness and the

ambience they provide (e.g., wood flooring and natural décor paintings), have a universal appeal across different hotel categories. Additionally, four-star hotels in the Waikiki area were observed to utilize natural form decorations more extensively than their counterparts in other regions, indicating a regional preference that possibly aligns with local cultural or environmental factors. Conversely, there was a noted reduction in the use of decorative images, natural ventilation, and other climate-open measures, highlighting a potential area for further exploration in relation to guest preferences and sustainability considerations.

8.1.4 BDIA-Index as a valid prediction tool

The validation of the BDIA-Index through its correlation with online platform ratings from Google, Expedia, and Booking.com reveals its efficacy as a predictive marker for hotel quality perception. The positive correlation between BDIA scores and higher ratings on these platforms not only indicates that a greater use of BDIA elements can significantly improve online scores and customer reviews but also suggests an increase in perceived value, comfort, and facility quality. This correlation underscores the BDIA-Index's relevance across varying hotel categories and geographical locations within Hawaii, establishing it as a robust and significant predictor of quality perception.

However, it is important to note that the perception of value for money did not show a significant correlation with BDIA scores. This finding suggests that while customers may appreciate higher quality and comfort levels, their perceptions of value are influenced by additional factors beyond interior design quality. Moreover, the observation that hotels with higher BDIA scores tend to rank higher in listings further validates the index as a useful tool for enhancing hotel competitiveness. By demonstrating the predictive validity of the BDIA-Index for hotel spatial quality perception, this study not only contributes to the academic discourse on interior design and hospitality management but also offers practical insights for hoteliers aiming to improve guest experience and market positioning through strategic interior design investments.

8.2 Research Significance and Importance

This study explores the application of indirect biophilic design in hotel interior design by integrating multiple methods, aiming to provide valuable insights for hotel designers.

(1) Promote biophilic design concepts

This study promotes the application and popularization of biophilic design concepts in hotel design practice by quantitatively evaluating the application of indirect biophilic elements. This helps raise public and industry awareness of biophilic design and its importance in improving quality of life and promoting environmental sustainability.

(2) Application of interdisciplinary methods in design research

This study combines methods and theories from multiple disciplines such as interior design, environmental psychology, and consumer behavior, demonstrating the value of interdisciplinary research. Through this collaboration, research can delve into the multidimensional effects of biophilic design from different perspectives, providing rich insights for future innovation and practice.

(3) Establishing a quantitative assessment framework involves

By establishing and improving the BDIA evaluation framework, this study not only provides a new quantitative tool for the field of hotel interior design, but also encourages designers to adopt scientific methods to innovate design practices. This quantitative approach helps designers more accurately evaluate and implement indirect biophilic design elements, thereby promoting innovation and development in quantitative design and design research. BDIA tools help industry professionals quantify the eco-efficiency of designs, ensuring design decisions are not only based on aesthetics but also take into account environmental and mental health considerations.

(4) Provide design and marketing strategy advice

The study provides an in-depth look at the effectiveness of indirect biophilic design elements in three interior spaces of the hotel using a variety of technical

methods, and provides practical suggestions in the form of design guidelines to help hotels enhance their market competitiveness through improved interior design. . These insights help hoteliers identify and implement differentiating elements in design to attract more customers and increase customer loyalty. These strategies not only enhance a hotel's physical environment but also serve as marketing tools to attract customers through design features that emphasize wellness and natural connections.

(5) Lay the foundation for subsequent research

The technical framework and evaluation model established in this study provide a reusable basis for subsequent research in related fields. This not only promotes the accumulation and dissemination of knowledge, but also provides future researchers with tools and methods to further explore and expand the applications of biophilic design.

(6) Promote sustainable development

This study supports the idea of sustainability by emphasizing the importance of natural elements in interior design. This design approach not only focuses on improving human well-being, but also considers reducing the environmental footprint, encouraging designers and owners to adopt more responsible and sustainable design practices.

8.3 Limitations and Futural Studies

8.3.1 Limitations

This study may have the following shortcomings, which may lead to deviations between the research data and the real situation, and should be optimized and explored in subsequent research:

(1) **Regional limitations**

Since this study mainly focuses on hotels in Hawaii, although hotels in Hawaii with different geographical locations and star ratings were included in the sample as much as possible, non-Hawaiian hotel samples were not used, so they may not fully represent hotels in different regions and star ratings. Hotel interior design situation.

(2) **Data Collection Methods**

Although participants were as demographically diverse as possible and pre-qualified, the online questionnaires and actual hotel customer feedback used may still be biased as they rely on participants' subjective evaluations and Participation is voluntary.

(3) **Comprehensiveness of the BDIA index**

Although the BDIA-Index provides a quantitative evaluation framework, due to limitations in time and data volume, this study focused on indirect natural experience elements and interior design, and failed to cover all related biophilia Design elements.

(4) **Subjectivity of the evaluation framework**

Although reliability and validity verification is used to avoid the subjective experience of raters from affecting the scoring, the development of BDIA-Index still involves expert review, which may bring subjective judgment and affect the objectivity of the evaluation.

(5) Limitations of the technical approach

The machine learning and NLP techniques used may have their limitations when dealing with complex natural language and scoring. The NLP sentiment analysis tool used in the study was not specifically designed for hotel reviews or biophilia, so it may have limitations; the NLP toolkit designed to quantify BDIA may not fully cover all related words.

(6) Possible effects of other variables

Due to time and funding constraints, some possible indirect influencing variables were not considered in this study. Research may not fully consider the differences in customer perceptions of biophilic design in different cultural backgrounds, the long-term impact of indirect biophilic design elements on customer experience, and fail to evaluate the cost-effectiveness of implementing indirect biophilic design elements.

8.3.2 Futural Studies

Based on the above shortcomings, future research should strive to expand the

sample scope to include more regions and hotels with different star ratings to enhance the broad applicability and generalizability of the study. At the same time, diversified data collection methods can be used, such as face-to-face interviews or field observations, which can reduce the bias of relying on participants' self-reported data, thus improving the accuracy and reliability of research results. In-depth analysis and improvement of BDIA-Index, exploring the inclusion of more biophilic design elements, while developing standardized and objective evaluation methods to reduce the impact of subjective judgment. Utilizing the latest artificial intelligence technologies, such as deep learning, can increase the depth and accuracy of data analysis. Consider differences in perceptions of biophilic design across cultural contexts, conduct long-term follow-up studies to assess the ongoing impact of these design elements on customer experience, and conduct cost-benefit analyzes to assess the economic feasibility of biophilic designs. In addition, exploring how design elements can better adapt to environmental changes and promote sustainable development, as well as how to combine biophilic design elements with emerging technologies to create a more interactive and personalized customer experience, will provide unprecedented opportunities for the fields of interior design and hotel management. insights and innovation opportunities.

APPENDIX

Please check the following appendix in:

https://github.com/WUChunya/Indirect-Biophilic-Attributes-in-Hawaii-Hotel-Interior-Design.git

1. 35000 Comments for NLP
2. BDIA rating form for 36 Hawaii hotels (for machine learning)
3. BDIA rating Scoresheet_Final Version
4. Customer Review Questionnaire
5. Waikiki 26 4-star Hotel Attributes NLP Analysis result by BDIA Keywords
6. Waikiki 26 4-star Hotel Attributes NLP Analysis result by Hotel Names
7. Waikiki 26 4-star Hotel review NLP analysis code

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